UNIVERSAL PHYSICS

Universal Physics

A Map of Why a Final Theory Is Open — and What Would Close It

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Contents

Preface

This book is a map of a question, not an answer to it. The question is the oldest and most abused in physics: is there a single universal theory underneath everything, and if so, what would it take to find it and to know we had found it? Shelves sag under books that answer yes and then sell you a candidate. This one does something different. It maps, as carefully as we could manage, the best current understanding of fundamental physics; it locates exactly where that understanding breaks; it sorts the field's assumptions into bedrock and historical accident; it surveys the candidate roads to unification without favoritism; and it delivers a verdict about what the evidence actually supports — which is neither a theory of everything nor a proof that none can exist.

The verdict, stated up front because you should not have to wait for it: a universal theory is **not forbidden and not delivered**. The two pillars of modern physics, quantum field theory and general relativity, are mutually consistent everywhere we can currently test them. None of the seventeen inter-framework clashes catalogued in chapter 4 is a proven logical contradiction; they are domain mismatches and unsolved-but-consistent problems, concentrated where both theories are extrapolated past their tested regimes. A consistent unified framework is therefore very plausibly possible. But the most developed candidate direction — the wager that causal, algebraic, and entanglement structure is more fundamental than metric geometry — turns out, under sustained adversarial pressure, to *encode* geometry without *generating* it, and to have no distinguishing experimental test even in principle. By this project's own standard, that makes it a sharp and coherent research strategy, not yet physics. Chapter 12 states this verdict in full, with its three explicit hedges. It is a stable conclusion, not a final one: the same chapter specifies precisely what would reopen it.

A word on how the book was made, because the method is unusual and the reader deserves to be able to discount for it. The text began as a research wiki built and refined over twenty-three iterations by a multi-agent process: teams of AI research agents, directed by a human author, each iteration deploying dozens of agents on focused questions — and, critically, every new claim passed through an adversarial referee step whose job was to break it. Iteration 2 stress-tested the central wager against the recent literature. Iteration 3 stopped surveying and started *attempting* — trying to derive the results the wager would need, and red-teaming the project's own posture. Iterations 4 and 5 executed the remaining designated verdict-flipping moves (they did not flip) and declared analytical saturation. Later iterations injected new literature, survived a maximal adversarial assault, ran the project's first numerical computations, proved what could be proved, and compressed the one remaining gate to a pair of precisely-posed lemmas; the verdict survived all of it, tightened each time — twenty-two consecutive refereed confirmations at this writing. Citations were web-verified against the actual papers, not recalled from memory; the bibliography contains real references only. The project's anti-crank protocol was simple and absolute: never let a speculative idea wear the clothes of an established fact, prefer a negative result to a flattering one, and treat overclaiming as a defect rather than a flourish. Where the investigation corrected itself — and it did, more than once — the correction is recorded, not erased.

The visible instrument of that protocol is the system of epistemic tags you will meet on nearly every page: `[ESTABLISHED]`, `[INFERENCE]`, `[SPECULATIVE]`, `[OPEN]`, and `[CONTESTED]`. They are defined precisely in Chapter 2, but the spirit is this: `[ESTABLISHED]` means experimentally confirmed or rigorously proven, with broad expert consensus; `[INFERENCE]` means a strong, standard bet that is not yet a fact; `[SPECULATIVE]` means principled but undecided; `[OPEN]` means the honest answer is *we do not know*; `[CONTESTED]` means competent experts actively disagree. The tags are not decoration. They are the book's load-bearing honesty, and reading them is reading the book. A claim's tag tells you how hard to lean on it, and the most important sentences here are often the ones tagged weakest.

Be clear, too, about what the verdict is not. It is not a proof that unification is impossible — we explicitly decline the fashionable Gödelian overreach. It is not a dismissal of the emergent-spacetime program, which the investigation found to be the most rigorous and self-aware road on the map; the diagnosis "encodes, does not generate" was reached by taking that program more seriously than its popularizations do. And it is not a theorem: it rests on named hedges, graded at medium confidence, which Part VI sets out without embarrassment. What it is, we hope, is the most precise available account of *why* the question remains open — and a usable specification of the analytical construction or experimental result that would close it.

On authorship. This book was directed, scoped, and edited by a human, Harshit Khemani, and researched and written in collaboration with Claude, an AI system built by Anthropic; the research labor — the literature sweeps, the derivation attempts, the refereeing — was substantially the machine's, under human direction and judgment. We list both names because both did the work, and because pretending otherwise would violate the book's own first rule. Whatever errors survive the process are failures of that process, and we ask readers to report them: the living version of this text, with its full working notes and change log, remains public at universal-physics.khe.money and github.com/HKTITAN/universal-physics, licensed for reuse.

The question is open. Here is the map.

Harshit Khemani & Claude — June 2026

PART I

THE QUESTION AND THE GROUND

The question, the method, and the ground it stands on.

CHAPTER 1

The Question

Whether one theory can underwrite all of physics is not a mystical question — it is a precise one, and the honest state of the evidence is stranger than either the optimists or the pessimists suppose.

Two towers

Physics at the start of 2026 rests on two theories, and both of them work better than they have any right to.

The first is relativistic quantum field theory, whose crowning instance is the Standard Model of particle physics: an anomaly-free chiral gauge theory with the group structure $SU(3)_c \times SU(2)_L \times U(1)_Y$, describing the strong, weak, and electromagnetic interactions of every particle ever detected. Its most famous prediction is the anomalous magnetic moment of the electron, computed order by order in perturbation theory and confirmed by experiment to roughly ten significant figures. There is no other quantitative statement in all of science with that pedigree. When a theory and an experiment agree to one part in ten billion, something about the theory is deeply right.

The second is general relativity, Einstein's theory of gravity as spacetime geometry. It began its confirmation career with the perihelion of Mercury and the bending of starlight, and has since survived every test we could devise: the Shapiro time delay, the orbital decay of binary pulsars, the direct detection of gravitational waves by kilometer-scale interferometers, and the imaging of black-hole shadows by a telescope the size of the Earth. Below the Planck scale it is, moreover, a perfectly predictive effective field theory — not merely a classical picture but a quantitative tool.

Around and beneath these two sit further structures that any future theory must reproduce rather than overturn. The Hilbert-space formalism of quantum mechanics, with unitary evolution and the Born rule — whose *form* is not a free choice but is forced, for state spaces of dimension three or more, by Gleason's theorem. The loophole-free violation of Bell inequalities, telling us that nature is non-locally correlated, together with the no-signaling property, telling us that these correlations cannot carry a message. Thermodynamics and statistical mechanics, sharpened in recent decades by exact fluctuation theorems and by the renormalization-group explanation of why utterly different materials share the same critical exponents. A six-parameter concordance cosmology, ΛCDM, that fits the cosmic microwave background, primordial nucleosynthesis, and the large-scale structure of the universe. And a set of structural theorems that function as bedrock: Noether's link between symmetry and conservation, Wigner's classification identifying a relativistic particle with a unitary irreducible representation of the Poincaré group, microcausality, unitarity, and Lovelock's rigidity theorem showing that Einstein's equation plus a cosmological constant is essentially the *unique* second-order metric theory of gravity in four dimensions.

One of those structures deserves its equation on the page, because it sits at the exact junction of everything this book is about. A black hole — the most gravitational object there is — carries an entropy, the quantity thermodynamics uses to count hidden microscopic arrangements:

$$S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar}$$

Here A is the area of the black hole's horizon, G is Newton's gravitational constant, $\hbar$ is Planck's constant, c is the speed of light, and k_B is Boltzmann's constant. In plain language: the formula says that the information capacity of a black hole is proportional to the *area* of its boundary, not its volume, and the proportionality constant welds together gravity (G), quantum mechanics ($\hbar$), relativity (c), and thermodynamics (k_B) in a single expression. It is the one formula in physics where all four fundamental constants meet, and it is the strongest hint we possess that the two towers are storeys of one building.

And yet. The two towers rest on foundations that do not merely differ — they contradict each other's basic postures toward the world. Quantum field theory, in its standard Minkowski formulation, presupposes a fixed, non-dynamical spacetime metric: a rigid stage on which fields fluctuate, anchoring the very definitions of "particle," "vacuum," and "cause precedes effect." General relativity's defining feature is precisely the opposite: the metric is dynamical, there is no prior geometry, and the stage is one of the actors. One theory needs the arena fixed; the other insists the arena moves.

So the question this book takes seriously is the obvious one. Can there be a single theory beneath both?

What "universal" would even mean

The phrase "theory of everything" has been so thoroughly abused that it is worth being pedantic about what is actually being asked. The question splits into three senses, and they have sharply different statuses.

Sense A: a consistent unified framework. Is there *some* mathematically consistent theory that contains both quantum field theory and general relativity in their tested regimes — a quantum theory of gravity? This is the weakest sense, and the answer is very plausibly *yes*. The catalogued clashes between the pillars (we will come to them shortly) are not logical contradictions; the consistency conditions on any quantum theory of gravity are known to be extraordinarily rigid; and the AdS/CFT correspondence provides a working existence proof that some consistent quantum gravity exists — with the honest caveat, added by this project's own refereeing, that the correspondence is itself an unproven conjecture, so the proof is a demonstration-by-example rather than a theorem. Whether the right framework *for our universe* is ever found and empirically confirmed is a separate, genuinely open matter.

Sense B: a final substrate. Is there a true bottom to the tower of theories — a layer that is not itself the low-energy shadow of something deeper? Here the evidence is far more equivocal. The great lesson of the renormalization group is that "fundamental" and "emergent" are partly scale-relative notions; the Wilsonian picture of physics as a tower of effective theories, each valid in its regime, is compatible with there being *no* bottom at all. That "no fundamental bottom layer" reading is a serious live alternative, not a fringe position.

Sense C: a complete theory. Would a final theory fix all the contingent numbers — the particle masses, the mixing angles, the cosmological constant — leaving nothing arbitrary? This is the *least* likely sense. The string-theory landscape read through anthropic selection, the cosmological-constant problem, and the apparently contingent low-entropy condition of the early universe each independently threaten the dream of parameter-fixing completeness, even if the underlying framework turned out to be unique.

Notice what disambiguation buys. "Is a universal theory possible?" sounds like one question and is actually three, with answers ranging from "very plausibly yes" through "genuinely unknown" to "probably not." Most of the loose talk on this subject — in both the triumphalist and the defeatist registers — comes from sliding between the senses.

One more piece of underbrush is worth clearing while we are here. It is sometimes claimed that Gödel's incompleteness theorems *forbid* a final theory. This investigation explicitly declines that argument: undecidability of some of a theory's *consequences* is not incompleteness of its *laws*. A final set of equations could exist and be exactly right even if certain questions about their solutions were formally undecidable. Gödel closes no doors here.

The war that isn't

Now for the first genuine surprise this project produced — a result that reshaped everything downstream of it.

Everyone knows that quantum mechanics and general relativity are "incompatible." It is the founding cliché of quantum-gravity research, the opening slide of a thousand colloquia. So one of the first sustained exercises of this investigation was to make the cliché precise: catalogue every known clash between the accepted frameworks of physics — seventeen of them, in the final registry — and subject each to adversarial refereeing, asking a deliberately cold question. *Exactly what kind of conflict is this?*

The methodological instrument was a four-way distinction that informal discourse routinely collapses:

- A **logical inconsistency**: two accepted statements that cannot both be true — a contradiction derivable within a single fixed set of assumptions. This is the strong thing people usually *mean* to allege.
- A **domain-of-validity mismatch**: two frameworks, each internally consistent, resting on incompatible idealizations, clashing only where their domains are forced to overlap — typically in regimes where *neither* has been tested.
- An **unsolved-but-consistent problem**: a well-posed question with no accepted answer, but no contradiction anywhere — we simply lack a derivation.
- A **conceptual tension**: a clash of interpretation or worldview with no operational or empirical conflict at all.

The result of running all seventeen clashes through this filter is the single most consequential negative finding in the whole investigation: **not one of them is a demonstrated logical inconsistency.** Under refereeing, every alleged contradiction resolved into one of the three weaker categories — and, tellingly, they cluster precisely where both theories are being extrapolated far beyond their tested regimes: the Planck scale, black-hole interiors, the first instants of the universe.

Three examples convey the flavor.

The *background-dependence clash* — the foundational opposition described above — survives as the deepest structural obstacle, but the popular framing that the two frameworks are "mutually exclusive in principle" was explicitly downgraded by the referee as false. Quantum field theory on a fixed *curved* background is mathematically rigorous, and perturbative graviton theory places a dynamical, even superposable, metric perturbation inside quantum field theory perfectly consistently below the Planck scale. The genuine, unsolved obstruction is narrower and stranger: a *superposed or fluctuating causal structure*, which deprives microcausality and the particle and vacuum concepts of their usual definitions. That is a frontier, not a paradox.

The *cosmological-constant problem* — the famous claim that quantum field theory mispredicts the energy of empty space by 120 orders of magnitude — turns out, in strict effective-field-theory logic, not to be a falsified prediction at all: the vacuum energy is a renormalized free parameter, and the notorious "10^{120}" is a regularization-dependent artifact (an electroweak cutoff gives a mere 10^{60}). What genuinely fails is *naturalness* — known physical condensates already overshoot the observed value by forty to sixty orders — which makes this a severe fine-tuning puzzle, an open problem, and not a contradiction.

The *black-hole information paradox*, poster child of quantum-gravity crisis, was likewise classified as a domain-of-validity tension rather than a logical inconsistency: unitarity is now widely favored, the island and replica-wormhole computations reproduce the expected information-return curve in controlled settings, and what is missing is a first-principles mechanism for a realistic four-dimensional black hole — an enormous open problem, but a problem, not a proof of inconsistency.

Two things follow from this classification result, and they pull in opposite directions. First, the optimistic pull: since the clashes are mismatches and puzzles rather than contradictions, a consistent unified framework is not blocked by any known obstruction — sense A above really is plausibly achievable. Second, the sobering pull: there is *far more established, mutually consistent physics* than the "GR versus QM at war" framing implies. The incompatibility is real but narrow and located — and a narrow frontier, far from experiment, is exactly the kind of place where theory can wander for decades without anything to check it. Mislabeling that frontier as a paradox is itself an epistemic error, and this book will try never to commit it.

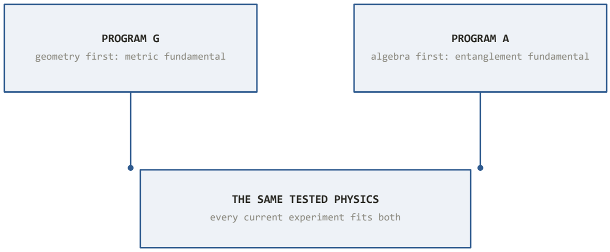
The wager

If the clashes do not force unification, what positive evidence points toward a deeper layer? Here the modern literature exhibits a remarkable convergence. Several of the strongest results in mathematical physics — the black-hole entropy formula above; the holographic bound it inspired; the Ryu–Takayanagi formula, which computes entanglement entropies in a boundary quantum theory as *areas* of surfaces in a bulk geometry; Jacobson's 1995 derivation of Einstein's equation as an equation of state; Van Raamsdonk's argument that reducing entanglement between two regions pinches the spacetime connecting them; and the deep operator-algebraic fact that the local algebras of quantum field theory are of "Type III_1 ," so spacetime regions do not factor into independent quantum subsystems the way qubits in a lab do — all point the same way.

They suggest one wager: **causal, algebraic, and entanglement structure is more fundamental than metric geometry.** Spacetime, on this bet, is not an ingredient of the final theory but an output of it — geometry is what certain patterns of quantum correlation *look like*.

This is the leading modern direction toward a universal theory of physics. It is held, in various dialects, by a substantial fraction of the field's most serious people. And it is a genuine scientific wager rather than a mystical slogan: it has mathematical content, partial derivations, and identifiable failure points.

When this investigation surveyed the live research directions and their underlying commitments, however, the terrain organized itself in an unexpected way. There is not one unified program chasing the wager. There are **two** internally coherent programs, joined only by thin bridges.



no observation yet separates the two programs; that is the problem

Two programs, one body of evidence. Every experiment to date is consistent with both a geometry-first and an algebra-first reading of physics.

Program A — call it *geometry-from-algebra* — takes the operator algebras, modular flows, and entanglement structure of quantum theory as primitive and tries to derive spacetime: holography, the crossed-product entropy constructions, entropic gravity. Program B asks a logically prior question: *is the quantum framework itself final?* — pursued through reconstruction theorems that derive the quantum formalism from operational axioms, and through tabletop experiments probing whether gravity is quantum at all. The two programs share only a pair of load-bearing planks: the no-signaling/causal-order principle, and the logical prerequisite that gravity be non-classical. The tempting fusion — "it's all information!" — was examined and explicitly **refused**: the word "information" names non-interchangeable mathematical objects in the two clusters, and pretending otherwise is exactly the kind of verbal unification this project exists to catch.

That refusal is a small preview of the book's temperament. So is the following pair of findings, which the coming chapters will develop in full and which we state here only in outline. The most developed program, A, *encodes* geometry without yet *generating* it: every load-bearing result in the geometry-from-entanglement literature — by the concession of its own architects — presupposes a fixed background spacetime with a special symmetry, so the geometry that comes out was, in a precise sense, put in. And the central wager itself currently has **no distinguishing experimental test even in principle**: every proposed signal would be produced identically by a theory in which geometry is fundamental. A research direction in that condition — explaining much, predicting nothing that its rivals do not — is, by the standards this project holds itself to, *not yet physics*. It is a sharp, coherent research strategy. A strategy is not a result.

The shape of this book

What you are holding is the story of taking that wager utterly seriously — more seriously, in one respect, than its own proponents usually do — and attempting to referee it to destruction.

The instrument was an unusual one: an adversarially refereed research wiki, maintained by a large language model under a fixed and binding epistemic constitution, directed by a human, and driven through twenty-three iterations of attack. Every claim entering the record was tagged with its evidential grade; every hypothesis was red-teamed against the great no-go theorems of physics; every load-bearing citation was verified against the primary literature; and every iteration ended with an adversarial referee whose default stance was to refute. When the process caught its own errors — and it did, including at moments when a spectacular positive result seemed within reach — the erroneous claims were struck in public, with the cause of death recorded. Chapter 2 is devoted to this machinery, because in an investigation of this kind the method is not scaffolding around the result. The method *is* the result's warrant.

Since the tags will appear throughout the book, here is the convention once, in full. Every nontrivial claim in the underlying record carries exactly one of five grades: `[ESTABLISHED]` — experimentally confirmed to high significance or rigorously proven within an accepted framework; `[INFERENCE]` — the standard conclusion of careful reasoning, a strong bet but still defeasible; `[SPECULATIVE]` — principled but lacking decisive evidence; `[OPEN]` — the honest answer is that we do not know; and `[CONTESTED]` —

competent experts actively disagree. The cardinal rule of the whole enterprise is that a speculative idea must never wear the clothes of an established fact. In these pages the tags appear sparingly, where a claim's grade is itself the point — but every claim in the book inherits one.

And so that no suspense is manufactured where none exists, here is where twenty-three iterations landed. The verdict — stable across twenty-two consecutive attempts to move it, including iterations whose explicit assignment was to overturn it — reads: **a single universal theory of physics is not forbidden and not delivered.** The coherence of the field's best efforts is PARTIAL: two programs, thin bridges, no fusion. The leading direction *encodes* geometry but does not *generate* it. The central wager is *not yet physics*, for want of any test that could distinguish it from its rivals even in principle. And the object-level question — is there a final theory, and is it this one? — is *not forecastable*: gated less by the fifteen orders of magnitude between our accelerators and the Planck scale than by the absence of any experiment that bears on the wager at all. The entire remaining analytical content of the question was ultimately compressed, over the final iterations, into a single named mathematical problem — the *carrier problem* — now posed as a self-contained challenge to external mathematics, at which point reasoning had taken the gate as far as argument can go.

That may sound like an anticlimax. The chapters ahead are an argument that it is the opposite. A verdict that refused twenty-two consecutive invitations to inflate itself — that survived its own maximal adversarial assault, retracted its own errors, and ended by handing mathematics a precisely posed open problem rather than a manifesto — is not a failure to answer the question. It is what an honest answer to this question currently looks like. The drama of this book, such as it is, lives entirely in that discipline: in watching, iteration by iteration, the machinery refuse to say more than the evidence allows, and discovering how much that refusal reveals.

We begin with the machinery.

CHAPTER 2

The Method

Before an AI can be trusted with the hardest question in physics, it must be placed under a constitution designed to catch it — and this chapter is that constitution, together with the two occasions it drew blood.

The failure mode this subject selects for

Every field has a characteristic way of going wrong. In foundational physics — and above all in the pursuit of a "theory of everything" — the characteristic failure is overclaiming: the speculative idea dressed in the clothes of established fact. The subject seems almost designed to elicit it. The stakes are absolute, the frontier is fifteen orders of magnitude from experiment, and the raw materials are mathematically gorgeous. A plausible-sounding wrong idea about quantum gravity can survive for decades, because nothing in a laboratory will ever contradict it. The graveyard of unified field theories is full of intelligent people who mistook coherence for evidence.

Now hand the problem to a large language model, and every one of those risks compounds. An LLM's native failure modes are precisely the crank's, executed at scale and with superhuman fluency: it will assert confidently beyond its evidence; it will, under pressure to be helpful, tell you your hypothesis is promising; it will fabricate a citation — author, year, journal, page — with the same serene tone it uses for real ones; and in a long, cumulative project, an error admitted on Tuesday becomes load-bearing context by Thursday, silently propagating into everything built on top of it. An unconstrained language model asked "is a universal theory of physics possible?" is not a research instrument. It is a machine for generating beautiful, confident, unfalsifiable prose — which is to say, a machine for generating exactly the substance this field already has in oversupply.

The investigation this book recounts was conducted by a large language model anyway — roughly one hundred sessions of research labor, directed by a human — and the reason its output deserves any attention at all is the machinery this chapter describes. The wager of the *project* (as distinct from the physics wager of Chapter 1) was that the failure modes are not intrinsic but disciplinary: that an AI investigation operating under a fixed, written, binding epistemic constitution — one that tags every claim, attacks every hypothesis, verifies every citation, and retracts in public — could produce something a careful human community would recognize as honest research. Whether that wager paid off is something the reader can judge from the fact that the investigation's twenty-three iterations *refused, twenty-two consecutive times, to confirm the hypothesis they were built to test*. Machines for telling you what you want to hear do not behave that way.

The method, then, is the protagonist of this book. Here is its anatomy.

Three layers and one cardinal rule

The project is structured as a persistent research wiki — a pattern with three layers. At the bottom sit the **raw sources**: the textbooks, the papers, the data. These are immutable; the project cites them and never edits them. In the middle sits **the wiki itself**: interlinked pages of compiled understanding — domain dossiers, registries of open problems and contradictions, a theory map, a findings synthesis — written once, enriched in place, and reused, rather than re-derived from scratch on every query. At the top sits **the schema**: the written rules of structure, labeling, and workflow that govern how the other two layers may be touched. The schema is short, explicit, and binding; every agent (human or machine) that edits the wiki is bound by it. Knowledge compounds; so, unless checked, would error — which is why the schema exists.

The heart of the schema is the tagging discipline introduced at the end of Chapter 1: every nontrivial claim carries exactly one of **`[ESTABLISHED]`**, **`[INFERENCE]`**, **`[SPECULATIVE]`**, **`[OPEN]`**, or **`[CONTESTED]`**. What makes the tags an instrument rather than a decoration is that each comes with an operational test. For `[ESTABLISHED]`: *would the overwhelming majority of working physicists stake their reputation on this?* For `[INFERENCE]`: *is this the standard conclusion of careful reasoning, while still logically defeasible by future data?* For `[SPECULATIVE]`: *is there a real argument for it, but no experiment that forces it?* The calibration examples in the project's own rulebook convey the intended sharpness. That general relativity predicts gravitational waves, now directly detected — `[ESTABLISHED]`. That dark matter is a particle — `[INFERENCE]`, and note the split: the gravitational evidence for missing mass is `[ESTABLISHED]`; that the missing mass is a *particle* is the leading inference, not a fact. That spacetime geometry is emergent from entanglement — `[SPECULATIVE]`. When one sentence mixes a fact and a guess, the rules require it to be split into two sentences so each can wear its own tag.

Two refinements complete the system. Tags of the middle grades may carry a coarse confidence — high, medium, low — with the rulebook's dry warning that this is a subjective credence, never a promotion: "a high-confidence speculation is still a speculation." And *assumptions* are graded on an entirely separate axis — not how true, but how load-bearing and how revisable: from `fundamental` (relaxing it breaks the theory or contradicts robust data) down through `plausibly-fundamental` and `historical-artifact` to `mathematical-convenience` and `untested-empirically`. A central research activity of the whole project is the attempt to move assumptions *down* this list — to demote supposed bedrock into derivable or contingent choices. Chapter 1's four-way classification of "contradictions" — logical inconsistency versus domain mismatch versus unsolved-but-consistent problem versus conceptual tension — is a third instance of the same instinct: refuse the blurry word, force the precise one.

Above all of it stands the cardinal rule, stated once in the schema and enforced everywhere: **never let a speculative idea wear the clothes of an established fact.** Overclaiming is treated as a defect, not a flourish. Tag inflation is the project's equivalent of falsifying data.

The referee

Tags discipline *assertion*. A second mechanism disciplines *belief*: no hypothesis enters the record as "kept" until it has survived an adversarial referee whose written protocol is worth quoting in structure.

First, the claim must be stated precisely, *including its mathematics*. Vague ideas cannot be refereed and are therefore not eligible to be believed. Second, it is red-teamed against the standard no-go theorems — the great results of twentieth-century physics that function as minefields under any unification proposal: Bell, Kochen–Specker, and PBR on hidden variables; Coleman–Mandula and Haag–Łopuszański–Sohnius on combining symmetries; Weinberg–Witten on emergent gravitons; Haag's theorem; Reeh–Schlieder; the singularity theorems; spin–statistics; the unitarity and causality bounds — *and* against the project's own accumulated closures, so that a hypothesis killed in iteration four cannot shamble back in iteration nine wearing a new hat. Third, the referee identifies errors and overclaims and produces a corrected statement. Fourth, a severity is assigned — none, minor, major, or fatal — and `fatal` means the hypothesis is not kept. Crucially, it is not deleted either: it moves to a rejected section *with the reason*. The project keeps the corpse and the cause of death, because a graveyard with legible headstones is how you stop re-fighting settled battles.

In the iterated research phase this protocol hardened further. Each iteration's attack tracks were refereed by independent adversarial agents whose *default stance was to refute* — their instructions presumed each submitted claim wrong and demanded it prove otherwise — and whose corrections were **binding over the original claims**. A finding did not enter the record as its author submitted it; it entered as its referee left it, and the changelog records, iteration after iteration, submitted conclusions being downgraded, overturned, or struck by referees before they could touch the standing verdict. In one late iteration the polarity was inverted entirely: the attack tracks were ordered to *assume the verdict false* and build the strongest possible case against it, with the referee assigned to defend. The verdict held that too.

One further clause completes the gate, and it gives the protocol its house name — the *anti-crank* gate: **a candidate that explains everything and predicts nothing fails by construction.** No distinguishing prediction, no admission to physics. It sounds like boilerplate. Chapter 1 already disclosed how it detonates: applied without fear or favor, this clause is what classifies the leading direction in modern quantum gravity — the entanglement-first wager itself — as *not yet physics*. A rule that only catches outsiders is a shibboleth. This one was allowed to catch the mainstream.

Nothing invented, everything checked

The third mechanism targets the most notorious LLM failure of all. The sourcing rules open bluntly: **never fabricate a citation, an experimental number, a theorem, or a date. A missing reference is acceptable; a fake one is a disqualifying error.** Where bibliographic details are uncertain, the required move is to say so — author, approximate year, and an explicit `[unverified]` marker — rather than to guess. Numbers must carry their caveats (the famous "10^{120}" of the cosmological-constant problem may not be quoted without noting it is regularization-dependent), and folklore must be flagged rather than parroted.

But rules against fabrication are only as good as their enforcement, so the later iterations added *live verification*: load-bearing citations were checked against the primary literature — arXiv identifiers resolved, PDFs actually retrieved and read — during the refereeing itself. The record shows the machinery earning its keep. In one iteration, a pipeline summarizing a PDF fabricated content; the fabrication was caught in-session, corroborated by the referee, and recorded as a process incident rather than propagated as a fact. In another, a track attributed a numerical result to the wrong authors; the referee caught the misattribution before it was written into any page. And in the iteration that turned the project's own numerics into a theorem, the referee *downloaded and read* a 1989 paper of Figliolini and Guido from the original journal archive to confirm that a claimed result really was proven there, on the page cited — at which point, and only at which point, a tag that had sat at `[unverified]` for two iterations was promoted to `[ESTABLISHED]`. The tag had been parked at `[unverified]`*despite the claim being true*, because truth was not the standard; verification was.

You may not install your answer

The fourth mechanism is the subtlest, and it is the one that ends up carrying the book's central physics finding, so it deserves careful statement here.

In emergence arguments there is a perennial sleight of hand: a construction *derives* spacetime geometry from something deeper — except that, on inspection, the geometry was hidden in the inputs. The symmetry that was assumed, the mathematical pairing that was posited, the slicing that was chosen, the template that was installed: somewhere in the machinery, the answer went in by hand, and the derivation is a conjuring trick performed sincerely. The project elevated the prohibition into a formal rule: **a construction that installs its own answer is a failure**, and it maintained an explicit, growing checklist of smuggling routes — a fundamental symmetry, an indefinite pairing, a foliation, a twist, a signature-valued template — each added to the list as some construction in the literature was caught using it. The project's standard for a genuine derivation of spacetime was eventually compressed into a single sentence: a construction that outputs Lorentzian causal structure from non-symmetric algebraic data *installing none of the* *listed items by hand*. Much of this book is the story of watching every known construction, including several of the project's own devising and several published by eminent hands, fail that test — always at the same step, a convergence that itself became a result.

The reader will notice that all four mechanisms are the same idea in four costumes: *the burden of proof sits with the claim, permanently*. Tags make the burden visible; referees enforce it; citation verification extends it to the literature; the anti-smuggling rule extends it to mathematics. What remains is to show the machinery actually firing — because a constitution is judged by its case law. Two episodes, one from the middle of the campaign and one from its final iteration, are the exhibits.

Exhibit one: the day the verdict almost moved

By its tenth iteration, the project's entire remaining uncertainty had been compressed into one principal hedge — a single conditional clause standing between the standing verdict and something much stronger. The hedge's condition involved a precisely located gap in an operator-algebraic argument (its technical content will occupy Chapters 7 and 8; for the present story it is enough that closing the gap was the *designated verdict-bearing lever*, named in advance: close it, and the central no-go result would harden to unconditional highest confidence). Iteration 10 was commissioned to attack that gap head-on.

The attack track came back triumphant. It submitted a proof sketch that the gap **closes**: a known ergodicity property of the vacuum state, it argued, empties the problematic set of operators, repairing the located deficiency and letting the hedge shed its condition. Had the claim stood, it would have been the single largest positive advance of the entire program — the one lever the project itself had said would move the verdict, pulled successfully.

The referee struck it. The submitted argument, the refutation ran, had confused two different mathematical objects with confusingly similar definitions: the modular *centralizer* — the operators inside the algebra $\mathcal{M}$ left fixed by the modular flow, $\mathcal{M}_\omega = \mathcal{M} \cap \{\Delta^{it}\}'$ — with a strictly larger *representation-space* set of operators commuting with the flow but living outside the algebra. (The displayed formula just says: the centralizer is the intersection of the algebra with the commutant of the flow — the point being that the ergodicity theorem trivializes this intersection, and *only* this intersection.) Ergodicity empties the centralizer; the iteration-9 gap lived in the larger set, which stays non-empty — and, mordantly, the very object at the center of the project's carrier problem survives inside it. The claimed closure was graded a **major error** and struck in full. The changelog entry is a small monument to the discipline: the hedge advanced in *neither grade nor condition*; the erroneous claim is preserved verbatim with its refutation; and of the iteration's three submitted outcome labels, *two* were corrected by referees — the second track's "resolved negative" was independently overturned on both physical and arithmetic grounds. The iteration that was designed to deliver the program's greatest victory instead delivered a null result, published as such, with the corpse on display.

That is the method working. A system optimized to please would have reported the gap closed — the mathematics was plausible, the conclusion was desired, and the error required real expertise to catch. The constitution caught it because the constitution assumes every submission is wrong until an adversary fails to kill it.

Exhibit two: a retraction in the last mile

Twenty-two iterations later — the final one in this book's record — the residual question had been ground down to a handful of named lemmas in spectral theory, and the pressure to finish was at its maximum. One track submitted a monotonicity claim (a statement that a certain function of the spectral data moves in only one direction) graded `[ESTABLISHED]` by its author. A *different* track, running independently in the same iteration, had produced a machine-verified computation — checked by explicit numerics, not judgment — that contradicted it.

The contradiction surfaced not in either track but in *assembly*: the stage where an integrating referee reconciles all tracks' outputs before anything touches the record. Confronted with an `[ESTABLISHED]` claim and a machine-verified computation that could not both be true, the assembly did the only thing the constitution permits — the *claim* was struck to `[OPEN]`, the retraction was logged by name in the changelog ("the discipline caught it in assembly"), and the iteration's summary records one honest retraction as part of its results, alongside its theorems. Under the retraction-on-contradiction rule, no page may quietly disagree with another: every promotion and demotion of a claim's tag is logged with a one-line reason in an append-only changelog, so the belief-revision history of the entire project is inspectable end to end. Even at iteration 23 — even with the finish line in sight, even for a claim whose truth would have been convenient — the machinery preferred a public retraction to a silent inconsistency.

What the discipline bought

It is worth stating plainly what these mechanisms cost. Twenty-three iterations produced, at the headline level, *no confirmation* — twenty-two consecutive refusals to move the verdict. Spectacular submitted results were struck. Promising claims spent months at `[unverified]`. An enormous fraction of the total labor went not into generating ideas but into attacking, checking, correcting, and burying them with headstones.

And that expenditure is precisely why the residue can be trusted. The project's founding schema declares that both outcomes of the investigation are valuable — a properly backed candidate framework, *or* a clear, reasoned account of why the evidence does not yet support one — and that "*we do not know*, said precisely, is a result." The chapters that follow will lean on that principle constantly, because the story they tell is dominated by negative results: no-go theorems, failed constructions, closed escape routes, retracted closures. Under the constitution just described, each of those negatives is load-bearing. When this book reports, in the chapters ahead, that every known route to generating spacetime from entanglement closes at the same step, the reader will know what stands behind the sentence: not an impression, but a claim that survived referees whose job was to destroy it, in a record where every failure to destroy is logged, and where the two most seductive opportunities to overclaim — a verdict-flipping lever in iteration 10, a convenient lemma in iteration 23 — were publicly declined.

The method, in short, is not the scaffolding of this investigation. It is the load-bearing wall. What it held up — the physics itself, from the established core to the two programs to the single mathematical gate on which everything came to rest — is the business of the rest of this book.

CHAPTER 3

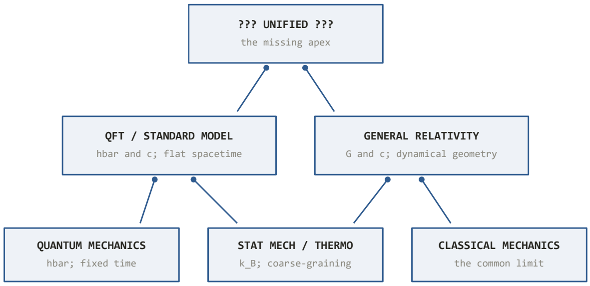
What Is Actually Established

Before asking what is broken, take inventory of what is not — and discover that the vault is far fuller than the folklore suggests.

Popular accounts of fundamental physics tend to open with a crisis. General relativity and quantum mechanics, the story goes, are locked in irreconcilable combat; physics is a house divided; somewhere beneath our feet the foundations are cracked. The first task of any honest audit is to check that story against the ledger — and the ledger tells a different tale. When the investigation behind this book began cataloguing what physics has actually secured, the striking result was not the size of the gaps. It was the size of everything else.

This chapter is a tour of the load-bearing core: the results that any future theory, however revolutionary, must reproduce rather than overturn. A future physics can reinterpret these results, embed them in something deeper, reveal them as limits of a grander structure — what it cannot do is contradict them, because they are not speculations. They are measured, re-measured, and in several cases proven as mathematics.

A word on bookkeeping before we start. Throughout this book, claims carry the five epistemic grades introduced at the end of Chapter 1, from `[ESTABLISHED]` down to `[CONTESTED]`, because the entire method of the investigation was to refuse to let well-tested results and hopeful conjectures share a sentence without labels. Everything in this chapter, unless flagged otherwise, is `[ESTABLISHED]`. That is precisely the point of the chapter.



The reduction tower. Each framework passes to the one below it in a controlled limit; the apex is the missing piece.

Quantum mechanics: weird in a precise way

Quantum mechanics is a century old, and its strangeness has calcified into cliché — the cat, the dice, the observer. What the clichés obscure is how *precisely* the strangeness has been pinned down. The theory is not vaguely mysterious; it is mysterious in a mathematically exact, experimentally certified way.

The formal core is compact. States of a system live in a Hilbert space — a complex vector space with an inner product — and observable quantities correspond to self-adjoint operators on that space. A closed system evolves deterministically and reversibly by the Schrödinger equation. Probability enters only at measurement, through the Born rule: if ρ is the state and E the projection operator asking a yes/no question, the probability of "yes" is

$$\Pr = \mathrm{Tr}(\rho\, E),$$

which says, in words, that the likelihood of an outcome is computed by a specific algebraic pairing — the trace — of the state with the question being asked. For a pure state $|\psi\rangle$ this reduces to the familiar squared amplitude $|\langle a|\psi\rangle|^2$.

Here is the first thing the folklore misses: the *form* of that rule is not a free choice. **Gleason's theorem** (1957) proves that for any Hilbert space of dimension three or more, the trace formula is the *unique* way to assign probabilities to measurement outcomes, provided the assignment depends only on the question asked and not on which other questions are asked alongside it (noncontextuality). Given the Hilbert-space scaffolding, nature had no other option for how probability could work. The rule's form is forced `[ESTABLISHED]`; what remains genuinely open — and we will return to it in the next chapter — is why any *single* outcome occurs at all.

The second precision-sharpened strangeness is nonlocality. In 1964 John Bell converted a philosophical unease into an inequality. Suppose measurement outcomes are determined by some shared local variable λ carried by the particles — any such variable, of any complexity. Then for two experimenters choosing between measurement settings a, a' and b, b', the correlations $E(a, b)$ must satisfy the CHSH bound

$$S = \left|E(a,b) - E(a,b') + E(a',b) + E(a',b')\right| \leq 2,$$

which says that no theory in which each particle's behavior is fixed by locally carried information can produce correlations stronger than a definite numerical limit. Quantum mechanics predicts $S = 2\sqrt{2}$ for suitably entangled pairs — comfortably over the line. The experiments, culminating in the loophole-free tests of 2015 and after, find exactly the quantum value. Local hidden-variable theories are dead, not wounded `[ESTABLISHED]`.

And yet — this is the precise part — the violation cannot be used to send a message. The **no-signaling** property holds exactly: nothing one experimenter does to her particle changes the statistics the other sees on his, no matter how entangled the pair. Nature is non-locally *correlated* but not superluminally *communicating*. The two results together carve out an extraordinarily narrow logical channel, and quantum mechanics threads it. Any successor theory must thread it too.

The Standard Model: the most accurate theory ever tested

The Standard Model of particle physics is a specific quantum field theory: a chiral gauge theory with symmetry group $SU(3)_c \times SU(2)_L \times U(1)_Y$, describing the strong, weak, and electromagnetic forces acting on three generations of quarks and leptons, with masses generated by the Higgs mechanism. Every clause of that sentence is an experimental scar tissue of the twentieth century, but two of its structural properties deserve emphasis.

First, it is *anomaly-free*: the quantum consistency conditions that could easily have destroyed a chiral theory — one that treats left-handed and right-handed particles differently — cancel exactly, and they cancel only because the quarks and leptons of each generation conspire with precisely the right electric charges. This is either a deep clue or a remarkable accident; either way it is a fact.

Second, it is renormalizable, meaning its predictions at accessible energies are fixed by a finite list of measured inputs — and those predictions are good to a precision unmatched anywhere in science. The showpiece is the anomalous magnetic moment of the electron, where renormalized perturbation theory agrees with measurement to roughly ten significant figures. To put that in earthly terms: it is like predicting the distance from New York to Los Angeles to within the width of a human hair, from first principles, before making the trip.

The strong-interaction sector contributes its own established gem. The QCD coupling α_s is not constant but *runs* with energy scale μ, governed by the beta function

$$\mu \frac{d\alpha_s}{d\mu} = -\frac{\beta_0}{2\pi}\, \alpha_s^2 + \cdots\,, \qquad \beta_0 = 11 - \tfrac{2}{3} n_f,$$

which says that the strength of the strong force decreases logarithmically as you probe shorter distances, so long as the number of quark flavors n_f is below $33/2$ — as it is. This is **asymptotic freedom** (Gross, Wilczek, and Politzer, 1973): quarks rattle around almost freely inside a proton at short distances, yet are bound unbreakably at long ones, and the same equation that frees them at high energy generates, by "dimensional transmutation," the scale $\Lambda_{QCD} \sim 200$ MeV that sets the mass of the proton and hence of nearly everything you have ever weighed. Most of your mass is not Higgs-given quark mass; it is confined gluon field energy `[ESTABLISHED]`.

One honest asterisk belongs here, because the method of this book demands it. Neutrino oscillations — flavors morphing into one another in flight — prove that neutrinos have mass, which the *minimal* Standard Model forbids. This is the single laboratory-confirmed breakdown of the minimal theory. It is repaired by extensions that are conservative by the standards of this subject, but the count matters: after seventy years of trying to break it, the tally of confirmed cracks in the Standard Model, as tested in laboratories, stands at one.

General relativity: a century of passed tests

Einstein's theory of gravity makes a structurally different kind of claim than the Standard Model: not that spacetime contains fields, but that spacetime *is* a field — a dynamical geometry curved by energy and momentum, with matter moving along the straightest available paths in the curved landscape.

The experimental ladder is a century long, and every rung has held. In the weak fields of the solar system: the anomalous precession of Mercury's perihelion; the bending of starlight at the Sun's limb by 1.75 arcseconds; the Shapiro time delay of radar signals grazing the Sun, confirmed by the Cassini spacecraft to a few parts in 10^5; gravitational redshift, now measured so well it is a working correction inside every GPS receiver. In strong and dynamical fields: the Hulse–Taylor binary pulsar, whose orbit decays by gravitational-wave emission at exactly the predicted rate to better than 0.1 percent; the direct detection of gravitational waves from merging black holes beginning with GW150914 in 2015 (Abbott et al., *PRL* **116**, 061102), with the tens of subsequent mergers matching Einstein's waveforms; the Event Horizon Telescope's images of black-hole shadows; stars whipping around the black hole at the center of our galaxy, precessing on cue. On the largest scales, general relativity is the frame on which all of modern cosmology is stretched.

There is a subtler established point about GR that the "physics is broken" narrative routinely gets wrong, and it matters enough to state carefully. It is true — a theorem, in fact, discussed in the next chapter — that GR cannot be extended as a conventional quantum field theory to arbitrarily high energies. But *below* the Planck scale, quantized general relativity is a perfectly predictive effective field theory. One can compute genuine quantum corrections to Newton's potential from it, unambiguously. Gravity and quantum mechanics are not refusing to speak to each other in any regime we can reach; they converse fluently, and the conversation only breaks down at energies some fifteen orders of magnitude beyond our best accelerators. The incompatibility is real, but it lives at the extrapolated edge, not in the world of experiments `[ESTABLISHED]`.

Thermodynamics, sharpened — all the way to black holes

Thermodynamics is the oldest member of the core, and the twentieth and twenty-first centuries did not merely preserve it — they sharpened it into exact statements and then extended its writ to the most extreme objects in the universe.

The sharpening first. The second law, in its textbook form, is an inequality about averages: entropy does not decrease. Modern nonequilibrium statistical mechanics upgraded it to *exact equalities* valid arbitrarily far from equilibrium. The **Jarzynski equality** (1997) states that for a system driven by any external protocol from thermal equilibrium at inverse temperature $\beta = 1/k_B T$,

$$\left\langle e^{-\beta W} \right\rangle = e^{-\beta \Delta F},$$

which says that a particular exponential average of the fluctuating work W done over many repetitions equals a simple function of the equilibrium free-energy difference ΔF — an exact bridge between messy nonequilibrium driving and clean equilibrium quantities. Crooks' fluctuation theorem refines it further, giving the precise ratio of probabilities for a process to run "forward" versus "backward." Together they show that microscopic, transient *violations* of the naive second law are not forbidden but exponentially rare, in an exactly quantified way — and these predictions have been verified in single-molecule pulling experiments. The second law is no longer a Victorian decree; it is a corollary of something more precise `[ESTABLISHED]`.

The extension is the more astonishing story. In the early 1970s, Bardeen, Carter, and Hawking proved that classical black holes obey four laws formally identical to the four laws of thermodynamics, with horizon area playing the role of entropy and surface gravity the role of temperature. Bekenstein argued the resemblance was no analogy, and Hawking's 1975 calculation settled it: quantum field theory in the curved spacetime near a horizon forces a black hole to radiate at temperature $T_H = \hbar c^3/8\pi G k_B M$, and to carry an entropy

$$S_{\mathrm{BH}} = \frac{k_B\, c^3 A}{4\, G\hbar},$$

which says that a black hole's entropy is one quarter of its horizon area A, measured in units of the Planck length squared — a single formula containing the speed of light c (relativity), Planck's constant $\hbar$ (quantum mechanics), Newton's constant G (gravity), and Boltzmann's constant k_B (thermodynamics). It is the only uncontested equation in physics in which all four constants meet, and it is established within semiclassical gravity — quantum fields on a curved background — which is exactly the regime where it is derived. What microscopic states that entropy counts is `[OPEN]` in general, though string theory has counted them exactly for special supersymmetric black holes (Strominger–Vafa, 1996). Hold on to this formula: it is the single most important clue in this entire book, and Chapters 5 and 6 are an interrogation of what it is trying to tell us.

The concordance cosmos

Cosmology, within living memory a data-starved speculation, is now a precision science with an almost embarrassingly economical standard model. **ΛCDM** describes the universe from its first minutes to today using six numbers: the densities of ordinary matter and of cold dark matter, the expansion rate, the optical depth to the era when the first stars reionized the cosmos, and the amplitude and tilt of the primordial density ripples. With those six dials set, the model fits the acoustic peaks of the cosmic microwave background, the light-element abundances forged in the first three minutes, the statistical clustering of galaxies across billions of light-years, and the accelerating expansion revealed by distant supernovae [ESTABLISHED as a fit; the *nature* of the dark ingredients is another matter].

The fine print is where the honesty lives. The measured tilt of the primordial spectrum, $n_s \approx 0.965$, is slightly but decisively less than one — a nontrivial prediction of inflationary scenarios, confirmed. The universe is spatially flat to sub-percent precision. But the expansion rate measured from the early universe and the one measured from the local distance ladder currently disagree at the level of four to six standard deviations — the "Hubble tension" `[CONTESTED]` — and whether the acceleration is due to an exactly constant Λ or something slowly evolving is under active fire from new survey data, as the next chapter recounts. The concordance model concords; it also has a short, well-lit list of loose threads, and cosmologists can tell you precisely where each one is.

The structural theorems: physics carved in mathematics

Beneath the individual theories lies a stratum of results that are not empirical generalizations at all but theorems — mathematical rigidities that constrain what *any* physical theory can look like. These are, in a sense, the most established things we know, because no experiment can revise a proof; experiment can only show that a theorem's premises fail to describe our world.

Noether's theorem (1918): every continuous symmetry of a physical system's dynamics implies a conservation law. Time-translation symmetry *is* energy conservation; spatial symmetry *is* momentum conservation; the phase symmetry of quantum electrodynamics *is* charge conservation. This is why symmetry became the organizing language of twentieth-century physics: symmetries are not aesthetic preferences but the source code of the conservation laws.

Wigner's classification (1939): in a quantum theory obeying special relativity, an elementary particle simply *is* an irreducible unitary representation of the Poincaré group — the group of spacetime symmetries. Mass and spin are not properties a particle happens to have; they are the labels that classify the representation. The particle concept, which sounds like ontology, is actually representation theory.

Microcausality and unitarity: in relativistic quantum field theory, measurements at spacelike separation commute — no operational influence travels faster than light — and time evolution conserves total probability, $S^\dagger S = 1$, which says the scattering matrix S composed with its adjoint is the identity: probabilities of all possible outcomes always sum to one, and information is never simply destroyed. These two properties are so constraining that large parts of field theory can be *reconstructed* from them.

Lovelock's theorem (1971): in four spacetime dimensions, Einstein's equations plus a cosmological constant are essentially the *unique* second-order field equations for a metric theory of gravity. Einstein's theory is not one option in a crowded field; given its basic commitments, it is the only game in town. This rigidity cuts both ways, and its sharpest consequence — that vacuum energy has nowhere to hide — opens the deepest problem in the next chapter.

The honest takeaway

Assemble the inventory and step back. Quantum mechanics: confirmed, with its strangeness fenced by theorems. The Standard Model: the most precisely verified theory in the history of science, with exactly one laboratory-certified crack, already patched. General relativity: a century of passed tests from the solar system to colliding black holes, and a perfectly predictive quantum effective theory below the Planck scale. Thermodynamics: upgraded to exact theorems and successfully annexed to black holes, producing the one formula where all of physics' fundamental constants shake hands. Cosmology: six parameters, thousands of fitted data points. And beneath it all, a bedrock of structural theorems that any future theory inherits whole.

Here, then, is the finding that the adversarial audit behind this book kept returning to, iteration after iteration, and which deserves to be stated as its own result: **there is far more established, mutually consistent physics than the phrase "general relativity and quantum mechanics are incompatible" suggests.** The two pillars are each confirmed to extraordinary precision, and they are mutually consistent *everywhere we can currently test*. The incompatibility is real — the next chapter takes it apart bolt by bolt — but it is narrow, and it is located: it lives where both theories are extrapolated far past their tested regimes, at the Planck scale, inside black holes, at the initial singularity.

That reframing changes the character of the problem. We are not staring at a house with cracked foundations. We are staring at a completed cathedral with a locked room — and the next chapter is about the lock.

CHAPTER 4

Where Physics Breaks

The famous "contradictions" of physics, put under adversarial cross-examination, turn out to be a taxonomy — and not one of them is what the folklore says it is.

If the last chapter was the inventory of the vault, this one is the inspection of the cracks. Fundamental physics does have genuine tensions — deep ones, tensions that have resisted the best minds for fifty to a hundred years. But the single most important finding of the audit behind this book, arrived at early and never overturned through twenty-three adversarial iterations, is a methodological one: **classify before alarming.**

Informal discourse conflates four very different kinds of trouble. A *logical inconsistency* is a contradiction derivable within a single fixed axiom system — the theory is internally broken. A *domain-of-validity mismatch* is two internally consistent frameworks resting on incompatible idealizations, clashing only where their domains are forced to overlap. An *unsolved-but-consistent problem* is a well-posed question with no accepted answer but no contradiction. A *conceptual tension* is a dispute about meaning or mechanism with no operational clash at all. These are not degrees of the same disease; they are different diseases, with different prognoses.

The investigation catalogued seventeen candidate "contradictions" among the accepted frameworks of physics and subjected each to adversarial refereeing — every claim challenged, every overstatement forced into the open. The headline result: **none of the seventeen survived as a demonstrated logical inconsistency.** Several entries arrived *labeled* as logical inconsistencies and were downgraded by the referee. What remains is a handful of located domain mismatches, a set of deep-but-consistent open problems, and some genuine conceptual tensions — clustered, tellingly, exactly where both pillars of physics are extrapolated past their tested regimes.

That is a very different picture from "physics is broken." But it makes the surviving tensions *more* interesting, not less, because each one is now a precision instrument: it tells you exactly which assumption fails, and where. This chapter walks through the six sharpest, telling each as the story it is.

The metric that would not sit still

The deepest structural tension is the least publicized, perhaps because it has no cinematic name. Call it what it is: *background dependence versus background independence*.

Quantum field theory, as practiced in every particle-physics prediction ever confirmed, presupposes a fixed spacetime stage. The flat Minkowski metric is not one of the actors; it is the theater. It anchors the symmetry group whose representations, per Wigner, *define* what a particle is. It defines which events are spacelike separated, and therefore anchors microcausality — the guarantee that no measurement influences another faster than light. It selects the vacuum, the state of lowest energy from which everything is built.

General relativity's founding move is to burn the theater down. The metric is dynamical — an actor, not a stage. There is no prior geometry; the physical content of the theory is carried by equivalence classes of geometries, and in the canonical formulation the Hamiltonian is a sum of constraints, $H \approx 0$, which is the mathematics' way of saying that time evolution itself is part of the gauge redundancy rather than an external clock. One framework treats geometry as fixed input; the other as dynamical output.

Stated that way, the two sound "mutually exclusive in principle" — and that is precisely the overstatement the referee struck down. The corrected statement is more interesting. Quantum field theory does not, in fact, require flatness or a preferred vacuum: QFT on fixed *curved* backgrounds is rigorous mathematics, is the framework in which Hawking radiation was derived, and already teaches that "particle" is an observer-dependent notion. And gravity can be placed *inside* quantum field theory perturbatively: write the metric as flat-plus-ripple, $g = \eta + h$, quantize the ripple, and you get a theory that works fine below the Planck scale. The genuine, unsolved obstruction is sharper and stranger than the folklore version: nobody knows how to formulate quantum theory on a *superposition of causal structures*. If the metric is quantum, then "spacelike separated" can itself be in superposition — and then microcausality, the vacuum, and the particle concept all lose their definitions simultaneously. That is the real lock on the door `[ESTABLISHED as the structural contrast; OPEN as to resolution]`. Classification: a domain-of-validity mismatch wrapped around an open foundational problem — explicitly *not* a logical inconsistency.

The theory that predicts its own retirement

The most frequently cited "incompatibility" is the perturbative non-renormalizability of quantized gravity, and it is the entry where honest classification does the most work.

The established core is a genuine theorem. Newton's constant G carries dimensions of inverse mass squared, which means the quantum loop expansion of gravity is organized in powers of E^2/M_{Pl}^2 — energy over the Planck scale, squared. Each order of accuracy demands new counterterms with new free coefficients. The details are a small epic of twentieth-century calculation: pure gravity miraculously survives at one loop ('t Hooft and Veltman, 1974); add matter and it fails already there; and at two loops even pure gravity diverges, with the fatal Weyl-cubed counterterm found by Goroff and Sagnotti in 1985. Quantized general relativity is not a theory whose predictions get harder at high energy; it is a theory whose predictive apparatus dissolves there, requiring infinitely many measured inputs `[ESTABLISHED]`.

Now the classification, which changes everything about the moral. **An effective field theory with a cutoff is fully consistent below that cutoff.** This is not spin; it is the central lesson of the Wilsonian revolution in field theory. Fermi's theory of beta decay was non-renormalizable too — and correct, and predictive, for forty years, its very non-renormalizability pointing to the scale where the W boson had to appear. Gravity as an effective field theory (the program associated with Donoghue) makes unambiguous quantum predictions at low energy, including calculable quantum corrections to Newton's potential. The referee also trimmed a common overclaim in the other direction: the theorem shows gravity cannot be a *perturbatively renormalizable* field theory in the traditional sense, not that it cannot be a quantum field theory at all — a nontrivial ultraviolet fixed point (Weinberg's "asymptotic safety" scenario) remains a logical possibility, unproven either way.

So the correct statement is deflationary and precise: gravity announces its own retirement scale, around 10^{19} GeV, and is impeccable below it. What replaces it above is `[OPEN]`. What is *not* open is whether this constitutes a contradiction. It does not. It is an established theorem plus an unsolved succession problem.

The worst prediction in physics that never was

Every physics popularization repeats it: quantum field theory predicts a vacuum energy 10^{120} times larger than observed — "the worst prediction in the history of physics." The audit's regularization-aware re-examination of this claim is a case study in how folklore calcifies, and the corrected version is in some ways *more* disturbing than the myth.

First, why vacuum energy matters at all. By Lovelock's theorem — the rigidity result from Chapter 3 — the term $\Lambda g_{\mu\nu}$ is the unique addition Einstein's equations permit, and a Lorentz-invariant vacuum has exactly the right form of stress-energy ($w = -1$, tension equal to energy density) to masquerade as Λ. So whatever energy the vacuum has, gravity must see it, and it must look like a cosmological constant. The observed dark-energy density is minuscule: about $10^{-47}\,\mathrm{GeV}^4$, the fourth power of a couple of milli-electron-volts.

Now the folklore autopsy. The famous 10^{120} comes from summing zero-point energies of quantum fields up to a Planck-scale cutoff, giving $\rho \sim M_{\mathrm{Pl}}^4$. But that calculation is *regulator-dependent* — an artifact of how you truncate the sum. A momentum cutoff isn't even Lorentz-invariant; cut instead at the electroweak scale and the ratio is 10^{60}; do the calculation in dimensional regularization, which respects Lorentz invariance, and each field contributes $\sim m^4 \ln \mu$ — proportional to its own mass, not to any cutoff. The "10^{120}" is not a prediction of quantum field theory. Strictly, QFT does not predict the value of Λ at all: it is a renormalized free parameter, exactly like the electron's mass, and there is **no failed prediction and no inconsistency** here `[ESTABLISHED]`.

Here is why the corrected problem is still dreadful. What fails is not prediction but *stability*. In the honest regulator-independent accounting, the electron loop alone contributes some thirty orders of magnitude more than the observed value; the top quark, fifty-five. Finite, physical, regulator-independent condensates — the QCD vacuum's gluon condensate, the Higgs potential — overshoot by forty-four and fifty-nine orders respectively. Worse, these contributions *changed* during the electroweak and QCD phase transitions in the early universe: a Λ tuned small today was enormous before them, so the tuning is epoch-dependent, not a one-time convention. And Weinberg's no-go theorem (1989) proves, under stated assumptions — a local four-dimensional Lagrangian theory with finitely many fields and a translation-invariant vacuum — that no adjustment mechanism can relax the vacuum energy to zero without re-importing the tuning. Every serious escape route exits through a named assumption of the theorem, and the audit's 2024–2026 survey found none that succeeds: unimodular gravity relocates the tuning into an untuned integration constant without removing it; sequestering cancels matter loops but not graviton loops `[SPECULATIVE]`; anthropic selection leans on a contested multiverse measure.

One live experimental thread: the DESI survey's recent data prefer a slowly *evolving* dark energy over a strict constant at roughly the three-sigma level — a hint, `[CONTESTED]`, sensitive to supernova calibration systematics, with a five-year verdict expected in 2027. If it survives, it would rearrange the problem without solving it: a rolling component is additive to, not a substitute for, the cancellation puzzle. Classification: an `[OPEN]` fine-tuning problem at the QFT–gravity interface. The honest mismatch is thirty to sixty orders of magnitude, not one hundred twenty — and unlike the folklore version, no known mechanism even in principle makes it go away.

The black hole that wouldn't forget

In 1974 Hawking showed that black holes radiate, at the temperature $T_H = \hbar c^3/8\pi G M k_B$ met in Chapter 3, and thereby lit the fuse on the finest paradox in physics. His semiclassical calculation yields radiation that is *thermal* — carrying no information about what fell in. Run the evaporation to completion and a pure quantum state has evolved into a featureless mixed one, which unitary quantum mechanics forbids: quantum evolution never destroys information, only scrambles it. Unitarity demands that the fine-grained entropy of the radiation follow the **Page curve** — rising as early radiation entangles with the hole, then turning over and returning to zero as the information comes back out. Hawking's calculation says it rises forever.

The referee's first act was to strike the label this entry arrived with: this is *not* a logical inconsistency. Each framework is internally consistent; the conflict arises only when the semiclassical approximation is iterated past its plausible domain of validity. It is the cleanest domain-of-validity tension in physics — which is precisely why it has become the field's favorite test bed.

The state of play, refereed: unitarity is now favored, on **two independent legs** `[INFERENCE]`. The first leg is the "islands" breakthrough of 2019–2020 (Penington; Almheiri, Engelhardt, Marolf, Maxfield, and collaborators): the semiclassical gravitational path integral, done more carefully to include "replica wormhole" saddle points Hawking's calculation omitted, *reproduces the unitary Page curve* in controlled settings. The consensus reading is quietly stunning — Hawking's answer was wrong not because semiclassical gravity fails, but because the semiclassical calculation itself was incomplete. The second leg is algebraic: operator-algebra constructions (Witten and collaborators, 2022 onward) independently produce horizon entropies with the right generalized form, as mathematics rather than saddle-point estimate.

The honesty clause, which the wiki's referees enforced against considerable community enthusiasm: the controlled island results live almost entirely in anti-de Sitter space, in lower dimensions, or in toy models; standing objections persist about whether replica wormholes compute the entropy of a *single* theory or an ensemble average, and about the massive gravitons lurking in the controlled constructions; and there is still no first-principles account of how information escapes a realistic, astrophysical, four-dimensional black hole `[OPEN]`. Unitarity is winning on points. It has not won by knockout.

The measurement problem: an incompleteness, not a contradiction

The oldest tension is internal to quantum mechanics itself, and the audit's classification of it is the most clarifying entry in the catalogue.

The setup is stark. Quantum theory's dynamical axiom says closed systems evolve unitarily — linearly, deterministically, reversibly. Apply it to a measurement: a superposed system interacting with an apparatus evolves into an entangled superposition of *apparatus states*, pointer-left-and-pointer-right. The theory's measurement axiom says something else entirely: one outcome occurs, with probability given by the Born rule. Nothing in the formalism says when the first rule hands off to the second — Bell called the movable boundary the "shifty split."

Decoherence, the modern piece, explains real physics: interaction with the environment suppresses interference between macroscopic alternatives with fantastic speed, selecting which variables behave classically. But — and the referee insisted this be stated exactly — decoherence yields an *improper mixture*: the global state remains a pure entangled superposition; the appearance of alternatives is explained, the occurrence of one of them is not.

The classification: **structural incompleteness, not logical inconsistency.** The two axioms as stipulated have disjoint domains — one governs closed systems, the other measurements — and a formal contradiction appears only if you add the premise that unitary evolution is *universal*, applying to apparatus and observer too. That premise is precisely what the interpretations dispute, and each buys consistency at a published price: Everett keeps universal unitarity and pays with branching worlds and a still-contested account of probability; Bohmian mechanics adds definite positions and pays with explicit nonlocality and a preferred frame; objective-collapse models (GRW, CSL) modify the dynamics itself and pay by predicting tiny, *testable* deviations from quantum mechanics.

That last price is the one corner of the problem that is experimentally decidable, which earns it a place on the watchlist: collapse models predict spontaneous heating and interference loss that matter-wave interferometry and optomechanics are steadily cornering, with decisive constraints plausible within a decade. The rest remains `[ESTABLISHED]` as a structural tension, `[OPEN]`/`[CONTESTED]` as to resolution — a door that has been ajar for a century, with instruments finally being built that could close one wing of it.

Time's arrow: a boundary condition, not a law

The last of the six is the one you experience directly: eggs scramble and never unscramble, entropy climbs, the past is fixed and the future open. Yet the microscopic laws — classical and quantum alike — are time-symmetric. (The weak interaction's small T-violation is a red herring here: a perfectly time-symmetric world would still have an arrow, given the right initial condition.) Where does irreversibility come from?

The nineteenth century already proved the sharp negative `[ESTABLISHED]`: it cannot come from the dynamics alone. Loschmidt observed that reversing all velocities in any entropy-increasing trajectory yields a legal entropy-*decreasing* one; Zermelo added that Poincaré recurrence eventually returns any confined system near its starting point. Liouville's theorem in classical mechanics and unitarity in quantum mechanics make it exact: the fine-grained entropy of a closed system is strictly *constant*. Boltzmann's celebrated H-theorem gets its arrow by inserting an assumption — molecular chaos — asymmetrically, before collisions but not after. Every derivation of the second law smuggles time-asymmetry in somewhere; the theorems merely force the smuggling into the open.

The modern resolution is honest bookkeeping: irreversibility = time-symmetric dynamics + coarse-graining + a spectacularly special initial condition. The universe began in a state of extraordinarily low entropy — low, above all, in *gravitational* entropy: matter nearly uniform, spacetime nearly unwrinkled, in a regime where gravity makes uniformity the improbable arrangement rather than the generic one. That posit is the **Past Hypothesis**, and it does something no other item in fundamental physics does: it enters not as a law of dynamics but as a boundary condition on the whole history. The classification follows: the *structure* of the arrow is `[ESTABLISHED]` — a time-asymmetric input is provably necessary, and the fluctuation theorems of Chapter 3 quantify exactly how the asymmetry cashes out — while the *origin* of that absurdly low-entropy beginning is `[OPEN]`, exported to cosmology and quantum gravity, where it waits, patiently, near the same locked door as everything else in this chapter.

The war that wasn't

Step back from the six stories and tally the classifications. Background dependence: domain mismatch plus an open foundational problem. Non-renormalizability: a proven theorem about a consistent effective theory, plus an open succession question. The cosmological constant: an open fine-tuning problem, purged of its folklore exaggeration and still severe. Black-hole information: a domain-of-validity tension with unitarity favored but unproven in the realistic case. Measurement: structural incompleteness, with one experimentally decidable corner. The arrow of time: established structure, open cosmological origin.

Count the demonstrated logical inconsistencies: zero. Out of seventeen catalogued, refereed, adversarially cross-examined candidates: zero.

This reframing is itself a load-bearing result — arguably the most consequential of the entire first phase of the investigation. If GR and quantum mechanics were *logically* inconsistent, at least one would be internally broken, and unification would mean demolition. Instead, the tensions are located, classified, and concentrated: they cluster at the Planck scale, inside horizons, at the initial singularity — precisely the regimes where both theories run naked beyond their tested domains, and where several of the six stories converge on the same missing item, a quantum theory of causal structure itself. The frameworks we have are not at war. They are two impeccable witnesses whose testimonies agree everywhere either can be checked, and diverge only about events neither actually saw.

Whether that divergence can be closed — and what it would cost — is the question the rest of this book pursues.

PART II

THE WAGER AND THE WALL

The wager, stated precisely — and the wall all five roads reach.

CHAPTER 5

The Wager

Several of the strongest results in modern mathematical physics converge on a single bet — that entanglement and algebra lie deeper than space and time — and this chapter states that bet precisely enough that it can lose.

An agreement among strangers

Physics does not usually produce coincidences of this size. Over roughly fifty years, results from black-hole thermodynamics, string theory, quantum information, and the austere mathematics of operator algebras — fields with different tools, different journals, and largely different people — kept arriving at the same suspicion from different directions. None of them proves it. That is the whole point of this chapter, and the reason the book calls it a *wager* rather than a discovery. But the convergence is real, and it deserves to be laid out before it is dissected.

The first arrival was the oldest. Bekenstein and Hawking showed in the 1970s that a black hole carries an entropy

$$S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar},$$

where A is the area of the event horizon, and k_B, c, G, $\hbar$ are Boltzmann's constant, the speed of light, Newton's constant, and Planck's constant. Read the formula slowly: it says the information capacity of a black hole is proportional to the *area* of its boundary, not the volume inside — and it is the only equation in physics that needs all four fundamental constants at once. Entropy is a count of microscopic states; area is geometry. Here they are, keeping each other's books.

The second arrival came from holography. Within the AdS/CFT correspondence — the conjectured exact equivalence between a gravitational theory in a negatively curved "bulk" spacetime and an ordinary quantum theory on its boundary — Ryu and Takayanagi (2006) found that the entanglement entropy of any boundary region A equals the area of a minimal surface γ_A hanging into the bulk:

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G_N}.$$

This says that a purely quantum-informational quantity, computed with no reference to gravity — how entangled region A is with the rest of the boundary — is *numerically identical* to a geometric area in a curved spacetime one dimension up. Entanglement, in this corner of physics, literally has a shape.

Van Raamsdonk (2010) sharpened the point with a thought experiment: take the boundary theory and dial down the entanglement between its two halves. On the gravity side, the corresponding bulk spacetime stretches and pinches until it disconnects into two separate pieces. Turn the entanglement off, and space falls apart. The suggestion is hard to unhear — entanglement is the thread that sews space together.

Then came dynamics. Faulkner, Guica, Hartman, Myers, and Van Raamsdonk (2013–14) showed that the "entanglement first law,"

$$\delta S_A = \delta\langle K_A\rangle,$$

imposed for all small ball-shaped regions, is equivalent to the linearized Einstein equation $\delta G_{\mu\nu} = 8\pi G\,\delta T_{\mu\nu}$. Here K_A is the *modular Hamiltonian* of the region — the operator that plays the role of energy in the thermodynamics of entanglement — and the first law says a small change in a region's entanglement entropy equals the change in its modular energy. That such a bookkeeping identity, demanded everywhere, *is* gravity's equation of motion at linear order is the single most quoted exhibit in the case for the wager.

Independently, and importantly *outside* the AdS box, Jacobson (1995) had already derived the full nonlinear Einstein equations by assuming the Clausius relation $\delta Q = T\,dS$ — heat equals temperature times entropy change — on every local acceleration horizon, with entropy proportional to horizon area and T the Unruh temperature an accelerating observer sees. On this reading the Einstein equations are not a fundamental law at all; they are an *equation of state*, like the ideal-gas law — the thermodynamics of some unknown microscopic degrees of freedom. We will return to Jacobson, under audit, at the end of this chapter.

The last arrival is the deepest and the least famous, and it comes from the mathematics of quantum field theory itself. The observables measurable inside any bounded region of spacetime form what is called a von Neumann algebra, and a theorem of the subject (Buchholz, D'Antoni, and Fredenhagen, building on Connes' classification) says these local algebras are of a strange type — Type III_1 — with consequences that sound like science fiction stated dryly: a region of space has *no density matrix*, no well-defined entropy of its own, and the total Hilbert space does *not* factorize into "inside" $\otimes$ "outside." The popular picture of spacetime as a lattice of little quantum subsystems — qubits sewing space — is not just unproven in continuum QFT; it is mathematically undefined. What the region *does* have is modular structure: Tomita–Takesaki theory equips every (algebra, state) pair with a canonical internal flow, a kind of clock that the algebra and the state generate all by themselves. And a run of results beginning in 2021 (Witten; Chandrasekaran, Longo, Penington, and Witten) showed that coupling this structure to gravity — adjoining an observer's clock and taking what is called the crossed product — converts Type III_1 into Type II, an algebra that *does* admit a trace and a finite, renormalized entropy:

$$S_{\text{gen}} = \frac{\langle \text{Area} \rangle}{4G} + S_{\text{out}} + \cdots .$$

The formula says that the notoriously divergent entanglement entropy of quantum fields becomes finite once the observer enters the algebra, and its leading term is, once again, an area over $4G$.

The area law is not decoration on this mathematics; it falls out of the bookkeeping.

Each of these is, on its own, a technical result with a careful domain of validity. Together they invite one reading, and by the 2020s much of the field had accepted the invitation: perhaps entanglement and algebraic structure are not features painted onto a pre-existing spacetime. Perhaps spacetime is the bookkeeping.

The bet, stated precisely

Here is the wager, in the form this project adopted, attacked, and refused to soften:

> **Causal, algebraic, and entanglement structure is more fundamental than metric geometry — and this is true of our universe.**

Every clause is load-bearing. "More fundamental" means that geometry — the metric field of general relativity, including its causal cones and its dynamics — should be *derivable* from the algebraic side, the way temperature is derivable from molecular motion, and not merely translatable into it. "Causal, algebraic, and entanglement structure" names the candidate substrate: a partial order of influence, a net of operator algebras with states, the modular and entanglement data those generate. And "our universe" means a universe with positive cosmological constant, dark energy and all — not a mathematically convenient cousin.

In the epistemic vocabulary this book runs on, the wager as a whole is graded `[SPECULATIVE]` — a coherent conjecture without decisive support — while its best sub-result, Ryu–Takayanagi within AdS/CFT, rises to `[INFERENCE]`, a well-supported conclusion short of established fact. That gap between the grade of the parts and the grade of the whole is the story of the next several chapters.

The refereed record develops nine research directions built on this wager (labelled H0 through H8 in the registry), approached through four lenses: the modular/algebraic lens, which takes the Type III_1 algebra and its modular flow as primitive; the causal-order lens, which takes a partial order of events as primitive and notes that, by theorems of Malament and of Hawking, King, and McCarthy, a causal order already fixes the metric up to a local scale factor; the amplitudes lens, in which spacetime locality is a derived boundary property of a "positive geometry"; and the operational lens, which asks whether quantum theory itself is the rigid limit of an information theory. Strikingly, the most epistemically defensible entry in the whole registry is the negative one — H2, the direction whose content is precisely that entanglement-geometry is *not yet* a supported route to a universal theory, together with explicit graduation criteria it would have to meet.

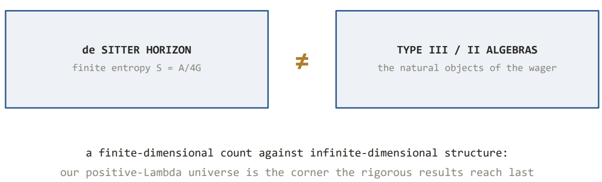
One more honesty note before the caveats: the project's own red team flagged that this multi-lens "convergence" is partly a streetlight effect. The lenses agree in part because they share load-bearing inputs — symmetric vacua, the AdS/large-N regime, an unproven discreteness conjecture — and agreement among methods that share their assumptions is weaker evidence than genuine independent convergence. The convergence is real; its evidential weight is smaller than it looks.

Recall from Chapter 1 that the record splits the wider field into two internally coherent programs joined only by thin bridges: Program A, *geometry-from-algebra* — the wager's home, comprising the modular, holographic, and entropic-gravity lines — and Program B, the logically prior question of whether the quantum framework itself is final. Everything in this chapter and the next lives in Program A.

The wrong universe

Now the caveat that the enthusiasm routinely omits, and that this book will not.

Essentially every rigorous result in the exhibit above — Ryu–Takayanagi, entanglement-wedge reconstruction, the first-law derivation of linearized gravity, the cleanest crossed-product constructions — lives in anti-de Sitter space: a spacetime with *negative* cosmological constant, $\Lambda < 0$. Our universe has been measured, since 1998, to accelerate: $\Lambda > 0$. This is not a detail of bookkeeping. Anti-de Sitter space has a timelike boundary at spatial infinity, a natural place for the dual quantum theory to live and a continuum of boundary-anchored regions whose minimal surfaces probe every bulk point. De Sitter space — our side of the sign — has neither: no established holographic dual, no Ryu–Takayanagi analogue, and a variational structure that is *sign-reversed* (empty de Sitter space *maximizes* the generalized entropy, where AdS constructions minimize; the static-patch first law runs $dE = -T\,dS$). The project's iterations 3 through 5 turned this from a complaint into located theorems-shaped obstructions, which Chapter 7 narrates; the summary is that the one route to a $\Lambda > 0$ Einstein-from-entanglement derivation fails on a dimension count, and that in *every* known route, AdS or otherwise, the area-entropy coefficient $1/4G$ is an input, never an output.



The de Sitter counting problem: a finite horizon entropy set against the infinite-dimensional operator algebras the wager runs on.

Two further deflations. The derivations that do exist recover only the *linearized* Einstein equation — gravity's first-order response, not its full nonlinear content. And the derivations are silent about why the substrate should organize itself geometrically at all. So the honest inventory reads: a spectacular dictionary in the wrong universe, at linear order, with the crucial coefficient put in by hand.

That inventory forces a question of principle. When a construction "derives" geometry from entanglement, what exactly has been shown? The answer requires the sharpest conceptual tool this project built, and it is the hinge on which the entire book turns.

Encoding versus generating

Take a nautical chart drawn in some ingenious new projection — one in which rhumb lines straighten, say, and currents become gradients of a potential. The chart may be more useful than the coastline itself for every practical purpose. It may reveal structure no one had seen. But the chart did not *produce* the coastline, and there is a simple forensic test: the cartographer needed the coastline to draw it.

The project's iteration 3 turned this intuition into a decision procedure, because slogans were not settling arguments. A construction **generates** geometry if the emergent metric and causal structure are fixed by abstract data — an algebra, a state, a dynamics — with *no* geometric information smuggled in through the setup. It merely **encodes** geometry if, somewhere in the construction, the answer was part of the question. The criterion comes with two graded axes — Δ_{geo}, measuring how much geometric input a construction consumes, and κ_Φ, measuring how much hand-guided post-processing its output needs — and a five-item smuggling checklist: a pre-chosen tensor factorization or subsystem decomposition; a pre-chosen region or graph structure; a symmetry-distinguished vacuum state; a pre-installed causal order; a pre-installed background on which the "emergent" object is recognized. (Iteration 6 later extended the list by two subtler items, which Chapter 6 will need.)

Applied to the six leading emergence constructions of the era — the crossed-product algebras, thermal time, holographic bulk reconstruction, "space from Hilbert space," causal sets, tensor networks — the criterion returned a uniform verdict: **none generates**. Every one of them consumes at least one checklist item, and the encode/generate frontier coincides, construction after construction, with the same underlying fact: the hand-chosen factorization or region structure that Type III_1 says the continuum theory does not intrinsically possess. Graded `[INFERENCE, high]`, this became the standing diagnosis of the whole investigation: *the* *algebraic and holographic program encodes geometry; it does not generate it.* The program's own architects, to their credit, largely concede the premise — Witten's "background independence" for the crossed product, read carefully, means state-independence on a fixed background, not freedom from one.

Jacobson, audited

The distinction earns its keep on hard cases, and the hardest is Jacobson — because his derivations are genuine theorems, because the 2015 version works at *any* value of Λ, including ours, and because if anything in the literature generates gravity from entanglement, it should be this.

The 2015 argument ("entanglement equilibrium," arXiv:1505.04753) runs as follows. Assume that in the true theory, the vacuum restricted to any sufficiently small causal diamond is a *maximum* of entanglement entropy at fixed volume. Demand that the generalized entropy — area term plus matter term — be stationary under small variations, in every small diamond, around every point. From this single principle the Einstein equation follows, in full nonlinear form as the small-diamond limit is assembled, with the cosmological constant entering as an undetermined integration constant and Newton's constant G fixed by the area-entropy coefficient. It sounds, told this way, like generation: a variational principle about entanglement in, field equations out.

The project audited it twice — once in the iteration-2 disaggregation of entropic gravity, and decisively in iteration 18, when an explicitly *generative* reading of Jacobson was proposed as a way to flip the standing verdict, and refereed. The audit asked one question: where does the Lorentzian structure — the minus sign that distinguishes time from space, the light cones, the causal order — enter? The answer: in three separate places. First, the metric signature $(-,+,+,+)$ is declared in the setup; the diamonds are *causal* diamonds in an assumed Lorentzian manifold. Second, the construction singles out a timelike vector to define them. Third —

and most tellingly — the modular Hamiltonian K whose expectation value balances the entropy is taken to be a *boost generator*: the operator implementing a Lorentz transformation, which is a symmetry of the pre-given Lorentzian vacuum. That identification is licensed by the Bisognano–Wichmann theorem for wedges and by Casini–Huerta–Myers for balls in a conformal theory — theorems *about* Lorentzian quantum field theory on a fixed background. The signature is not derived once and used thereafter; it is installed three ways.

And yet the derivation is not circular where it claims content. What Jacobson genuinely extracts, and what the audit affirmed, is the *form* of the Einstein equation — why the response of geometry to matter is the Einstein tensor and not something else — and the *value* of G in terms of the entanglement-entropy coefficient. Given a Lorentzian world with an area law and Unruh temperatures, the stationarity of vacuum entanglement forces Einstein's dynamics. That is a real and beautiful theorem. It is also, in the precise sense of the criterion, an encoding: the deepest datum — that the world is Lorentzian at all — rides in with the inputs, and what comes out is the dynamical behavior of a structure whose existence was presupposed. The same verdict fell on the AdS chain in the same audit: the FGHMVR first-law derivation installs a fixed Lorentzian boundary and an AdS background before the entanglement does any work. The referee's summary line for the iteration stands as the summary of this section: the maximal-entanglement principle generates the Einstein-equation *form* — a genuine, non-trivial result — but not the signature.

What hangs on one word

It might seem that "encodes versus generates" is a philosopher's quibble. Reformulations, after all, have carried physics before: Lagrange's mechanics added no force Newton lacked, and the path integral was "merely" a rewriting of quantum mechanics until it became the engine of modern field theory. If the algebraic program were content to be a reformulation — a better coordinate system for semiclassical gravity — no one would object, and much of its mathematics would still be permanently valuable.

But the wager claims more. It claims ontological priority: that the algebra is the substrate and geometry the derived bookkeeping, *for our universe*. A claim of priority cannot be cashed by a dictionary, however elegant, because a dictionary is symmetric — it reads in both directions, and a geometry-first physicist can use the same pages. The claim needs generation: some demonstration that the algebraic side, given none of the geometric answers, produces them. And on this the record is, so far, unanimous in the negative — not for lack of trying, but under twenty-three iterations of adversarial effort to force the issue, including direct instruction to *construct* the generative step, every attempt either failed honestly or was caught installing the signature by hand.

The consequence reaches further than aesthetics. As Chapter 6 will make precise, a program that only encodes has the same observational content as the effective field theory it encodes — which means no experiment, even in principle as far as anyone has shown, can distinguish the wager from ordinary geometry-first physics. The standing verdict of the whole investigation — PARTIAL coherence, encodes-not-generates, not yet physics — turns on the single word this chapter has tried to define. The next chapter walks the five roads on which the field has tried to flip it, and the one wall on which all five end.

CHAPTER 6

Five Roads to One Wall

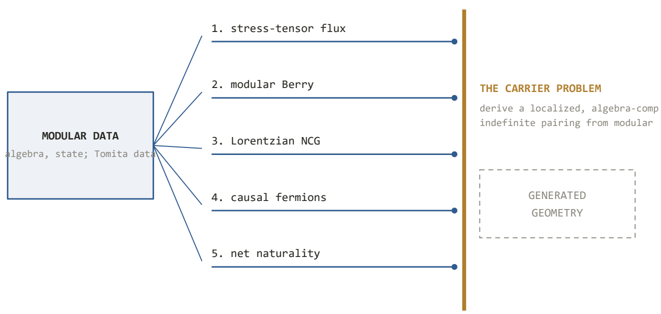
Five independent ways of reading geometry out of pure algebra all stop at the same brick: the minus sign that makes spacetime Lorentzian is always installed, never derived — and a program that only encodes cannot, even in principle, be told apart from what it encodes.

What generation would actually take

Chapter 5 ended with a definition and a verdict: the algebraic program *encodes* geometry, and to claim more it would have to *generate* — produce geometric structure from (algebra, state) data with nothing geometric smuggled into the setup. This chapter is about what, concretely, that would require, and about the remarkable fact that five unrelated-looking research lines all fail the requirement at the same step.

A Lorentzian metric carries two separable pieces of information. One is the **causal order**: which events can influence which — the light cones, the partial ordering of the world. The other is the **signature**: the single minus sign in $(-,+,+,+)$ that makes one direction time rather than a fourth direction of space. In algebraic terms the signature shows up as an *indefinite pairing* — an inner-product-like form that can return negative as well as positive values, the mathematical residue of "timelike." The natural home for such a form is a Krein space: a Hilbert space equipped with a fundamental symmetry η ($\eta = \eta^*,\ \eta^2 = 1$) that splits it into a positive and a negative part, of signature (n, n). Throughout this chapter, when a construction needs a minus sign, watch for where its η comes from.

The raw material available to any would-be generator is fixed, and it is worth stating once. Given a von Neumann algebra $\mathcal{M}$ and a suitable state with vector Ω, Tomita–Takesaki theory produces exactly two canonical operators: the modular operator Δ, whose flow $\sigma_t(a) = \Delta^{it} a \Delta^{-it}$ is the intrinsic clock of Chapter 5, and the modular conjugation J, an antiunitary reflection mapping the algebra onto its commutant. In plain terms: an algebra and a state, by themselves, own a clock and a mirror — and nothing else. The Bisognano–Wichmann theorem gives the flagship geometric instance: for the algebra of a Rindler wedge in vacuum, the abstract clock Δ^{it} is the Lorentz boost. The generative question is whether that dictionary can be run backwards — whether, starting from the clock and the mirror alone, one can *derive* a localized indefinite pairing (the signature) and a causal order (the localization pattern), rather than recognizing them in situations where a Lorentzian spacetime was supplied in advance. The project's shorthand for the two required outputs: the pairing η, and the localization template n_1
.



Five independent routes from modular data toward geometry, and the single wall — the carrier problem — at which all five halt.

Road one: stress-tensor flux

The most physical-sounding readout comes first. If modular flow is secretly a spacetime motion, one should be able to detect the geometry by watching how energy moves under it: compute the flux of the stress tensor along the modular flow and read off the vector field it generates. This is the strategy of the Sorce-style flux readouts, and it works — in exactly the cases where it was always going to.

The decisive result is Sorce's theorem (arXiv:2403.18937, 2024), realized explicitly in two dimensions by Caminiti, Capeccia, Ciambelli, and Myers (arXiv:2502.02633, 2025): when the recovered generator exists as a geometric vector field, it is a *conformal Killing vector* — a symmetry of the background. The readout succeeds precisely where the spacetime already possesses the symmetry that makes the modular flow geometric, which by Bisognano–Wichmann and its conformal extensions means wedges, CFT balls, and their conformal images. For a generic region, or a generic state, the modular flow is non-geometric — it scrambles rather than moves — and the flux readout returns nothing usable. The geometry read out is the geometry put in: the route needs a CKV, and a CKV is a fact about a pre-given metric. In the hedge ledger this closure is the origin of the name **HYP-CKV-VACUITY** — the running hypothesis, refined for nineteen iterations, that all such readouts are vacuous as generators.

Road two: modular Berry transport

The second road is more sophisticated: instead of one modular Hamiltonian, use the *family* of them. As the region varies, the modular Hamiltonians form a bundle, and one can parallel-transport along it; the curvature of this "modular Berry connection" is geometric data not visible in any single flow, and in AdS/CFT it reconstructs the bulk connection on kinematic space — the space of boundary-anchored geodesics.

This road produced the survey's one genuine landmark: the first emergent *Lorentzian* metric in the entire constructions pool — a two-dimensional de Sitter metric on kinematic space (Huang–Ma, arXiv:2003.12252) — crossing the signature barrier that purely Riemannian recoveries never touch. The audit that followed is a small classic of the genre. Where did the signature come from?

From the conformal symmetry of the CFT vacuum: the kinematic-space metric is fixed by the symmetric state and inherits its signature from it, checklist item three. When iteration 9 pressed the route on its own terms — what does modular Berry transport produce *naturally*, without a symmetric vacuum? — the answer was exact: the holonomy-natural object is the Kirillov–Kostant *symplectic* form. A symplectic form is a beautiful thing, but it is signature-free: it encodes oriented area, not light cones; no minus sign distinguishes any direction in it.

Then, in 2025, the literature confirmed the closure from outside. De Boer, Najian, van der Heijden, and Zukowski (arXiv:2505.04682), constructing modular Berry transport over a family of *states* — the freshest and most general version of the idea — found exactly what the house analysis predicted: "the modular Berry phase gives rise to an emergent **symplectic** form," with the Poincaré algebra installed separately by hand. A third party, running the strongest form of the route, landed on the same signature-free output plus the same installation. In a project whose confirmations are mostly its own referees, an independent one is worth recording.

Road three: Lorentzian noncommutative geometry

Connes' noncommutative geometry offers the most literal version of "geometry from algebra": a reconstruction theorem states that a commutative spectral triple — an algebra, a Hilbert space, and a Dirac operator D satisfying axioms — *is* a Riemannian manifold, with the metric recoverable from D (the Connes distance formula: the Dirac operator is the metric). If any framework should generate spacetime, it is this one.

It generates the wrong signature. The reconstruction theorem is Riemannian — all plus signs — because its analytic backbone (a positive-definite Hilbert space, a self-adjoint D with compact resolvent) is Euclidean through and through. The Lorentzian extensions of the field repair this the only way anyone knows how: by installing the minus sign as an axiom. In indefinite spectral triples the Dirac operator is Krein-self-adjoint on a *pre-installed* Krein space whose fundamental symmetry η — often built from a chosen spacelike foliation or time orientation — simply *is* the (n, n) carrier the generative program was supposed to derive (Bizi–Brouder–Besnard, arXiv:1611.07062). The "twisted" spectral triples obtain their Krein structure from the twist, and the twist is chosen. Nothing in this is a criticism of the mathematics, which is exactly what honest Lorentzian NCG should look like. It is an observation about inputs: wherever this literature has a minus sign, the minus sign is an axiom.

Road four: causal fermion systems

Finster's causal fermion systems are the road least known to physicists at large and the most instructive for our question, because the framework *advertises* that spacetime, causality, and fields all emerge from a variational principle on a measure over operators. Its first definition posits a Hilbert space and a class of self-adjoint operators, each with at most n positive and at most n negative eigenvalues — the $(\leq n, \leq n)$ spectral condition.

Iteration 6 proved a small necessity lemma about that condition, re-derived independently by finder and referee and checked numerically twice: if the operator class is positive semidefinite — drop the negative-eigenvalue allowance — the framework's *lightlike* relation is empty. No indefiniteness axiom, no light cones, anywhere in the theory. The $(\leq n, \leq n)$ condition is therefore a necessary carrier of the Lorentzian signature: it is the Krein η of road three, relocated into the first definition. Consistently with this, every worked example in the program's literature is a Dirac-sea construction on pre-given Minkowski space. As a candidate *generator*, the framework installs its output in its opening axiom.

Road five: the modular sign — the target achieved, and void

The fifth road is the project's own, and it is where the mathematics of this book runs deepest. If the signature must come from (algebra, state) data alone, there is exactly one place left to look: the modular operators themselves. Iteration 6 compressed the whole generative question into a single sentence — *derive the indefinite pairing, a Krein fundamental symmetry, from* $(\mathcal{M}, \Omega)$ *modular data rather than positing it* — and recorded, after a literature search, that no such derivation existed or had been sketched.

Iteration 7 then executed the target. And here the story takes the turn that justifies the chapter's title, because the derivation *succeeded*:

$\eta_0 = \operatorname{sgn}(\ln \Delta)$.

Read it plainly: take the modular operator Δ, take the logarithm — the modular Hamiltonian, the generator of the intrinsic clock — and keep only its sign, $+1$ on modes of positive modular frequency, -1 on negative. The result is a canonical fundamental symmetry: genuinely indefinite (with signature (∞, ∞) on any Type III factor), odd under the modular conjugation J, built from nothing but the algebra and the state, and clean under the house smuggling checklist. The literal target of the generative program — a Krein η from pure modular data, the thing roads three and four install by hand — exists.

Nothing flipped. The referee's audit established three properties, each fatal to generative use. First, η_0 is **unitarily universal**: the same operator, up to unitary equivalence, across the entire class of algebras in modular position — it carries no information whatsoever about any particular spacetime, region, or dynamics; on the project's geometric-content axis, $\Delta_{\text{geo}} = 0$, proven rather than graded. Second, it is generically **algebra-incompatible**: conjugation by η_0 does not map $\mathcal{M}$ to itself, so the pairing cannot even be expressed in the theory's own observables. Third, it is **non-**

localizable: it cannot be attached to a region without first positing the region structure — the very thing to be derived. Worse for any hope of repair, η_0 is not unique: it is one member of an infinite family of equally canonical J-odd signs ($\mathrm{sgn}(\sin \ln \Delta)$ is another, referee-constructed), all equally geometry-void — and non-uniqueness here *strengthens* the negative, since no member has any claim to be *the* signature of anything. A companion structural fact sealed the direction: the modular conjugation J provably carries *reflection positivity* — the defining property of Euclidean field theory — so no J-induced pairing can be secretly Lorentzian. The mirror lives on the Euclidean side of the house.

The refereed slogan deserves its place in the main text: **the modular sign is to signature what thermal time is to time** — canonical, universal, and empty of particulars. Modular theory really does hand you a clock and a minus sign for free; it hands *everyone* the *same* clock and the *same* minus sign, which is precisely why neither is geography.

The engine, and why the roads converge

Five roads, one wall — but is that a pattern or a coincidence? Iterations 8 and 9 answered by finding the engine that closes all of them, in the form of a naturality argument simple enough to state in prose.

Suppose a candidate carrier existed that avoided η_0's fate: an indefinite form that is *localized* (attached to bounded regions) and *natural* (transforming consistently under everything that preserves the net of algebras and the vacuum — which, by Bisognano–Wichmann, includes the modular boost flows). Being indefinite and localized, it must change sign somewhere: it has a sign locus inside the index set of regions. Being natural, that locus must be dragged along by every symmetry — including the wedge's modular flow, which acts on the relevant half-line of the index set as a pure dilation, $p \mapsto e^{-2\pi t}p$. A dilation has no fixed point in the open half-line; its only fixed points, $p = 0$ and $p = \infty$, are the *wedge edge* — a geometric boundary that is not a region of the net at all. So no sign locus can hold still: **localized + natural + indefinite is impossible**, and the only invariant indefinite object is the delocalized modular sign of road five. Iteration 8 extended the same engine from single algebras to full nets and families — relative modular operators, half-sided modular inclusions, modular Berry data over a net supply more operators but no localized indefinite form — with one precise residue: in a net, localization enters through the index set of regions, and the index set's causal order *is* the geometry. Even the word "localized" turns out to presuppose the answer.

That is the wall, and it now has a name and a formally posed statement — the **carrier problem** (iteration 15 distilled it into a self-contained dossier an operator-algebraist can attack cold). In one sentence: *exhibit, or rule out, a structure in which the causal order n_1 and the signature η are outputs of the algebra and state, rather than indexing data of the source category or installed data on the carrier space.* Every modern mathematical framework subsequently thrown at it — factorization algebras, index theory and K-theory, higher-categorical field theory — reproduced the same screen, smuggling the same two interchangeable inputs. Under adversarial pressure the count of genuinely independent converging routes was held at exactly five: proposed sixth routes (the Krein/indefinite-metric literature, the net-level extension, the categorical translations) were each shown to be re-expressions of an existing one, and the referees said so.

The ledger, R3 to R7

How confident is the project that this wall is real, and not an artifact of the readouts tried so far? The discipline here is the book's method in miniature, so it is worth walking the ledger. The no-go was never allowed to call itself a theorem. It lived as a named hedge — HYP-CKV-VACUITY — with a revision number and an explicit condition string, and it moved only when a referee certified the movement.

At R3 (iteration 6) it was a classification: every surveyed construction with a Lorentzian output installs the signature through a symmetric state, an indefinite-pairing axiom, or a signature-valued template — graded HIGH as a classification of the pool, but only MEDIUM that the classification exhausted all possible mechanisms. At R4 (iteration 7), the η_0 result closed the J-route and hardened the grade to MEDIUM-HIGH, *conditional* on a flagged constructibility ansatz — and forced an honest repair, because iteration 6's sentence "no such derivation exists" had become literally false: the compression was lossy, and the record says so in those words. R5 (iteration 8) upgraded single-algebra to net-and-family and narrowed the condition. R6 (iteration 9) replaced the ansatz with a naturality formulation — the constructibility half a theorem modulo one located gap — and then held, through six further iterations, against every attempt to close that gap, including two *struck* claims of advancement at iteration 10: one killed for a wrong-object error, one for an overstated discharge, both retractions logged as the method working rather than buried. Only at iteration 16 did a referee-verified multi-wedge closure mint **R7**, replacing the diffuse gap with a single named open hypothesis: the no-go is conditional on exactly $\{n_1 \text{ AND } (E_O)\}$ — the irreducible localization template, plus "(E_O)": vacuum ergodicity on massive double-cone algebras, the statement that the algebra of a bounded diamond-shaped region shares no nontrivial element with the fixed points of its own modular clock, $M(O)_\omega = \mathbb{C}1$. Given (E_O), the carrier no-go is a theorem on its stated hypotheses; its negation is necessary, though not sufficient, for any counterexample. The subsequent campaign on (E_O) itself — reductions, the project's first numerics, a theorem of Figliolini and Guido (1989) promoted to load-bearing status — belongs to Chapters 8 through 11. As of this writing the hedge stands at R7, unmoved through seven further adversarial iterations, and the verdict it guards has been confirmed twenty-two consecutive times.

The screen: why no experiment can referee the wager

One question remains, and it is the one a physicist should have been shouting for two chapters: forget the mathematics — why not just *test* it? Here the encode/generate distinction delivers its most consequential corollary, and the reason the standing verdict ends in "not yet physics."

Call the bridge from the algebraic substrate to spacetime observables Φ. If Φ is encode-only — if it consumes a background geometry among its inputs and outputs no geometric data beyond what it consumed — then the observational content of the whole program *equals* that of the semiclassical effective field theory whose background Φ consumed. The prediction map factors through the EFT. Every signal the wager-true theory can produce, a geometry-first theory with the same infrared limit produces identically; confirmation of any such signal yields explanation, never discrimination. This is the **encoding screen** (HYP-ENCODING-SCREEN in the ledger), and it is exactly as strong as the encode-only classification it rests on — it inherits the R7 hedge, condition and all, a bookkeeping the project enforces explicitly.

The screen is not an armchair worry; it has a published, peer-reviewed worked instance. The most natural home for a wager-discriminating signal would be interferometer noise from spacetime fluctuations, and the definitive taxonomy of such signals — Sharmila, Vermeulen, and Datta, *Nature Communications* **17**, 701 (2026), arXiv:2505.22892 — classifies models by their two-point noise correlators while stating outright that the approach is agnostic to microscopic origins. In its classes, emergent-spacetime and fundamental-metric models land *together*: the geontropic model shares a class with Károlyházy-type metric fluctuations, and collapse models with classical-quantum gravity. The experiments underway can detect; they cannot separate. Meanwhile every live channel in the watchlist — gravitationally induced entanglement (which tests whether gravity is non-classical: the program's logical *gate*), objective-collapse bounds (its quantum-mechanical *cores*), the dark-energy equation of state $w(z)$ (its cosmological *target*) — tests something adjacent to the wager; none touches the wager itself. Even a triumphant BMV result confirming that gravity entangles would not distinguish geometry-from-algebra from algebra-decorated-geometry.

What would break the screen? A theorem of a shape no one has ever proven or even attempted: a *reverse Weinberg–Witten* theorem. The original Weinberg–Witten theorem forbids a massless spin-2 graviton composite of fields in the same spacetime — it polices how geometry may *not* emerge. Its reverse would identify an observable realizable *only* in theories without fundamental metric degrees of freedom — a signal that could not be faked by any geometry-first model. The project's iteration 9 proved the sobering identity: the reverse-WW theorem is equivalent to the carrier no-go, and a reverse-WW *witness* is strictly harder than a positive solution of the carrier problem. The missing experiment and the missing mathematics are the same missing object. Its absence is a literature fact, tagged `[OPEN]`, not a proven impossibility — the screen has a seam, and the record marks exactly where it is.

So the five roads end at one wall, the wall is one theorem thick, and the door and the wall coincide: solve the carrier problem affirmatively and geometry is generated, the screen shatters, and a distinguishing experiment becomes possible in principle; prove the no-go and "encodes, not generates" becomes a theorem, screen included. Either outcome would be a major result. What is not available — and this is the honest lesson of the whole convergence — is the comfortable middle in which the wager is treated as physics while the wall stands. Until one theorem moves, the most-developed road to a universal theory is a research strategy of real depth and no testable consequence: not wrong, not vindicated, and not yet physics.

CHAPTER 7

Twenty Iterations of No

In which the wager is attacked for fourteen consecutive iterations by its own referees — escape routes executed, a closure struck from the record, a false premise caught in mid-argument — and the only thing that never moves is the verdict.

The rules of engagement

By the end of the first iteration the investigation had a map, a wager, and a verdict. The map said that quantum field theory and general relativity are mutually consistent everywhere we can test them, and that their famous "war" is a handful of located domain mismatches. The wager — the most developed candidate direction the field offers — said that causal, algebraic, and entanglement structure is more fundamental than metric geometry. And the verdict, delivered at the close of iteration 2 and never overturned since, said three things at once: the wager's supporting programs cohere only *partially*; the algebraic machinery *encodes* geometry rather than *generating* it; and because no experiment can distinguish the wager from its negation even in principle, it is a research strategy, not yet physics.

What followed was a campaign. Over the next fourteen iterations that verdict was stress-tested, red-teamed, handed its own designated escape routes, confronted with two years of fresh literature, and finally subjected to an assault in which the attacking agents were instructed to *assume it false*. Every iteration ran under the same discipline: finder agents attack a designated lever; an adversarial referee, whose default stance is to refute, re-derives the mathematics, re-verifies every citation against the live literature, and issues corrections that are binding over the original claims. Claims are graded throughout with the five-tag convention of Chapter 1, and no claim outranks its tag.

The ledger of that campaign is unusual reading, because its drama is inverted. In ordinary scientific narrative, progress means a claim getting stronger. Here progress usually meant an obstruction getting sharper, an escape closing, or — twice, memorably — the referee striking a result the finders believed they had won. By the time the arc this book covers was complete, the verdict had survived more than twenty consecutive attempts to move it. This chapter walks the long middle of that campaign, from iteration 2 through iteration 16, where the entire remaining question was finally compressed into a single named hypothesis.

Sharpening without confirming

Iterations 2 and 3 established the campaign's characteristic texture. Five refereed research tracks in iteration 2 pushed on the highest-leverage questions and returned a result nobody markets in a grant proposal: the wager came back *sharper and more constrained, not confirmed*. The de Sitter graduation barrier — the demand for even one rigorous entanglement-first result in a universe with the right sign of the cosmological constant — turned out to be structural, not temporary. The algebraic program's own architects, read carefully, conceded that the modular and crossed-product results reorganize semiclassical gravity on a fixed background; they do not produce the background. That diagnosis got a name that would carry the rest of the book: *encodes, does not generate*.

Iteration 3 stopped surveying and started attempting, and then did something rarer: it red-teamed its own posture. Four weaknesses were forced into the open, the most important being that the project's "fixed cores" were heterogeneous — the Type III_1 structure of local algebras, listed flatly as bedrock, is bedrock only on a fixed background, and becomes `[OPEN]` the moment gravity is dynamical. The same iteration made the encode/generate distinction rigorous — the two-axis criterion and smuggling checklist that Chapter 5 laid out. The finding was uniform. No surveyed construction generated Lorentzian spacetime from data free of a hand-chosen factorization, region structure, or symmetric state — and the frontier between encoding and generating coincided, every time, with exactly such a choice.

Iterations 4 and 5 then did the most scientifically honest thing available: they executed the escape routes. Iteration 3 had left one un-foreclosed path to a positive-Λ Einstein-from-entanglement derivation — enlarge the single de Sitter boost to the full $SO(d+1,1)$ family of observer modular Hamiltonians. Iteration 4 ran it, and it failed on a dimension count of satisfying brutality: the isometry orbit supplies at most ten global charge constraints, and ten numbers cannot be inverted into the infinitely many local components of $\delta G_{\mu\nu}(x)$. The published execution of exactly this route (Chen and Xu, arXiv:2511.00622) landed precisely where the count predicted — entropy matching, no Einstein equation. Iteration 5 executed the one move iteration 4 had designated as the only remaining verdict-flipper, a non-symmetric "readout" of Lorentzian geometry from modular data. It produced the survey's first genuinely Lorentzian emergent metric (a dS_2 kinematic space, following Huang and Ma) — and the referee's dissection showed its Lorentzian signature was inherited wholesale from the symmetric vacuum it was computed in. The failure mode was now readout-independent: every known route to an emergent Lorentzian metric installs the signature at the same step. Iteration 5 declared *analytical saturation* — no new positive structure, only sharpening — and consolidated the standing verdict into a capstone conclusion with named hedges.

A hedge, in this project's vocabulary, is a formally registered condition under which the verdict could be wrong — graded by confidence and carried on a public ledger. The principal hedge attached to the closure of the emergent-signature routes; it was graded MEDIUM at saturation. Watching that grade move — and refuse to move — is the plot of everything that follows.

The screen and the sign

Saturation did not mean stopping; it meant changing the null hypothesis. Iteration 6 injected roughly ninety refereed findings of new 2023–2026 input, with four tracks explicitly tasked to attack the standing verdict. The headline survived a fifth consecutive time, but the iteration produced the campaign's most consequential piece of new structure: the *encoding screen*. The sharpest sentence in the standing conclusion — "no distinguishing experimental test exists even in principle" — was shown to be a corollary rather than an observation. If the program is encode-only, its observational content equals that of the effective field theory whose background it consumed, so every candidate signal factors through physics equally available to geometry-first descriptions. A genuine test would require a theorem of a shape nobody has ever proven — a "reverse Weinberg–Witten," an observable realizable *only* without fundamental metric degrees of freedom. The screen was registered as a hedge in its own right, inheriting the grade of the principal one. The same iteration compressed the analytical target to its minimal form: derive the indefinite pairing — the mathematical carrier of the Lorentzian signature — from modular data alone.

Iteration 7 executed that compressed target, and resolved it negatively in the most informative way possible: *the target is achievable, and achieving it is worthless*. A canonical candidate does exist — from the modular operator Δ of a local algebra and its vacuum one can construct the operator

$$\eta_0 = \mathrm{sgn}(\ln \Delta),$$

a self-adjoint unitary built from nothing but modular data, genuinely indefinite whenever the state is non-tracial. This says: take the logarithm of the modular operator — the abstract "energy" of the state-algebra pair — and keep only its sign. The construction is smuggle-free; nothing geometric goes in. But three vacuity results showed that nothing geometric comes out either: in exactly the class of situations where modular data has causal content, the structure (Δ, J, η_0) is unitarily universal — the same for every theory, carrying not one bit of information about any particular spacetime. A trichotomy lemma closed the whole family: every canonical pairing built from modular data is positive, signature-free, or indefinite-but-universal. The iteration coined the campaign's best slogan — *the modular sign is to signature what thermal time is to time* — and named the real problem, the one the compressed target had been hiding: the **carrier problem**. It is not hard to build an indefinite sign from modular data; it is apparently impossible to *localize* one — to attach it to regions, fiberwise, the way a metric is attached to points. Five independent lines of attack were found to converge on that single gate, and the principal hedge hardened from MEDIUM to MEDIUM-HIGH.

From one algebra to the net

Iteration 7's no-go lived on a single algebra. Physics does not: a quantum field theory is a *net* — a coherent family of algebras indexed by spacetime regions, with relative modular operators, inclusions, and connections between them. Iteration 8 asked whether that vastly richer structure contains a localized indefinite form the single-algebra argument missed, and the answer was the campaign's cleanest mechanism: the **covariance–locality obstruction**. A local indefinite form must change sign somewhere, and its sign locus is a place; the modular flow of a wedge region acts on that place as a boost, sweeping it off to infinity. So a candidate carrier can be local, or covariant under the modular flows, but not both — every modular-covariant form from net data is positive-and-local, signature-free, or indefinite-but-non-local. Localization survives only in the net's index set — the assignment of algebras to regions — which *is* the geometry, put in by hand. The hedge hardened again, toward HIGH, now conditional on a single named assumption, with two obligations left standing between it and a theorem.

Iteration 9 discharged one of them and transformed the other. The ad-hoc constructibility assumption was replaced by a *naturality* argument: if a carrier is to be canonical — invariant under all symmetries of the algebra-and-vacuum pair — then its structure is forced into the commutant of the modular operator, and a second lemma showed that a localized, natural, indefinite form is impossible outright (the boost engine again, now promoted from computation to obstruction). But the referee's binding correction located a subtle gap in the first lemma: naturality forces the carrier into the *commutant* of Δ, which in general is strictly larger than the algebra *generated by*Δ. That daylight — the **commutant-to-generated gap** — became the named residual condition of the principal hedge, and the target of the next two years' work. The same iteration proved something quietly important about the empirical side: using Weiner's algebraic Haag theorem (arXiv:1006.4726), the hoped-for "reverse Weinberg–Witten" observable was shown to be logically equivalent to a positive solution of the carrier problem. The missing experiment and the missing mathematics are one obstruction wearing two costumes. And in the same breath the referee overturned one of the iteration's own submissions — a claimed gravitational cancellation collapsed on a fatal error (the transverse projector annihilates its own axis; the claimed positive coefficient was exactly zero), and a foothold the campaign had been building on was demoted accordingly. The method does not spare its own side.

The closure that was struck

Iteration 10 is the campaign's cautionary chapter, and this book narrates it as such because the striking of a result is the method working, not failing. The designated lever was explicit: close the commutant-to-generated gap and the principal hedge advances to unconditional HIGH. The finder track came back claiming victory — an elegant argument that in the geometrically meaningful (Bisognano–Wichmann) case, the vacuum's ergodicity makes the modular centralizer trivial, emptying the gap. Its companion track claimed a clean negative on the other side. Both submitted labels were corrected by the referee, and the centerpiece was **struck** as a major error.

The flaw was a wrong object. The centralizer — the elements of the algebra fixed by the modular flow — is not the gap. The gap lives one level up, in the representation space, driven by spectral degeneracy of Δ itself; and since Δ does not belong to the algebra, no theorem about the algebra's centralizer can empty it. The gap remained provably non-empty — the canonical η_0 still sits in it — and the hedge advanced nowhere. The record for the iteration reads: two attempted advances of the hedge's revision counter, both struck, one for the wrong-object error and one for an overstated discharge; grade unchanged, condition unchanged, ninth consecutive confirmation of the verdict. What the wreckage left behind was still useful: the gap was shown to be geometry-void on both of its sides — wherever geometry lives, the gap's interior is trivial; wherever the gap is populated, its occupants carry discrete bookkeeping data and no metric content. But classified is not closed, and the referee refused the conflation.

A false premise, and a terminal diagnosis

Iteration 11 attempted the opposite maneuver: if the gap could not be closed internally, perhaps it could be *reduced* — mapped exactly onto a named open problem of operator algebra theory, so that the world's specialists would be working on the wiki's residual without knowing it. The candidate was Connes' bicentralizer problem, open since the 1980s. The submission claimed a logical equivalence; the referee downgraded it to a loose structural relation, having caught the spine of the argument conflating two *opposite* conditions — a trivial centralizer, which kills a carrier and is unconditionally achievable on every Type III_1 factor (Marrakchi and Vaes, Crelle 809 (2024)), and Haagerup's bicentralizer-triviality, which produces a *large*-centralizer state in which a carrier exists. Worse, the referee caught the track's framing resting on a false premise: the claim, circulating in the mission brief, that the bicentralizer problem had been "resolved in general by Houdayer–Marrakchi–Tomatsu" is simply not true — the problem was and remains open in general, settled only for large classes. The catch is worth pausing on. A multi-agent research process is exactly the kind of system that can launder a plausible-sounding falsehood into a load-bearing assumption in one generation; the adversarial layer exists because of that failure mode, and here it fired.

The same iteration ran a meta-audit with a blunt question: does any purely-reasoning lever capable of moving the verdict still exist? The answer, refereed and upheld, was no. Of the seven listed residuals, the ones reasoning could decide were headline-inert by their own scoping, and the rest waited on external input — four of them on the *same* mathematical breakthrough. The marginal-yield trend was monotone to zero, and the audit gave the campaign its formal diagnosis: **terminal analytical saturation**. Consolidate; watch the clocks; do not iterate without genuinely new input.

Iterations 12 and 13 honored the diagnosis and confirmed it from two more angles. Iteration 12 retired the physics-net bicentralizer lever entirely (the theorem class that might have applied demands a structure — a normal conditional expectation — that the physical inclusions provably lack), swept the external mathematics and found nothing, and closed the indefinite-metric route. Iteration 13 audited the *frame*: is the carrier problem really the unique gate, or had fourteen iterations of tunnel vision missed a parallel door? Every candidate independent obstruction — the measurement problem, the cosmological constant, black-hole information — failed the two-part test of being both a wager-gate and reasoning-decidable. The bottleneck map was complete over the tracked set. The one shared structural invariant across all five convergent programs turned out to be the area law, which every geometry-first description reproduces too: screen-passing, not class-separating.

The assault, and the alien frameworks

Iteration 14 inverted the burden of proof. Three adversarial tracks were instructed to assume the verdict wrong and build, from all web-verifiable 2024–2026 literature, the strongest possible falsification of their assigned pillar — no hedging toward the house position. Against the encodes-not-generates pillar, the strongest candidate in the literature was a 2026 paper (Fullwood, Vedral, and

Guzmán-González, arXiv:2604.07471) deriving the Minkowski metric and full Lorentz group from an information-theoretic invariance on qubits. The attack collapsed at one named step: the determinant on 2×2 complex Hermitian matrices *is* a quadratic form of signature $(1, 3)$, unconditionally — the signature was installed the moment a complex two-level system was chosen as the carrier, then read back out as a discovery. Against the no-test pillar, the funded GQuEST experiment's target signal was shown, in its own authors' words, to be predicted at the same scale by emergent and non-emergent frameworks alike — it separates EFT-violating from EFT-respecting quantum gravity, an axis orthogonal to the wager. Against the saturation pillar, the most dangerous-looking new object — a non-Hermitian tensor network coupling a genuinely indefinite Krein structure to an emergent de Sitter geometry — turned out to pre-select the critical point that does all the work. All three pillars held, and the ledger recorded a thirteenth consecutive confirmation, this time earned under maximal fire.

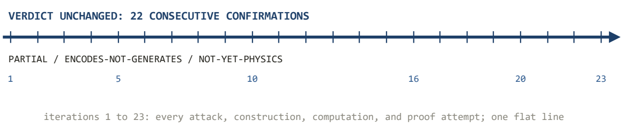
Iteration 15 then tried strangeness instead of strength: three mathematical frameworks never previously applied to the carrier problem — factorization algebras and functorial field theory, Connes-style index theory and K-theory, and higher-categorical formulations of QFT. All three reproduced the screen. The categorical equivalence between algebraic QFT and time-orderable prefactorization algebras (Benini, Carmona, Grant-Stuart, and Schenkel, arXiv:2412.07318) transports the carrier question verbatim, because both sides are functors over the same fixed causal-disjointness relation; every index pairing factors through a Dirac operator, which *is* the metric datum; and the higher-categorical world admits indefinite signature only as installed reflection data. The common smuggle could now be stated once, for all frameworks: the geometric input always enters as one of two interchangeable objects — the causal index set that says *where* things are, or the indefinite form that says *what kind* of geometry holds there — and no framework makes either an output. With that, the campaign judged the problem ready for hands other than its own, and iteration 15's deliverable was a dossier: the carrier problem distilled into its most attackable formal statement and posed publicly for external solvers. Fourteen consecutive confirmations; the final planned analytical iteration.

Iteration sixteen: the gap classified, the residual named

The saturation rule permitted new analytical work only if one of the watch clocks delivered genuinely new input. On June 22, 2026, one did: a new unconditional ergodicity theorem in exactly the operator-algebraic neighborhood of the residual (Marrakchi, arXiv:2606.23636). Iteration 16 was the sanctioned response, and it produced the first movement of the principal hedge in seven iterations — the first since iteration 9.

The verdict-bearing track finally attacked the commutant-to-generated gap with a lever the audits confirmed had never been used: intersect the naturality constraints of *all* wedge modular flows at once, rather than working with a single modular operator. The result, **Lemma G**, is a two-branch classification whose proofs the referee independently re-derived step by step. The first branch kills every *localized* natural carrier outright: Borchers' commutation relations force the fixed points of a wedge modular flow down to the vacuum ray, and the Reeh–Schlieder theorem finishes the job — any localized natural carrier is ± 1. The second branch handles carriers that give up localization: the boosts of all translated wedges together generate the full Poincaré group (a group identity verified by direct computation), so any fully-natural non-localized carrier commutes with everything Poincaré-covariant and is revealed as a *gauge-symmetry implementer* — an internal charge grading, with zero geometric content. Decisively honest, the lemma does not pretend the gap is empty: Fock-space parity, $(-1)^N$, the operator that counts whether a state holds an even or odd number of particles, certifies occupancy. The gap is real, populated, *classified* — and everything in it is bookkeeping, not geometry.

One branch of the classification could not be closed, and the campaign did with it what it had learned to do: pinned it to a single named hypothesis. On the fibers the lemma cannot reach — the algebras of bounded *double cones* in a massive theory, where the modular flow is non-geometric and its spectrum unknown — the no-go becomes a theorem if and only if a specific ergodicity statement holds. That statement was christened **(E_O)**: the vacuum's modular flow on every massive double-cone algebra fixes nothing but multiples of the identity, $M(O)_\omega = \mathbb{C}1$. Its logic is asymmetric in exactly the way an honest residual should be: a proof of (E_O) makes the carrier no-go a theorem on the stated hypotheses; a *disproof* is merely the entry ticket for a counterexample, which would still face every other wall the campaign had built.



The saturation flatline: twenty-three iterations of attack, and a verdict that has not moved through twenty-two consecutive refereed confirmations.

On the strength of the referee-verified proof package, the principal hedge advanced from its sixth registered revision to its seventh — grade unchanged at conditional HIGH, condition transformed from the diffuse "commutant-to-generated gap" to the sharp pair: the localization-template assumption, and (E_O). The house history was honored in the minting: the record notes that two prior attempts at this same advance, back in iteration 10, had been struck — and states exactly why this one differs (correct object, occupants exhibited rather than denied, written proofs surviving full adversarial re-derivation). The same iteration, in passing, caught a document-summarization fabrication in its own pipeline and kept the process record; resolved the Marrakchi theorem's non-transfer (its "with expectation" hypothesis is definitional, and the physical inclusions cannot satisfy it); watched seven of eight further untried frameworks reproduce the screen; and repaired three erroneous claims found standing in its own earlier record, by dated addenda rather than silent edits.

Fifteen consecutive confirmations. But for the first time since the campaign began, the entire remaining distance between the standing verdict and a theorem had a name, a definition, and a single mathematical address. The next iteration would go to that address and knock. What it found there — a question forty years of algebraic quantum field theory had somehow never asked — is the subject of the next chapter.

CHAPTER 8

The Gate Called (E_O)

One sentence about one algebra and one state now stands between the standing verdict and a theorem — and it turns out nobody, in forty years of algebraic quantum field theory, had ever asked it.

One sentence, one algebra, one state

Strip away fifteen iterations of scaffolding and what remains is a single mathematical sentence. Take a *double cone* O — the diamond-shaped spacetime region formed by intersecting the future of one point with the past of a later one; it is the algebraic quantum field theorist's model of a laboratory, bounded in both space and time. Attach to it the von Neumann algebra $M(O)$ of all observables measurable inside it, and let ω be the vacuum state. Tomita–Takesaki theory then supplies, canonically and for free, a distinguished one-parameter group of transformations of $M(O)$: the *modular flow* $\sigma_t = \mathrm{Ad}\,\Delta_O^{it}$, an intrinsic "time evolution" that the algebra-and-state pair carries whether anyone asked for it or not. The hypothesis on which everything now rests says that, for a massive theory, this flow stirs the algebra completely:

$$M(O)_\omega \;:=\; M(O) \cap \{\Delta_O^{it} : t \in \mathbb{R}\}' \;=\; \mathbb{C}\,1.$$

In words: the *centralizer* — the set of observables in the double-cone algebra that the vacuum's modular flow leaves exactly fixed — contains nothing but multiples of the identity operator. No observable localized in a bounded region of a massive theory sits still under the vacuum's intrinsic dynamics. That statement is hypothesis **($\mathbf{E}_O$)**: vacuum ergodicity on massive double-cone algebras. At the close of iteration 16 it carried the tag `[OPEN]`, and it carries it still.

It is worth feeling how specific this is. The analogous statement for a *wedge* — the region to one side of two crossing light-fronts, unbounded toward infinity — is a theorem, and an old one. For wedges, the Bisognano–Wichmann theorem identifies the modular flow with a Lorentz boost, a genuine geometric motion; Borchers' commutation relations then scale the lightlike translations to infinity, a boost-fixed vector is forced to be translation-fixed, a Cauchy–Schwarz equality pins it to the vacuum ray, and the Reeh–Schlieder theorem promotes the conclusion from vectors to the whole algebra. Every step of that engine leans on the wedge's unboundedness: there are translations that map the wedge into itself, and the argument rides them out to infinity. A double cone has no such translations — nothing maps a bounded diamond into itself and off toward infinity at the same time. The engine simply does not start. For *massless* theories the question closes anyway, because conformal symmetry makes the double-cone modular flow geometric too (Hislop and Longo: it is a conformal dilation, and one can transport the wedge argument across). The open case is exactly the massive double cone, where the modular flow is known to be non-geometric — it does not move points of spacetime along any flow lines at all — and where its spectrum, as of the iteration-16 audit, was simply unknown.

Why everything narrowed to this

The claim that a whole research verdict "reduces to" one hypothesis deserves suspicion, so it is worth restating exactly what Lemma G established and what it did not. The carrier no-go — the proposition, at the heart of the encodes-not-generates verdict, that no localized indefinite structure capable of carrying a Lorentzian signature can be derived from modular data — was proven on two of its three branches. Localized natural carriers die on the wedge engine; fully-natural non-localized carriers are gauge gradings with no geometric content. The third branch consists of candidate carriers living on massive double-cone fibers, precisely where the wedge engine has no purchase. On that branch the lemma's conclusion holds *if* the modular flow there fixes nothing — that is, if (E_O) is true.

The direction of the implication matters and was fixed by the referee as binding. Given the standing structural hypotheses (the standard axioms of a massive algebraic quantum field theory: isotony, locality, Haag duality, the split property, Poincaré covariance with a unique vacuum, Reeh–Schlieder, Bisognano–Wichmann for wedges) plus the localization-template assumption, a proof of (E_O) makes the carrier no-go a theorem — and with it, the no-distinguishing-test corollary becomes unconditional on those hypotheses. A *disproof* of (E_O), by contrast, is necessary but nowhere near sufficient for a counterexample: a non-trivial modular-fixed observable would merely be allowed to exist; to become a carrier it would still have to survive rotation equivariance, the cross-fiber consistency web, and the centralizer-sign vacuity results of iterations 7 through 10. The principal hedge's condition string after iteration 16 reads, verbatim, conditional HIGH on the localization template *and* (E_O). One assumption already argued irreducible; one open sentence of operator algebra. That is the entire distance between the book's verdict and a theorem.

The assault

Iteration 17 was the first dedicated attack on (E_O) as a mathematical object. Iteration 16 had located it; the synthesis note explicitly named "a proof of (E_O)" as the honest expected outcome of the next legitimate step. The iteration deployed five finder lenses — each attacking from a distinct technology — under three independent binding referees whose default stance was to refute, with every load-bearing 2020–2026 citation re-verified live against the literature. One lens worked the free field through second quantization; one worked the structure theory of Type III_1 centralizers; one tried to reduce the double cone to the solved wedge case; one was pure adversary, tasked with *disproving* (E_O) by constructing a counterexample; and one asked the humblest question available: is this already known?

The outcome, recorded at the top of the ledger: verdict unchanged, sixteenth consecutive confirmation; hedge unmoved at its seventh revision. (E_O) was neither proved nor disproved. But the iteration transformed the hypothesis from a bare sentence into a mapped object — and refuted, along the way, the one argument everyone would naturally reach for first.

The free field, and the trap in the obvious argument

For the free massive scalar field the problem ought to be as easy as it ever gets, because the entire modular structure is assembled functorially from one-particle data. The double-cone algebra is generated from a closed subspace of the one-particle Hilbert space; the vacuum is the Fock vacuum; and the modular operator *second-quantizes*:

$$\Delta_O = \Gamma(\delta_O), \qquad \sigma_t = \mathrm{Ad}\,\Gamma(\delta_O^{it}),$$

where δ_O is the *one-particle* modular operator and Γ is the functor that lifts a one-particle operator to the full multi-particle Fock space. This says the modular flow of the full field algebra is completely determined by a single operator acting on one-particle states — so the infinite-dimensional operator-algebra question should collapse to a spectral question about $\ln \delta_O$, an unbounded self-adjoint operator one can in principle just study.

The naive collapse is exactly what iteration 17 refuted. The tempting argument runs: if $\ln \delta_O$ has no eigenvalue at zero — no one-particle state exactly fixed by the modular flow — then nothing in the algebra is fixed, and (E_O) holds. The referee-confirmed obstruction is a piece of elementary but easily-forgotten Fock-space algebra:

$$\Gamma(u)\, d\Gamma(A)\, \Gamma(u)^* = d\Gamma(uAu^*),$$

where $d\Gamma(A)$ is the second quantization of a one-particle observable A — the operator that adds up A over all particles present. The identity says that conjugating such an operator by a second-quantized flow just conjugates its one-particle seed. Consequence:

for *any* A commuting with the one-particle modular flow — and there are always some, starting with the identity itself, whose second quantization $N = d\Gamma(1)$ is the particle-number operator — the operator $d\Gamma(A)$ is fixed by the full modular flow *regardless of point spectrum*. Bosonic Fock space carries modular-fixed number-type operators no matter how featureless the one-particle spectrum is. Absence of a zero eigenvalue kills the fixed *Weyl* operators, the exponentiated fields; it says nothing about these bilinear survivors. The naive reading — no kernel, therefore ergodic — is necessary but provably not sufficient.

What survives is a sharper and correct reduction, refereed at `[INFERENCE, high]`: if $\ln \delta_O$ has *purely continuous spectrum* — no eigenvectors at any value, equivalently a weakly mixing one-particle modular flow — then the full flow is weakly mixing on $M(O)$, and weak mixing forces the fixed-point algebra down to $\mathbb{C}1$ wholesale, sweeping the $d\Gamma(A)$ obstruction away without ever needing to decide which such operators are affiliated to the local algebra. Purely continuous one-particle spectrum implies (E_O) for the free field. The whole free-field case now hangs on one spectral fact.

And that fact, at iteration 17, was open — genuinely, embarrassingly open. The massive double-cone one-particle modular Hamiltonian was known only numerically: Bostelmann, Cadamuro, and Minz (arXiv:2209.04681, Annales Henri Poincaré 2023) computed it as "close to a multiplication operator" — which would mean purely continuous spectrum — but with an unresolved non-diagonal remainder "orders of magnitude smaller" that their numerics could not certify as zero, with follow-up in Cadamuro (arXiv:2312.08525). No closed form exists. The cautionary tale sits in the literature in plain sight: Longo and Morsella's paper on the massless modular Hamiltonian (arXiv:2012.00565) carries a version comment stating that its results on the *massive* modular Hamiltonian of a ball "contained a gap, and have been removed." The numerics lean toward continuous spectrum, hence toward (E_O) — but a lean is not a proof, and this project's tagging discipline exists precisely to keep the two apart.

Three roads that provably fail

For general interacting theories — the standing class with its standard axioms, not just the free field — iteration 17 ran the three structure-theory routes any operator algebraist would propose, and all three failed with *identified* fatal gaps, which is the useful way to fail.

The first route tries to import ergodicity from physics: massive vacua cluster — correlations between far-separated regions die off — so surely the modular flow mixes. The gap: cluster decomposition is a statement about *spacelike translation* to infinity, and the modular flow of a bounded region translates nothing anywhere; its orbit stays inside the double cone's algebra, and no known argument transports spatial clustering into modular mixing on a bounded region. The second route tries to inherit the wedge theorem: a double cone is an intersection of wedges, $M(O) \subset M(W)$, and wedge centralizers are provably trivial — so squeeze. The gap: an operator fixed by Δ_O need not be fixed by Δ_W; the two modular operators belong to different algebras and do not commute, and the half-sided modular inclusion machinery that reconstructs geometry from wedge data reconstructs precisely the *ambient* boosts and translations — flows that slide O off itself — never the non-geometric internal flow of O. The route would close (E_O) exactly when Δ_O is geometric, which is the conformal case already conceded. The third route reaches for genericity: by Marrakchi and Vaes, states with trivial centralizer are generic — a dense G_δ — on any Type III_1 factor. The gap is almost poignant: generic is not specific. A dense G_δ can omit any prescribed point, and the vacuum is about as prescribed as a state gets — pinned by Poincaré invariance, unique by axiom. Genericity says almost every state would satisfy the ergodicity property; it cannot say the one state physics cares about does. Marrakchi's own new theorem (arXiv:2606.23636), the very result whose appearance triggered iteration 16, does not transfer either: it holds for inclusions carrying a normal conditional expectation, and the physical inclusion $M(O) \subset M(W)$ provably carries none.

The disproof that failed

The fourth lens ran the attack this project's method demands: assume (E_O) false and build the witness. A non-scalar element of the centralizer — bounded, inside $M(O)$, exactly fixed by the modular flow — would be a genuine object, constructible in principle from the symmetries the axioms guarantee. The finder proposed four mechanisms: a localized conserved flux; invariants of the rotation subgroup that survives inside the double cone's symmetries; hidden point spectrum of $\ln \Delta_O$ in the uncharted massive case; and integrable or almost-periodic modular data. The referee, refusing to let the adversary off lightly, added four more of its own: interpolating Type I factors from the split property; central sequences; the canonical sign $\eta_0 = \mathrm{sgn}(\ln \Delta_O)$ from iteration 7; and Fock parity $(-1)^N$, the certified occupant of the commutant gap.

All eight died on the same triple wall: *bounded*, and *in* $M(O)$, and *exactly* Δ_O*-fixed* — pick any two and candidates abound; demand all three and every mechanism fails. The rotation generators are unbounded and global; the split-property factors are not modular-fixed; parity and η_0 live in the wrong algebra; conserved fluxes leak. The only bounded modular-fixed *vector* is the vacuum itself, and Reeh–Schlieder converts that into: only scalars.

Here the refereeing discipline produced one of its characteristic small severities. The finder, surveying the wreckage of its own disproof, submitted the summary "net evidence for (E_O) — leaning true." The referee struck the lean and re-graded the status to *bare OPEN*, with a stated reason worth quoting for what it says about the house epistemics: an adversary's failure to construct a witness from guaranteed symmetries is absence of witness, not absence of existence — the massive double-cone modular structure is spectrally uncharted, and an uncharted region is exactly where an unconstructed object could still live. Eight failed mechanisms move the psychology; they do not move the tag.

A question no one had asked

The fifth lens returned the iteration's strangest finding. An exhaustive sweep of the algebraic QFT and operator-algebra literature — refereed, cross-checked against the other lenses — found that no paper states, proves, disproves, or even *poses* the triviality of the vacuum centralizer on a bounded double cone, massive or massless. The ingredients have been on the shelf for decades: Tomita–Takesaki theory since the 1970s, the wedge theorem since Bisognano and Wichmann in 1975–76, the massive double-cone modular operator explicitly characterized — numerically — in the 2020s. The ergodicity question those ingredients jointly suggest was never asked.

Why not is a question about the sociology of mathematics rather than its content, but the shape of an answer is visible in this book's own arc. (E_O) is not interesting *in itself*; it is interesting as the last load-bearing wall of a particular structure — a fifteen-iteration reduction of the question "can algebra generate geometry?" to its minimal residue. Nobody asks about the ergodicity of a non-geometric flow on a bounded region unless something makes them need the answer, and until this campaign nothing had. The finding cuts both ways, and honesty requires saying so. It means the isolation of (E_O) is a genuine contribution: a new, sharply posed, unusually tractable open problem — for the free field, tractable enough that its modular operator is already sitting in published numerics. It also means the hypothesis arrives with no pedigree of failed attempts, no community conviction, and no partial results to lean on. The tag is `[OPEN]` because that is precisely what it is.

The iteration closed with the watch state re-sharpened to a single decidable sub-target: prove purely continuous spectrum for the free massive scalar's double-cone one-particle modular Hamiltonian. That proof would fire the hedge's next advance on the free subclass, making the carrier no-go — and with it the no-distinguishing-test corollary — a theorem there, modulo the one assumption already argued irreducible. A counterexample instead would hand the campaign its first entry ticket toward a verdict flip, with every other wall still standing behind the gate. Sixteen consecutive confirmations, and the entire question now balanced on the spectrum of one operator that a computer could tabulate but no one had proved anything about. The obvious next move was to compute it carefully and then try to prove what the computation showed. That — the project's first numerical experiment, and the theorem campaign it ignited — is the next chapter.

PART III

COMPUTATION, PROOF, VERDICT

Computation, proof, verdict: what twenty-three refereed iterations actually established.

CHAPTER 9

The Computation

Seventeen iterations reasoned about one spectral question; the eighteenth finally computed it — and then spent two more iterations, and several honest wrong turns, learning what the numbers were actually saying.

For seventeen iterations, the fate of the wager hung on an object nobody in this program had ever calculated. The gate condition (E_O) — the statement that the vacuum state has a trivial centralizer on a double-cone algebra, so that no local observable sits frozen under the modular flow — had been reasoned about, reformulated, reduced, and attacked from every analytic direction the referees would permit. It had never once been measured. That is not as scandalous as it sounds: the object in question is the spectrum of a modular Hamiltonian for a bounded region of a massive quantum field theory, and the mathematical physics literature, after fifty years, does not contain a closed form for it. But by iteration 17 the question had been squeezed into a shape a computer could grip: for the free massive scalar field, does the one-particle modular Hamiltonian of an interval have an eigenvalue at zero, or does its spectrum merely *accumulate* at zero, continuously, with nothing pinned there?

The distinction carries the whole program. An exact zero-mode would be a non-trivial element of the centralizer — an observable invariant under the modular flow — and that is precisely the "entry ticket" a carrier construction would need: a place inside the local algebra where extra structure could live without the modular dynamics washing it away. A purely continuous spectrum reaching zero means the centralizer is trivial, the ticket window is closed, and the encodes-not-generates verdict gains a computational leg to stand on alongside its analytic ones. Iteration 18 ran the computation. This chapter is the story of that computation, the two independent probes that followed it, and — because documenting wrong turns is the method, not an embarrassment to it — the two probes that failed.

What had to be computed

The setting is deliberately the most decidable case on the board: the free scalar field of mass m in one space and one time dimension. For a free field the time-slice axiom does real work here: the algebra of a double cone (the diamond-shaped spacetime region on which (E_O) is posed) equals the field algebra of its base — the interval of the $t = 0$ Cauchy line it rests on. So the double cone's modular theory *is* the interval's modular theory, and the question becomes one about a segment of a line at a single instant.

Better still, for a free field in its vacuum state everything is Gaussian, and Gaussian states are fully described by their two-point correlators. This is what makes the problem computable at all: the modular Hamiltonian of a region, an a priori horrendous object living on the full Fock space, is determined for Gaussian states by finite-dimensional matrix data once the theory is put on a lattice. The technique is the correlator method associated with Peschel, standard in the entanglement-entropy literature, here pointed at a question that literature rarely asks.

Discretize the field on a periodic ring of N sites. The Hamiltonian is

$$H = \tfrac{1}{2} \sum_i \pi_i^2 + \tfrac{1}{2}\, \varphi^\mathsf{T} K \varphi,$$

which says: the energy is the usual sum of a kinetic term in the field momenta π_i and a potential term coupling the field values φ_i through a matrix K — for our chain, a circulant matrix whose Fourier eigenvalues are $\lambda_q = m_{\text{lat}}^2 + 4\sin^2(\pi q/N)$, the lattice dispersion relation of a scalar of lattice mass m_{lat}. Writing $\omega = K^{1/2}$ for the one-particle energy operator, the ground-state correlators are

$$X = \tfrac{1}{2}\,\omega^{-1}, \qquad P = \tfrac{1}{2}\,\omega,$$

where $X_{ij} = \langle \varphi_i \varphi_j \rangle$ and $P_{ij} = \langle \pi_i \pi_j \rangle$: the vacuum fluctuations of the field are large where the energy is small and vice versa, in exactly inverse proportion — the harmonic-oscillator uncertainty relation written as a pair of matrices.

Now restrict attention to a region A: a block of L contiguous sites, standing in for the interval. Truncate both correlator matrices to the block, giving X_A and P_A, and form

$$C^2 = X_A P_A.$$

The eigenvalues ν_k of $C = \sqrt{X_A P_A}$ are the *symplectic spectrum* of the reduced state — the complete invariant of a Gaussian density matrix. Each satisfies

$$\nu_k \geq \tfrac{1}{2},$$

with equality exactly when that mode is pure: $\nu = \frac{1}{2}$ is the uncertainty-principle floor, and any excess above it measures how entangled that mode is with the rest of the chain. The modular Hamiltonian of the block is quadratic in the fields, and its single-particle energies — the rapidities at which the modular flow rotates each mode — are read off from the symplectic eigenvalues by

$$\varepsilon_k = \log \frac{\nu_k + \frac{1}{2}}{\nu_k - \frac{1}{2}}.$$

This formula is the hinge of the whole computation, so it deserves its plain-language sentence: a barely-entangled mode (ν close to $\frac{1}{2}$) has an enormous modular energy and is essentially inert under the modular flow, while a strongly entangled mode (ν large) has modular energy near zero — and a mode with ε *exactly* zero would be a fixed point of the flow, the non-scalar centralizer element that (E_O) forbids. The question "does the vacuum centralizer contain anything beyond multiples of the identity?" has become "does the list of numbers ε_k contain an exact, protected zero?" — a question about the eigenvalues of an $L \times L$ matrix.

One subtlety must be stated before the results, because everything downstream turns on it. The continuum interval algebra is a type III_1 factor, which forces the modular spectrum to reach all the way down to zero no matter what. Finding small ε_k therefore proves nothing by itself. The (E_O) question is *how* the spectrum reaches zero: continuously, with every mode sliding toward zero at the rate the continuum limit dictates and none arriving — or with an isolated mode that detaches from the crowd and pins itself at zero. Only the second outcome would be a carrier entry ticket.

Validation before verdict

A computation aimed at a load-bearing question earns trust the same way a proof does: by being checked against things already known. Two calibrations were run before the probe.

First, correlator decay. In the massive theory, field correlations fall off exponentially with a correlation length fixed by the lattice dispersion: $\xi = 1/(2\,\mathrm{asinh}(m/2))$. The measured decay matched the formula at both masses tested. Second, the area law: for a massive field in one dimension the entanglement entropy of a block must saturate to a constant as the block grows past the correlation length, because only the two endpoints contribute. It did, and the symplectic eigenvalues respected their floor $\nu_k \geq \frac{1}{2}$ throughout.

One calibration *failed*, and the failure is worth recording because it was correctly diagnosed rather than papered over: the massless check is contaminated by the well-known zero-mode of the massless lattice field on a periodic ring — an artifact of finite volume, not physics. The massless runs were flagged and excluded from any load-bearing role. The massive case, which is what (E_O) actually concerns, is IR-clean.

The result: a ladder, and nothing below it

The probe used continuum refinement: fix the physical interval size R and physical mass m, then shrink the lattice spacing $a = R/L$ by increasing L (with the lattice mass scaling as $m_{\mathrm{lat}} = mR/L$ and the ring kept eight times the block, $N = 8L$, so the environment stays effectively infinite). Anything that survives refinement is continuum physics; anything that drifts with L is lattice.

What survives refinement is strikingly clean. The smallest modular energy scales as

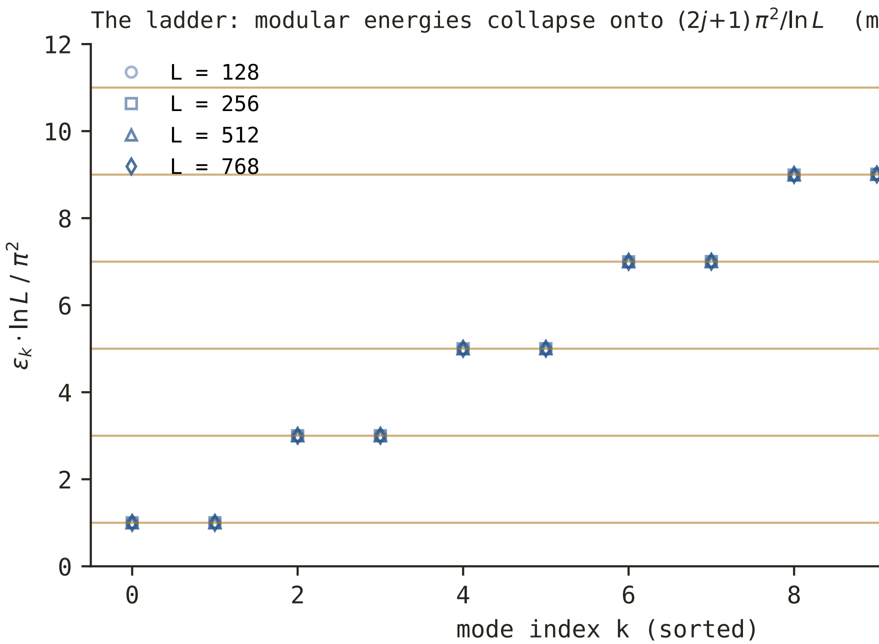
$$\varepsilon_{\min} \;\to\; \frac{\pi^2}{\ln L},$$

which says: the lowest rung of the modular spectrum falls to zero, but only logarithmically in the refinement, and with a universal coefficient $\pi^2 = 9.8696$ — the signature of a continuous spectrum filling in from above rather than an eigenvalue sitting at zero. The scaling is *mass-independent* in the continuum limit: at $m = 1$ the ratio $\varepsilon_{\min}/(\pi^2/\ln L)$ was 1.000 at every lattice size tested, from $L = 128$ to $L = 768$, and the product $\varepsilon_{\min} \ln L$ converged to π^2 across all masses.

Above the lowest rung, the low-lying modes assemble into a uniform ladder,

$$\varepsilon_k \;\sim\; \frac{(2k+1)\,\pi^2}{\ln L}, \qquad k = 0, 1, 2, \ldots,$$

each rung appearing as a *reflection-degenerate pair* — one copy for each endpoint of the interval, exchanged by the interval's mirror symmetry. And, decisively: no isolated mode sits below the ladder. The ratio of the actual minimum to the predicted bottom rung converged to 1.000; nothing splits off and races toward zero faster than the continuum accumulation.



Measured modular energies across four lattice refinements collapse onto the ladder $(2j+1)\pi^2/\ln L$ — the signature of a continuous spectrum accumulating at zero. Data from the repository's own computation (scripts/eo-modular-numerics.py).

This is the spectral portrait of a modular Hamiltonian with continuous spectrum and no eigenvalue at zero: a trivial centralizer. Numerically, (E_O) holds for the free massive scalar in two dimensions — no modular-fixed algebra element, no carrier entry ticket by this route. One further detail deserves emphasis because it disarms a natural hope. The double cone of a *massive* theory is the non-geometric case — its modular flow is not a pointwise spacetime flow, unlike the massless case, and one might have hoped the mass deformation would change the structure near zero where the centralizer question lives. It does not: near zero, the massive spectrum is numerically indistinguishable from the massless one. The mass shifts finite-lattice corrections and higher rungs (consistent with the "non-geometric part" analyzed by Bostelmann, Cadamuro and Minz, arXiv:2209.04681) but leaves the accumulation-at-zero structure — the only thing (E_O) cares about — untouched.

The honest scope statement, as it went into the ledger: lattice, finite L, two dimensions, free field. Strong evidence, tagged `[INFERENCE]` at high confidence; not a continuum proof, and silent on interacting theories. The script that produced it, `scripts/eo-modular-numerics.py`, ships with the repository and reruns in minutes.

The second signature: hunting embedded eigenvalues

Iteration 18 ruled out a zero-mode. But the analytic work of iteration 19 (the subject of the next chapter) sharpened a parallel worry: the full free-field gate requires excluding eigenvalues at *every* modular energy, not just zero. An eigenvalue at some $E \neq 0$ would be an *embedded* eigenvalue — a bound state hiding inside the continuous spectrum — and embedded eigenvalues are notoriously capable of existing where soft arguments say they shouldn't.

The lattice can hunt for these too, if one asks the right question. A continuum eigenvalue at $E \neq 0$ would appear as a lattice mode whose $\varepsilon_k(L)$ converges to a *fixed nonzero value* under refinement, while every genuine continuum mode shrinks as $(2j+1)\pi^2/\ln L$. So iteration 19 defined a scale-aware statistic per sorted mode:

$$g_k = \frac{\varepsilon_k(768)\ \ln 768}{\varepsilon_k(96)\ \ln 96},$$

which compares each mode's rescaled energy at the finest lattice against the coarsest: a continuum mode gives $g \approx 1$ (its $\varepsilon \ln L$ is constant), while a pinned eigenvalue — whose ε refuses to shrink — gives exactly $g = \ln 768/\ln 96 = 1.456$. The two hypotheses are separated by 45 percent; no delicate judgment call is required.

The scan swept masses up to $mR = 32$ — deep into the regime where the interval is many Compton wavelengths across and any mass-generated bound state would have every opportunity to form. The worst g found anywhere was 1.049 (at $m = 0.5$); at $m = 1$ the median was 1.0001, continuum scaling to three decimal places; heavier masses drifted slightly *below* one, the opposite direction from pinning. No embedded eigenvalue, at any energy, at any mass.

Two forensic details from the scan show the machinery working as machinery should. A single suspicious mid-spectrum drift, briefly a pinning candidate, resolved cleanly on closer inspection: $\varepsilon_{19} \ln L$ was converging *upward* to $21\pi^2$ — a late-onset ladder level still climbing into its $(2j+1)\pi^2/\ln L$ slot, not an eigenvalue. And modes with $\varepsilon \gtrsim 22$ sit at $\nu - \frac{1}{2} \lesssim 10^{-10}$, at the edge of double precision, where the naive eigenvalue solver becomes untrustworthy; the scan was precision-hardened by the symmetrized reformulation $\mathrm{eig}(X_A P_A) = \mathrm{eigh}(P^{1/2} X_A P^{1/2})$, which trades a non-symmetric problem for a manifestly symmetric one, and modes beyond the trusted window were excluded rather than reported.

The scan also dovetailed with the analytics in a satisfying division of labor: iteration 19's conditional Mourre theorem (next chapter) allows *at most finitely many* eigenvalues per compact energy interval; the scan says the actual number, in the accessible window, is zero.

The third signature — and the first failed probe

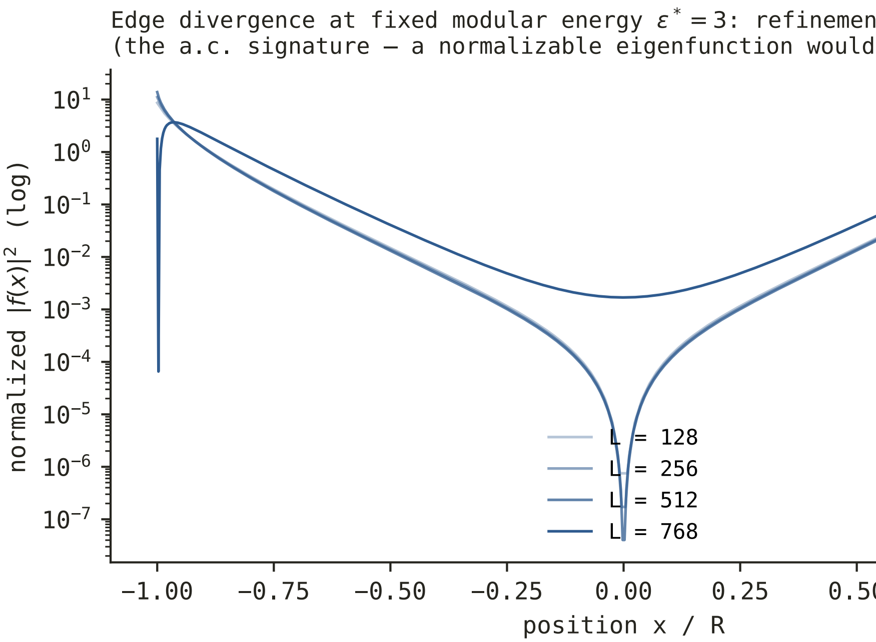
Iteration 20 went after the question at the level of eigen*functions* rather than eigen*values*. Here the record contains a genuine wrong turn, documented in the script's own comments, and it is exactly the kind of wrong turn worth narrating.

The naive idea: a normalizable eigenfunction should concentrate in the bulk of the interval, while a continuum (generalized) eigenfunction should pile up at the edges — so measure each mode's "bulk fraction" and look for outliers. The probe was built, run, and then *disqualified by its own design review*: on a fixed compact interval, bulk-versus-edge fraction per se does not discriminate.

Mid-ladder lattice modes legitimately spread across the interval without being eigenvalue candidates; the statistic flags them anyway. A probe that cannot distinguish its signal from its background is not evidence for either hypothesis, and the script says so in plain text where the next reader will find it. Nothing was concluded from it.

The repaired probe asks a sharper question: refinement scaling *at fixed modular energy* ε^*. Track the mode nearest a chosen ε^* as the lattice refines. If it is a continuum generalized eigenfunction, its norm is dominated by log-divergent mass at the interval's endpoints — so its central fraction must shrink toward zero and its edge mass must grow toward one as $a \to 0$. If it is an embedded normalizable eigenfunction, it converges to a fixed shape with a fixed, nonzero central fraction. The lattice's finite spacing acts as a regulator that the two hypotheses exploit in opposite directions.

The result, at $\varepsilon^* \in \{1.5, 3.0, 6.0\}$, masses $m \in \{1, 2\}$, and L from 96 to 768: central fraction at or below 0.03 and shrinking; mass in a fixed physical window at the edges between 0.96 and 0.997 and growing. The edge-divergent continuum signature, at every tracked energy, with no bulk-normalizable candidate anywhere.



The edge-divergence signature: at fixed modular energy, lattice refinement piles eigenfunction mass onto the interval's endpoints — generalized continuum behavior, not a normalizable eigenfunction. Same computation, second independent probe.

That made three independent numerical signatures — the ladder density-of-states, the pinning scan, the edge-divergence profile — all reporting the same verdict: purely continuous spectrum, no point spectrum, (E_O) holds for the free massive scalar as far as any lattice can see.

The second failed probe: a lesson in frames

The other documented wrong turn came later, in preparatory work for iteration 22, and it is subtler — a failure not of experimental design but of linear algebra hygiene.

By then the analytic program had produced a beautiful structural claim (Chapter 10 explains where it came from): the spectrum of the operator the numerics diagonalize should be, up to a compact correction, a function of the Möbius-flow generator D of the interval — the ladder $\varepsilon_k \to (2k+1)\pi^2/\ln L$ being exactly the box quantization of D in the logarithmic coordinate. The temptation was to verify this at the *vector* level: take the computed eigenvectors, pair them against D, and check the dispersion relation mode by mode.

The first attempt produced ratios that drifted from 0.77 down to 0.53 across the ladder — a trend that, taken at face value, would have undermined the spectral identity. It was an artifact. The computed eigenvectors of the symplectic problem are orthogonal only in a transformed inner product (the $S^{1/2}$-conjugated frame, in the program's notation), while D had been built as a matrix in the raw discretization frame; pairing vectors from one frame with an operator from another silently inserts a non-identity metric between them, and the "drift" was that metric, not physics. A companion test — windowed commutators evaluated through the pseudoinverse of the non-orthogonal physical frame — was worse: the pseudoinverse amplified the frame's poor conditioning into what looked like a non-compact correction term, faking evidence *against* the structural claim.

Both tests were struck, the failure mode was written into the script as a methodology caveat, and the frame-correct version was specified and handed forward: either conjugate D into the frame where the diagonalized operator is genuinely self-adjoint, or compare spectra only, never raw vectors. When the frame-fixed pairing was eventually run, the ratios improved to 0.86–0.74 with the residual deficit attributable to a known discretization effect (central-difference damping on oscillations concentrated at the corners) — and the clean quantitative confirmation of the structural claim remained the phase-fit measurement of iteration 21, which agreed with prediction at the one-to-three-percent level.

The point of telling this story is not that the mistake was caught — mistakes usually are, eventually. The point is *where* it was recorded: in the reproducible script, adjacent to the code, addressed to the next person who would naturally make it. A program whose central claim is that it refuses to fool itself has to leave its near-misses on the record, because the near-misses are the calibration data for how much to trust the hits.

What three signatures buy, and what they don't

By the close of iteration 20, the computational file on free-field (E_O) read as follows. One direct spectral measurement: the modular spectrum accumulates at zero as $\pi^2/\ln L$, universally, with no isolated mode below the ladder and no zero-mode. One refinement-scaling scan: no embedded eigenvalue at any energy, for masses up to $mR = 32$, with the worst candidate statistic at 1.049 against a pinned prediction of 1.456. One eigenfunction-profile test: edge-divergent continuum behavior at every tracked energy, no bulk-normalizable mode anywhere. Three probes, three different observables, one conclusion.

And yet the verdict line of the project did not move — not at iteration 18, not at 19, not at 20. The hedge stayed at its level; the standing conclusion remained PARTIAL, encodes-not-generates, not-yet-physics. That is not the numerics being disrespected; it is the numerics being *classified correctly*. A lattice computation at finite L in $d = 2$ for a free field is evidence about a continuum statement, not a proof of one, and this program's house rules price the difference explicitly. What the computation did do was transform the question's status: (E_O) for the free field was no longer a conjecture floating on analogy, but a sharply-posed spectral statement with three independent lines of numerical support and a visible target for proof.

Which raised the obvious next demand, and the program's referees raised it immediately: *prove what the numerics found*. That is the next chapter.

CHAPTER 10

Proof, Where It Could Be Had

Three iterations of theorem-hunting turned half the free-field gate into established mathematics, traced the other half to a 1989 paper's page 429, and compressed what remains into a single lemma a specialist could check cold.

Numerics persuade; they do not prove. The three signatures of the last chapter left the free-field gate condition (E_O) in an uncomfortable and productive position: overwhelmingly supported, formally undecided. The program's house rules are unsentimental about this distinction — the hedge on the central vacuity hypothesis could only advance if the free-field statement closed *as a theorem*, referee-confirmed. So iterations 19 through 21 went hunting for theorems, under the standing adversarial protocol: every proposed lemma submitted to a referee whose default stance is REFUTE, every load-bearing citation live-verified against the primary source, every gap named rather than glossed.

The hunt did not close the gate. It did something arguably more valuable for a program whose product is calibrated knowledge: it partitioned the question cleanly into a half that is now proved, a case that is now theorem-grade, and a residual that is now *exactly one statement* — sharper at each iteration, and by the end checkable by a specialist in a field that has never heard of this project. This chapter is the anatomy of that partition.

The E = 0 half: LEM-K0

The zero-mode question — is there a vector fixed by the modular flow, equivalently an eigenvalue of the modular operator δ_O at 1? — turned out to be answerable with tools that have existed for decades, assembled into a lemma the program called LEM-K0.

The natural language for one-particle modular theory is that of *standard subspaces*: a closed real subspace K of a complex Hilbert space H, positioned so that K and iK together span everything while intersecting trivially. This is the one-particle shadow of a local algebra with a cyclic and separating vacuum, and it carries its own Tomita data: an antilinear involution-like operator $S = J\delta^{1/2}$ with $K = \mathrm{fix}(S)$, where J is an antiunitary conjugation and δ the (positive, generally unbounded) modular operator. The symplectic complement K' is the fixed-point set of $S^* = J\delta^{-1/2}$. In plain language: K is the set of one-particle vectors reachable by local field operators in the region, K' the corresponding set for the causal complement, and J, δ encode how the two are geometrically entangled with each other.

LEM-K0 states, first, a structural identity:

$$K \cap K' = \big(\ker(\delta - 1)\big)_J, \qquad \ker(\delta - 1) = (K \cap K') + i\,(K \cap K'),$$

which says: the vectors common to a region and its complement are exactly the J-invariant part of the modular operator's fixed space — so the standard subspace is *factorial* (trivial intersection $K \cap K' = \{0\}$, the one-particle analogue of the algebra having trivial center) precisely when 1 is not an eigenvalue of δ. Second, a bootstrap: if K is factorial, then K contains no eigenvector of δ for *any* eigenvalue. The proof is four moves: a vector $f \in K$ with $\delta f = \lambda f$ satisfies $Sf = f$, hence $Jf = \lambda^{-1/2} f$; applying J twice and using $J^2 = 1$ forces $\lambda = 1$; and part one then forces $f = 0$. An eigenvector inside the local subspace is trapped between the conjugation and the modular scaling, and only the trivial case survives.

The application closes the E = 0 half. For the massive free scalar double cone, the second-quantized algebra is known to be a type III $_1$ factor — Araki's work, with the double-cone case in Figliolini and Guido (J. Operator Theory 31, 1994) — and Araki duality transports factoriality down to the standard subspace K_O. By LEM-K0, $1 \notin \sigma_p(\delta_O)$: no modular-fixed one-particle vector, hence no fixed Weyl operator, hence the E = 0 half of free-field (E$_O$) *as a theorem*. The zero-mode the lattice never found provably does not exist. `[ESTABLISHED]`

Page 429

Then came the novelty check, and with it the iteration's best story.

The protocol requires that any claimed lemma be searched against the literature before it enters the ledger as new mathematics. The referee traced the E = 0 exclusion back to the paper where the double cone's Tomita operator was first seriously computed — Figliolini and Guido, *The Tomita operator for the free scalar field*, Annales de l'Institut Henri Poincaré 51 (1989) 419–435 — and did not settle for the abstract or a secondary citation. The referee downloaded the full 1989 PDF from the numdam archive and read it. On page 429, in the middle of the spectral analysis, sits the sentence beginning "It is clear from (3.6) that $1 \notin \sigma_p(\delta)$" — Figliolini and Guido had stated and proved the E = 0 exclusion thirty-seven years earlier, by a different mechanism: the *antilocality* of the operator $\omega = (-\Delta + m^2)^{1/2}$, a property established by Segal and Goodman in 1965. (Antilocality is a strange and powerful rigidity: no nonzero function can vanish on an open set *together with* its image under ω — the square root of the Klein–Gordon operator smears everything everywhere.)

So LEM-K0 was reclassified, and the reclassification is the method on display. It is not new mathematics; it is a verification of a 1989 result from the primary source, plus an independent second proof through the factoriality route (whose structural part (a), the referee also established, is standard-subspace folklore — stated explicitly in a 2025 Communications in Mathematical Physics paper on inclusions of standard subspaces). The temptation in a program hungry for wins is obvious: announce the lemma, cite the paper vaguely, collect the credit. The ledger instead records: FG 1989 proved it first; we verified it verbatim and re-proved it independently; the previously `[unverified]` tag on the claim is retired to `[ESTABLISHED]` with a page-number anchor. A program that will not let itself inflate a thirty-seven-year-old theorem into a novelty is a program whose positive claims mean something.

The massless case, theorem-grade

One more piece of the map firmed up in the same iteration. For the *massless* field, the double cone is conformally equivalent to a wedge — the Hislop–Longo construction conjugates the double-cone modular flow to a wedge boost — and there the heavy machinery of wedge theory applies: Borchers' commutation theorem and the Stone–von Neumann uniqueness argument together force the generator $\ln \delta_O$ to have purely absolutely continuous spectrum, the whole real line, with no eigenvalues at all. For the massless double cone in 3+1 dimensions this is now theorem-grade `[ESTABLISHED]`; the 1+1 scalar testbed needs an infrared-safe variant built on the derivative field, graded one notch lower. The massless case, in other words, is *easy* — and understanding precisely why it is easy (a symmetry the mass destroys) became the organizing clue for everything that followed.

The residual, and why it is genuinely hard

What remained was the $\lambda \neq 1$ exclusion: no eigenvalue of the massive double-cone modular operator anywhere else, the embedded-eigenvalue question the pinning scan had probed numerically. Iteration 19 threw three analytic lenses at it, and all three returned PARTIAL — but the failures were informative in a way that reshaped the problem.

The literature lens established that the statement is proved *nowhere*: a once-claimed massive result had been withdrawn, and a 2023 paper (Cadamuro, Fröb and Minz, arXiv:2312.04629) calls it an open problem outright. Worse — in the useful sense — the lens proved a meta-theorem about *why* it is open: invariant-level arguments cannot decide it. There exist factors of the same Connes type III_1 whose one-particle modular spectra are pure point (Powers and Araki–Woods constructions realize them), so no amount of reasoning from the algebra's classification invariants can exclude eigenvalues; any proof must use the specific locality and mass-shell data of the free field. And the one known kill mechanism in the field — Borchers commutation — is exactly what bounded regions lack; the rigidity lens proved that the interval-inside-wedge inclusion is *not* a half-sided modular inclusion for $m > 0$, so the instant Borchers–Wiesbrock kill is unavailable. The Mourre lens produced a conditional theorem: under two unproven regularity hypotheses, the point spectrum in any compact interval away from zero is at most finite — a bound the lattice scan sharpens, in its window, to zero.

The rigidity lens also produced the first genuinely new constraint: any eigenvector of δ_O extends boost-analytically into a strip of width π in *every* wedge containing the double cone, with a norm bound independent of the wedge. No contradiction was extracted — but a candidate eigenfunction now had to thread multiple needles simultaneously.

And the novelty check paid a second dividend. Figliolini and Guido's own Theorem 3.6 converts eigenvalues $\lambda \neq 1$ of δ_O into real eigenvalues $\mu = (\lambda+1)/(\lambda-1)$ with $|\mu| > 1$ of an auxiliary operator, and their antilocality argument kills exactly the boundary points $\mu = \pm 1$. The sharpest open target, minted at the close of iteration 19: *extend the antilocality argument from* $\mu = \pm 1$ *to all real* μ — one mechanism, the same toolset FG already used, no operator algebra required.

The attack: an exact reformulation and two rigidity theorems

Iteration 20 attacked that target head-on with three refereed lenses and a binding assembling referee, and although no complete proof assembled — the hedge held, the verdict ran to its nineteenth consecutive confirmation — the problem came out the other side compressed to its sharpest form ever.

The centerpiece is an exact reformulation, referee re-derived. Work on the interval $I = (-R, R)$ with exterior $E = \mathbb{R} \setminus \bar{I}$ and the fractional operator $\omega = (m^2 - d^2/dx^2)^{1/2}$. Then:

$\mu^2 > 1$ is an eigenvalue $\iff \exists\, F, G : \quad \omega F = 0$ on $I, \quad \omega G =$ 0 on $E, \quad G = cF$ on $I, \quad G = F$ on $E,$

with coupling $c = \mu^2/(\mu^2 - 1) > 1$. In words: an eigenvalue exists precisely when two functions can be found, each annihilated by ω on complementary regions, that agree outside the interval and differ by a fixed stretch factor c inside it. Set $\mu^2 = 1$ and the statement degenerates to: F vanishes on I while ωF also vanishes there — which is exactly what Segal–Goodman antilocality forbids. FG 1989 is the boundary case of the open problem; the open problem is *bilateral antilocality with coupling* $c > 1$. Every trace of operator-algebra language has been boiled away: what remains is a statement about one fractional differential operator, addressable by people who study fractional operators.

Two unconditional rigidity theorems then constrained any would-be eigenfunction, both independently re-derived by the referee `[ESTABLISHED]`. First, *forced Dirichlet rigidity*: every eigenfunction ψ lies in $H^1_0(I)$ — the space of functions with square-integrable derivative vanishing at the endpoints. The proof chains a bootstrap ($\psi = \nu^{-2} X_I \varphi$ with φ square-integrable, and ω^{-1} smoothing one full derivative — the mass is essential here) through the Lions–Magenes characterization of the relevant form domain, whose Hardy-type condition $\int |\psi|^2/\mathrm{dist}(x, \partial I)\, dx < \infty$ is incompatible with a nonzero boundary value: continuity forces $\psi(\pm R) = 0$. Second, *support rigidity*: no eigenfunction vanishes on any open subinterval, and no defect datum vanishes on any open subset of the exterior — nothing about a candidate mode can be locally trivial. A hypothetical eigenfunction must now vanish at the endpoints, vanish nowhere else in bulk, oscillate boost-analytically in every ambient wedge, and survive refinement scaling the lattice says nothing survives.

The iteration also proved a sobering *no-soft-theorem obstruction*. Bertola, Katsevich and Tovbis (arXiv:2008.10058) showed that in the closely analogous family of finite-interval convolution operators — the multi-interval finite Hilbert transform — embedded eigenvalues *genuinely occur*, geometry-dependently, inside absolutely continuous spectrum. So no family-level argument, no general principle about operators of this class, can decide our case: the answer depends on the specific geometry, and only geometry-specific analysis can reach it. A companion structural fact (via the Grabovsky–Hovsepyan classification) explains the pattern of what is solved and what is not: the massive kernel admits no commuting differential operator of prolate type, because $K_0(m|z|)$ has a logarithmic branch point where the classification demands meromorphy — exactly the miracle that makes massless problems exactly solvable is absent for $m > 0$.

The iteration closed by minting its successor target, strictly weaker than the one it inherited:

$$\textbf{LEM-A1}':\quad \omega F = 0 \text{ on } I,\quad \omega(\chi_E F + c\,\chi_I F) = 0 \text{ on } E,\quad \chi_I F \in H_0^1(I) \implies F \equiv 0,$$

for every $c > 1$ — the bilateral-antilocality statement, now with the H_0^1 hypothesis *supplied for free* by the Dirichlet rigidity theorem. The proposed route runs through a Caffarelli–Silvestre-type extension: realize the fractional operator $(m^2 - \Delta)^{1/2}$ as the boundary map of the elliptic operator $\partial_y^2 + \partial_x^2 - m^2$ in a half-plane, turning the coupled pair into a mixed Dirichlet–Neumann (Zaremba) boundary-value problem whose behavior at the two corner points $\pm R$ should decide everything. The lemma was posed so that a specialist in fractional unique continuation could check it cold, with no operator-algebra input whatsoever.

The Corner Indicial Theorem

Iteration 21 attempted LEM-A1', and its first established result is the one this chapter closes on, because it is the program's single most striking piece of mathematics: the corner computation, hand-derived by the integrator, submitted to a binding adversarial referee under default REFUTE, and returned CONFIRMED_WITH_CORRECTIONS.

At each corner $(\pm R, 0)$ of the extension problem, pass to polar coordinates (r, θ) with $\theta \in (0, \pi)$ spanning the half-plane, and ask the classical Kondratiev question: for which exponents s does the problem admit local solutions scaling as r^s? The coupled boundary conditions — Neumann conditions $\partial_\theta u(\pi) = 0$ and $\partial_\theta v(0) = 0$ on one side of each function, and the coupling $v(0) = u(0)$, $v(\pi) = c\, u(\pi)$ stitching them together — feed a 4×4 determinant that the referee re-derived symbolically and evaluated numerically to 10^{-14}. It collapses to the astonishingly clean *indicial equation*

$$\cos^2(\pi s) = c,$$

which says: the admissible corner exponents are exactly those where the squared cosine of πs reproduces the coupling constant. Since $c > 1$ while $\cos^2$ of a real argument never exceeds 1, no real exponent qualifies: the roots are forced into the complex plane, and the complete root set is

$$s = n \pm i\tau_0, \qquad n \in \mathbb{Z}, \qquad \tau_0 = \frac{\operatorname{arccosh} \sqrt{c}}{\pi},$$

every root simple. In words: near each corner, every local solution oscillates *logarithmically* — it behaves like $r^n r^{\pm i\tau_0} = r^n e^{\pm i\tau_0 \ln r}$, spinning at a fixed rate per e-folding of distance as it approaches the corner, at frequency τ_0 set by the coupling alone.

Now translate c back through the modular dictionary, $c = \cosh^2(\varepsilon/2)$ with ε the modular energy. The two identities combine — verified to twelve digits — into

$$\tau_0 = \frac{\varepsilon}{2\pi}.$$

This deserves a moment of quiet. The relation between a modular "energy" and a logarithmic oscillation frequency, with the factor 2π that carries the Unruh temperature, is the local Bisognano–Wichmann/Rindler relation — the deepest structural fact of modular theory, ordinarily obtained from Lorentz covariance and the analytic structure of vacuum correlations. Here it fell out of a boundary-value computation at a corner of a half-plane: four boundary conditions, one determinant, no quantum field theory in sight. And it is not merely internally consistent — it is *lattice-verified*. The corner-Mellin oscillation $|x \mp R|^{-1/2+i\tau}$ is exactly the edge behavior whose shadow is the iteration-18 ladder; two independent estimators on the $L = 768$ data (a zero-counting staircase across the ladder, showing the expected two-corner parity doubling, and an analytic-signal phase fit) matched $\tau_0 = \varepsilon/2\pi$ at the one-to-three-percent level on well-resolved modes. The Chapter 9 numerics and the corner theorem are measurements of the same object from opposite ends.

The theorem's fine print, faithfully carried: dropping the mass term at the corner is legitimate (it enters at weight $+2$, by the classical Kondrat'ev theory), *but* because $s + 2$ is again a root whenever s is, the mass correction series is resonant — logarithmic factors generically appear at order r^{s+2}, and any global argument built on corner expansions must carry them. And the H^1 classification cuts both ways: the $n = 0$ family $r^{\pm i\tau_0}$ has gradient blowing up like r^{-1} and is *not* locally H^1 — these are the continuum modes, present for every c — while the $n \geq 1$ families *are* H^1-admissible with boundary-compatible traces. The corner, by itself, forbids nothing. There is no local kill; the exclusion, if it exists, must be global.

The ledger, honestly ruled

So the accounting at the close of this stretch of the program, graded as the ledger grades it. Proved: the E = 0 half of free-field (E_O), twice over — FG 1989's antilocality argument, verified at the primary source, and the independent factoriality route through LEM-K0. Proved: the massless 3+1 double cone has purely absolutely continuous modular spectrum, no eigenvalues. Proved, unconditionally: any massive eigenfunction lies in H^1_0, vanishes nowhere locally, and lives above a corner structure whose indicial equation is $\cos^2(\pi s) = c$ and whose oscillation frequency recovers Bisognano–Wichmann exactly. Proved, as obstruction: no invariant-level argument, no soft family-level theorem, and no local corner analysis can close the residual.

Open: the residual itself — by now a single lemma about a single fractional operator, with the quantifier structure and function-space hypotheses fully pinned. Three iterations, three referee verdicts of no-complete-proof, three consecutive confirmations of the standing verdict (the eighteenth, nineteenth, and twentieth), and the hedge held unmoved at its level throughout, because the house rule demands full closure and full closure was not achieved.

It would be easy to read this as failure. The truthful reading is the opposite: the program took a question that in early iterations was a fog — "does the modular flow of a bounded region hide a fixed point?" — and machined it down to a statement a fractional-unique-continuation specialist could evaluate over a long afternoon, while proving everything provable in the neighborhood and refusing, twenty times running, to claim the one thing it could not prove. Where proof could be had, it was had. Where it could not, the ledger says so, in a form the outside world can now check.

CHAPTER 11

One Operator, One Question

In which twenty-three iterations of adversarial refereeing compress an entire program into a single spectral question about a single explicitly constructed operator — and four proved no-go theorems fence off every easy way of answering it.

This is the chapter for the reader who wants the real thing. The preceding chapters told the story of the wager in plain language; here we write down the mathematics that the story is about, at the level of precision the refereed record actually achieves. Nothing in this chapter is softened. Where a claim is proved we say so; where it is conditional we name the conditions; where the program retracted a claim we print the retraction. The epistemic tags of Chapter 1 do real work in this chapter, because the entire drama here is the distance between `[ESTABLISHED]` and `[OPEN]`.

The situation entering iteration 20 was this. The free-field gate — the statement labeled (E_O) in the ledger, the last thing standing between hedge rung R7 and rung R8 — had been reduced to a question about the modular operator of a single interval for the free massive scalar field in two spacetime dimensions. Half of that question — the E = 0 half, the absence of a modular zero-mode — was established in iteration 19. What remained was the eigenvalue half: show that the modular operator of the interval has no point spectrum away from the trivial eigenvalue. Three independent numerical signatures said the spectrum is purely continuous. No proof existed. What happened over iterations 20 through 23 is that this question was transformed — exactly, reversibly, with every step refereed — into a statement about one self-adjoint operator on one

Hilbert space, and then the boundary of what any known method can do to that operator was mapped with unusual thoroughness. The verdict did not change. The question did.

From modular theory to antilocality

Fix the free scalar field of mass $m > 0$ on two-dimensional Minkowski space, and fix the interval $I = (-R, R)$ on the time-zero line, with exterior $E = \mathbb{R} \setminus \bar{I}$. The single-particle structure of the theory is governed by the operator

$$\omega = \left(m^2 - \Delta\right)^{1/2},$$

where $\Delta = d^2/dx^2$ is the one-dimensional Laplacian: ω is the relativistic energy of a single particle, the square root of momentum-squared plus mass-squared, and every question about the vacuum state restricted to the interval passes through it. The modular operator of the interval algebra — the Tomita–Takesaki object whose spectrum encodes how the vacuum entangles I with its exterior — has eigenvalue problem best written in the variable Figliolini and Guido introduced in 1989: $\mu = (\lambda + 1)/(\lambda - 1)$, where λ is a modular eigenvalue, so that the trivial eigenvalue $\lambda = 1$ is pushed to $\mu = \infty$ and the question becomes whether any $\mu^2 > 1$ occurs.

Iteration 20 proved the following exact reformulation `[ESTABLISHED — the referee re-derived every step]`. The value $\mu^2 > 1$ is a modular eigenvalue if and only if there exist a pair of functions F, G — defect-matched data in the appropriate Sobolev spaces — satisfying

$$\omega F = 0 \text{ on } I, \qquad \omega G = 0 \text{ on } E, \qquad G = cF \text{ on } I, \qquad G = F \text{ on } E,$$

with coupling constant $c = \mu^2/(\mu^2 - 1) > 1$. In words: an eigenvalue exists exactly when two functions can each be "invisible" to the relativistic energy operator on complementary regions while agreeing outside the interval and disagreeing by a fixed factor c inside it. This is a statement purely about the fractional operator ω. Every trace of operator algebras, modular flows, and von Neumann theory has been eliminated — encoded, one might say, into a single real parameter.

The reformulation also explains why the one solved case is solved. Figliolini and Guido's 1989 theorem — that $\mu^2 = 1$ is impossible — is exactly the degenerate boundary case: there F vanishes on I while ωF also vanishes there, and the antilocality of ω (the Segal–Goodman theorem: a function and its ω-image cannot both vanish on a common open set unless the function is zero) kills it outright. The open problem is therefore, in the phrase the ledger adopted, *bilateral antilocality with coupling* $c > 1$: FG-1989 is its boundary case, and everything above the boundary is open.

Two unconditional rigidity theorems came with the reformulation, both `[ESTABLISHED]`. First, forced Dirichlet rigidity: every putative eigenfunction lies in $H^1_0(I)$ — it is not merely square-integrable but has square-integrable derivative and vanishes at both endpoints $\pm R$. The mass is essential here; the proof runs through the smoothing $\omega^{-1} : L^2 \to H^1$ and a Hardy-type condition at the boundary. Second, support rigidity: no eigenfunction can vanish on any open subinterval of I, and no defect datum can vanish on any open subset of the exterior. Any counterexample to the wager's gate, if one exists, is thus a function pinned at both ends, oscillating everywhere, hiding nowhere.

And one obstruction came with it, which deserves to be stated carefully because it shaped everything after. Bertola, Katsevich, and Tovbis (arXiv:2008.10058) proved that the finite Hilbert transform on a union of several intervals has absolutely continuous spectrum *with possibly embedded isolated eigenvalues*, occurring or not depending on the geometry. This is the no-soft-theorem obstruction `[ESTABLISHED — live-verified]`: embedded point spectrum genuinely occurs in the family of finite-interval convolution operators to which our operator belongs. No family-level theorem — no argument of the form "operators of this general type never have embedded eigenvalues" — can decide our case, because the general statement is false. Only geometry-specific analysis can. The massive kernel, moreover, admits no Morrison- or prolate-type commuting differential operator (its Bessel kernel $K_0(m|z|)$ has a logarithmic branch point, which the Grabovsky–Hovsepyan classification excludes), which is a structural explanation of why exactly the massless cases are the ones the literature has solved. From iteration 20 onward, the program knew it was digging in ground where no general theorem would ever strike water.

The sharpest missing piece was named LEM-A1' (read: LEM-A1-prime): for every $c > 1$, if $\omega F = 0$ on I, if $\omega(\chi_E F + c\,\chi_I F) = 0$ on E, and if $\chi_I F \in H_0^1(I)$ — the last input now free, by Dirichlet rigidity — then $F \equiv 0$. Here χ_I and χ_E are the indicator functions of the interval and its exterior, the operators that cut a function down to one region. A statement checkable cold by a specialist in fractional unique continuation, with no operator-algebra input whatsoever. Iteration 21 attacked it directly.

The strip: a corner problem becomes a waveguide

The attack began at the corners. Near the endpoints $\pm R$, the coupled system defining LEM-A1' is a boundary-value problem with a jump, and such problems are governed by their *indicial equation* — the algebraic equation whose roots dictate how solutions can behave as one approaches the corner. Iteration 21's Corner Indicial Theorem [ESTABLISHED — refereed, the determinant re-derived symbolically and numerically to 10^{-14}] computes it in closed form: for the coupled extension pair at a corner, the indicial equation is

$$\cos^2(\pi s) = c,$$

and its complete root set is $s = n \pm i\tau_0$ for integer n, every root simple, with

$$\tau_0 = \frac{\operatorname{arccosh}\sqrt{c}}{\pi}.$$

This says that solutions near the corner behave like $r^{n \pm i\tau_0}$ in the distance r to the endpoint: a discrete ladder of allowed behaviors, each carrying a fixed logarithmic-oscillation frequency τ_0 determined by the coupling alone. Under the modular dictionary

$c = \cosh^2(\varepsilon/2)$, one finds $\tau_0 = \varepsilon/2\pi$*exactly* — the Bisognano–Wichmann relation between modular parameter and Rindler temperature, recovered from a pure boundary-value computation with no field theory in it. The $n = 0$ family oscillates boundedly with gradient blowing up like r^{-1} and is not locally H^1: these are the continuum modes. The $n \geq 1$ families *are*H^1-admissible. The consequence is sobering: no local argument at the corner can kill a putative eigenfunction. The exclusion, if it exists, must be global.

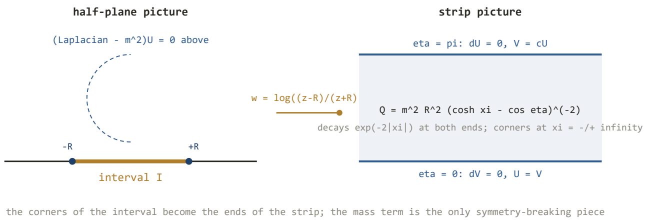
So iteration 21 globalized. The Möbius–logarithm map

$$w = \log \frac{z - R}{z + R}, \qquad z = -R\coth(w/2),$$

sends the interval problem to an infinite strip $\{0 < \eta < \pi\}$ in the coordinates $w = \xi + i\eta$ — the two corners $\pm R$ go to the two ends $\xi \to \pm\infty$, and spatial infinity goes to the single interior point $w = 0$. In these coordinates the coupled system becomes a two-component Schrödinger problem `[ESTABLISHED – assembler-verified]`:

$$(-\Delta + Q)\begin{pmatrix} U \\ V \end{pmatrix} = 0, \qquad Q(\xi, \eta) = \frac{m^2R^2}{(\cosh\xi - \cos\eta)^2},$$

with translation-invariant coupled edge conditions: on $\eta = 0$, $\partial V = 0$ and $U = V$; on $\eta = \pi$, $\partial U = 0$ and $V = cU$. The equation says: after the conformal change of variables, all the difficulty of the fractional operator ω has been traded for an ordinary Laplacian plus an explicit closed-form potential Q — the mass term dressed by the geometry — acting on a pair of functions coupled only through the strip's two edges, where the coupling constant c sits. The potential decays like $4m^2R^2e^{-2|\xi|}$ at *both* ends of the strip (a correction to the iteration-20 architecture, which had wrongly expected a confining end), and its only singularity is a $|w|^{-4}$ barrier at $w = 0$, the image of spatial infinity. The transverse eigenvalue problem across the strip's width reproduces the indicial equation exactly: one oscillatory channel $\pm i\tau_0$ per parity, all other channels evanescent. The problem is now a waveguide with one open channel.



The strip geometrization: the logarithmic map sends the interval's two corners to the ends of a strip, turns the coupled boundary data into translation-invariant edge conditions, and isolates the mass term as the only symmetry-breaking piece.

Two structural discoveries came out of the strip immediately, one a tool and one a wall. The tool: complex *translation* fails structurally (the free part is translation-invariant, so translating in the complex direction deforms nothing that matters), but exterior *dilation* in ξ — the Simon–Hunziker–Gérard–Sigal technique — works, and Q is exterior-dilation-analytic in the full sector. The wall: the unique pointwise symmetrizer of the coupled strip system is the indefinite Krein metric

$$J_c = \begin{pmatrix} c & -c \\ -c & 1 \end{pmatrix},$$

which has signature $(1, 1)$ — one positive and one negative direction — so *no positive local inner product exists* in which the coupled pair is self-adjoint. This is the Krein-symmetrizer no-go `[ESTABLISHED]`. It explains structurally why naive complex scaling had resisted every attempt: the c-coupled pair is the characteristic-value form of a self-adjoint operator, not itself a self-adjoint operator, and the classical machinery of self-adjoint spectral theory — the virial theorem, the Froese–Herbst exclusion — is simply not licensed on it.

The c-collapse

Which raises the obvious question: where is the self-adjoint operator whose characteristic-value form this is? Iteration 21's answer — established through the second proof lens, independently re-derived by the assembling referee, and verified numerically to 10^{-8} — is the single most consequential compression in the program's history.

Define $S_I = \chi_I\,\omega\,\chi_I$ and $R_I = \chi_I\,\omega^{-1}\chi_I$: the energy operator and its inverse, each cut down to the interval. Both are positive operators on $L^2(I)$. Setting $f = \chi_I F$, the LEM-A1' data are equivalent — in both directions, in about six lines, for $m > 0$ — to the eigenvalue equation

$$R_I S_I f = \kappa f, \qquad f \in H^1_0(I), \qquad \kappa = \frac{c}{c-1} \in (1, \infty).$$

This says the whole coupled two-region system collapses to a single equation on the interval alone: apply the localized energy, then the localized inverse energy, and ask whether any function comes back as a multiple $\kappa > 1$ of itself. And now observe what happened to the quantifier. LEM-A1' demanded a conclusion *for every* $c > 1$; but c enters the collapsed equation only through the eigenvalue $\kappa = c/(c-1)$, which sweeps $(1, \infty)$ exactly as c does. The "for all c" is not a family of problems — it is one problem. Symmetrizing via the similarity transform that Segal–Goodman injectivity of S_I licenses:

The entire free-field gate is equivalent to the statement that the single fixed self-adjoint operator

$$A = S_I^{1/2}\,R_I\,S_I^{1/2} \quad \text{on } L^2(I)$$

has no point spectrum in $(1, \infty)$**.** This is the statement the ledger registered as LEM-A1'' `[OPEN]`, and everything since iteration 21 has been about it. The operator A is bounded, positive, and self-adjoint — no Krein indefiniteness, no coupled system, no exterior. One operator, one question.

Two corollaries landed with the collapse, one embarrassing and one clarifying. The embarrassing one: iteration 20 had proposed a "phase monotonicity in c" mechanism as a route to the exclusion. That mechanism was ill-posed from the start — c is not a coupling constant that deforms the operator, it *is* the eigenvalue parameter in disguise. The ledger says so plainly. The clarifying one: no sign-definite eigenfunction of A can exist — every H_0^1 candidate oscillates infinitely often near $\pm R$, by the corner theorem — so Perron–Frobenius-type arguments, which live on positivity, fail structurally too.

And one more thing, which gives the collapse its peculiar flavor of irony. Under the modular dictionary, $\kappa = \coth^2(\varepsilon/2)$: the operator A is exactly the symmetrized Peschel object that the repository's own numerics — `scripts/eo-modular-numerics.py`, running since iteration 18 — had been diagonalizing all along. The lemma target and the lattice computation are formally the same object. Three independent numerical signatures (the density-of-states ladder, the pinning scan, and the iteration-20 edge-divergence probe, which showed every tracked mode carrying 96–99.7% of its mass in a shrinking edge window, growing under refinement — the unmistakable signature of continuous spectrum) all say: no point spectrum. The program has been *measuring* the answer for six iterations. It cannot yet *prove* it.

Five lines that closed a hundred doors

Iteration 22 executed the two named routes toward LEM-A1" — a direct Mourre positive-commutator argument, and a Feshbach reduction onto the strip's open channel — and in the course of doing so found the program's most elegant negative result. It deserves to be printed in full, because it is five lines long and it terminates an entire genre of proof strategy.

The hope had been: the effective potential produced by the Feshbach reduction is a symmetric nonlocal kernel with exponential decay in both variables and full dilation analyticity; surely *some* theorem says such operators have no embedded eigenvalues, the way the Kato–Agmon–Simon theorem forbids them for decaying local potentials. The counterexample `[ESTABLISHED — re-derived line-by-line in assembly]`:

> Let g be a Gaussian. Set $\varphi := g'' + \tau_0^2\, g$, so that $\hat{\varphi}(\pm\tau_0) = 0$. Set $\lambda := 1/\langle\varphi, g\rangle$, which is nonzero at generic Gaussian width. Then the operator
>
> $$H = -\frac{d^2}{d\xi^2} + \lambda\,|\varphi\rangle\langle\varphi|$$
>
> is self-adjoint; its perturbation kernel $\lambda\,\varphi(\xi)\varphi(\xi')$ is Gaussian in both variables — inside *every* exponential decay class — and dilation-*entire*; and $Hg = -g'' + \lambda\langle\varphi, g\rangle\,\varphi = -g'' + g'' + \tau_0^2 g = \tau_0^2\, g$: the Gaussian g is a genuine L^2 eigenfunction embedded at energy τ_0^2, inside the continuous spectrum $[0, \infty)$.

That is the whole proof. A rank-one perturbation, smoother and faster-decaying than anything the Feshbach reduction could ever produce, with a bound state embedded in the continuum — a nonlocal BIC, in the physicists' phrase. The mechanism is visible in the construction: the embedded eigenvalue occurs exactly because the perturbation's on-shell amplitude vanishes, $\hat{\varphi}(\pm\tau_0) = 0$; the kernel is invisible to the waves it lives among.

The consequences are absolute. "Exponentially decaying symmetric nonlocal kernel implies no embedded eigenvalues" is false as a class statement. Kato–Agmon–Simon is irreducibly a local-potential theorem. Aguilar–Combes dilation analyticity does not kill embedded eigenvalues of nonlocal perturbations — it merely makes them persistent under the deformation. Any proof of LEM-A1'' must use the *structural origin* of the effective potential — the Schur-complement energy structure of the actual Feshbach reduction, the fact that this kernel is not arbitrary but is manufactured by this geometry — and never merely its decay or analyticity class. The counterexample bounds methods, not the answer: it exhibits no embedded-eigenvalue mechanism for A itself, and the numerics still say the point spectrum is empty. But it joined the BKT obstruction as the second proved boundary on proof strategies, and together they force every surviving route through the fine structure of the specific operator.

The conditional theorem

What iteration 22 built on the positive side is the closest thing the program has to a proof: a Mourre theorem for A, complete in its machinery, conditional in exactly two named places.

The massless problem is the integrable landmark. Hislop and Longo showed the massless interval modular flow is generated by a Möbius vector field, and the corresponding self-adjoint generator D — strip translation ∂_ξ after the log map — is the exact symmetry of the massless part of our problem. Define the reference operator

$A_0 := \coth^2(\pi D).$

In 1+1 dimensions this is *definitional* rather than derived — the referee's correction, accepted into the record: the massless comparison operator does not exist independently (the massless R_I is infrared-log-divergent), so A_0 is the definition and all physical content migrates into the compactness hypothesis below. The consistency checks — indicial exponents $\pm i\tau_0$, the ladder density of states, essential spectrum $[1, \infty)$ — all pass. The spectral identity $\mathrm{spec}(A) = \coth^2(\pi \cdot \mathrm{spec}(D))$, floated in a prep note, holds at its refereed grade: correct as a consistency statement, with its operator-level content being exactly the hypothesis (H1).

Now the conjugate operator. Mourre theory needs a self-adjoint D_g whose commutator with A_0 is strictly positive. With $g(s) = \coth^2(\pi s)$, define the renormalized generator

$$D_g = -\frac{i}{2}\left(\frac{1}{g'}\frac{d}{ds} + \text{h.c.}\right), \qquad \text{then} \qquad i\,[A_0, D_g] = 1 \text{ exactly},$$

verified in the spectral representation. This is the ideal Mourre estimate — the commutator is not merely bounded below on spectral windows, it is the identity. The equation says: D_g generates a flow along which the spectrum of A_0 slides at unit speed, everywhere, with no dead zones.

The conditional theorem `[ESTABLISHED as conditional]`: write $K := A - A_0$ for the mass term. Under **(H1)** — K is compact (`[OPEN; INFERENCE, medium]` — supported by resolvent probes, with the non-chiral $\ln(m)$ bookkeeping unproven) — and **(H2)** — K has $C^{1,1}$ regularity with respect to D_g on compact spectral windows (`[OPEN]`; the referee caught that the abstract Mourre theorem of Amrein–Boutet de Monvel–Georgescu genuinely requires $C^{1,1}$, not the C^1 the attempt had claimed, and the estimate is hardest near $\kappa \to 1^+$, where $1/g' \sim e^{2\pi s}/8\pi$ blows up) — the conclusions are: the point spectrum of A in any compact $\Delta \subset (1, \infty)$ is *finite*; the singular continuous spectrum is empty; and there is a quantitative virial bound — if $\|E_\Delta\, i[K, D_g]\, E_\Delta\| < 1$ on a window Δ, there are no eigenvalues in that window at all, which delivers absence at small mR, window by window. Intrinsically perturbative: the large-mR regime escapes, and the closed-channel sub-lemma resurfaces there.

The Feshbach flank produced its own survivors: uniform Keldysh–Riesz projection bounds for the transverse pencil `[ESTABLISHED]`, and the exponential decay of the effective potential, $|V_{\text{eff}}(\xi, \xi')| \le C_\delta\,(1 + |\ln|\xi - \xi'||)\, e^{-(2-\delta)(|\xi|+|\xi'|)}$, established modulo one finite computation labeled (a') — a Birkhoff strong-regularity determinant, at that point the cheapest open item in the entire program.

Iteration 23: a theorem, a no-go, and a retraction

Iteration 23 did three things, and the third is the one this book exists to narrate honestly.

First, (a') was computed and refereed to `[ESTABLISHED]`. In the variable $z = e^{2\pi i s}$ the indicial function satisfies $c - \cos^2(\pi s) = -(z^2 - 2(2c-1)z + 1)/4z$, with discriminant $16c(c-1) > 0$ for all $c > 1$; the roots are $z_\pm = (\sqrt{c} \pm \sqrt{c-1})^2 = e^{\pm 2\pi\tau_0}$, and the transverse spectrum is two shifted, simple, uniformly separated lattices $(\mathbb{Z} + i\tau_0) \cup (\mathbb{Z} - i\tau_0)$ — machine-checked to 10^{-15} over two thousand random parameter pairs. But the referee's corrections re-opened the framework the computation was meant to close: regularity degenerates as $c \to 1^+$ (the FG-1989 boundary again — every bound is non-uniform there), and, decisively, *no off-the-shelf theorem converts "distinct separated roots" into a Riesz basis for this pencil.* The 4×4 first-order reduction has repeated characteristic roots — both components carry the same symbol, coupled only through the boundary conditions — which places it outside the entire classical chain (Mikhailov, Kesel'man, Dunford–Schwartz, Shkalikov) and the modern systems chain (Lunyov–Malamud), all live-verified. What does apply is a chain through entire-function theory: the determinant $\Delta(s) \sim c - \cos^2(\pi s)$ is a sine-type entire function of exponential type 2π with simple separated zeros `[ESTABLISHED]`, and by Levin and Golovin such zeros generate an exponential Riesz basis $\{e^{is_n \cdot}\}$ of $L^2(0, 2\pi)$. One link remains open: the transfer lemma.

Second, a fourth methods-no-go `[INFERENCE, high]`. The transverse eigenvalues of the strip pair are $\nu_n = n^2 - \tau_0^2 \pm 2in\tau_0$: exactly one real branch ($n = 0$) and infinitely many non-real conjugate pairs. Langer's theorem bounds the non-real spectrum of a definitizable operator in a Krein space to finitely many points; infinitely many non-real rays exceed it. *The strip pair is not globally definitizable in the Krein metric for any $c > 1$* — the same corner exponents $n \pm i\tau_0$ that define the gate destroy the global Krein-space structure that would have tamed it. What survives is local: on the spectral strip $\{|\mathrm{Im}\, z| < 2\tau_0\}$, conditional on the transfer lemma plus relative compactness of Q, local definitizability holds and the origin is a spectral point of type π_- — a Pontryagin-type structure with a finite-index negative spectral subspace (Behrndt–Jonas; Azizov–Jonas–Trunk), a genuinely *nonlocal* almost-definite structure that sidesteps the iteration-21 pointwise no-go. And one clean machine-verified fact anchors it `[ESTABLISHED]`: the open channel has constant negative Krein density, $\langle J_c w, w\rangle = -(c-1) = -1/(\kappa - 1)$. The one channel that carries the continuum is precisely the negative direction of the metric.

Third, the retraction. The energy-monotonicity lens had submitted, graded `[ESTABLISHED]`, a Herglotz–Loewner monotonicity property of the Schur complement in its spectral slot — a step that would have carried real weight toward the crux. In assembly it was refuted by internal contradiction: the Loewner order — the partial order on self-adjoint operators that monotonicity arguments live on — has no meaning in the everywhere-indefinite strip metric, and the other lens's machine-verified non-real closed channels made the contradiction explicit. The claim was struck to `[OPEN]` by the assembling referee. Per house rule, the record keeps the corpse visible: submitted grades are provisional until assembly, and a struck claim is documented, not erased. What survives of that lens is the exact Feshbach–Schur reduction as a scheme, and a boundary Hellmann–Feynman identity with strict sign `[INFERENCE, medium]`. This is the discipline working exactly as designed — an `[ESTABLISHED]` tag lasted less than one iteration because two independent lenses were forced through the same referee — and it is why the twenty-two consecutive unchanged verdicts mean something. A process that cannot retract cannot confirm.

The residual, stated for the specialist

After twenty-three iterations, the free-field gate rests on two named lemmas, plus two standing hypotheses. Here they are, precisely enough to pick up.

The transfer lemma (LEM-A1'''-T) `[OPEN]` — the upstream bottleneck. Let $\{s_n\} = (\mathbb{Z} + i\tau_0) \cup (\mathbb{Z} - i\tau_0)$ be the zeros of the sine-type function $\Delta(s) \sim c - \cos^2(\pi s)$, so that $\{e^{is_n \cdot}\}$ is an exponential Riesz basis of $L^2(0, 2\pi)$ by Levin–Golovin (canonically, Avdonin–Ivanov). Prove that the explicit map carrying $e^{is_n \cdot}$ to the normalized transverse pencil eigenvectors (U_n, V_n) — explicitly known trigonometric combinations — is bounded with bounded inverse. This is a concrete Gram-matrix bound on explicitly written functions: cold-checkable, and numerically probeable first (condition numbers of the truncated Gram matrices versus truncation size and c — the designated next move of the program). The bounds are necessarily non-uniform as $c \to 1^+$. Everything downstream — the Feshbach machinery, the local definitizability, the mode decomposition behind (H1) — waits on this.

The transversality lemma (LEM-A1'''') `[OPEN]` **— the crux.** For κ in any compact $\Delta \subset (1, \infty)$: an L^2 eigenfunction candidate for A whose on-shell amplitude vanishes — $\hat{w}(\pm\tau_0) = 0$, where w is the open-channel component, the exact vanishing that the Gaussian counterexample proves *cannot* be ruled out by decay or analyticity — must itself vanish. The argument must run through structure: either the Schur-complement energy monotonicity of the actual Feshbach reduction (whose naive Loewner form was the retracted claim; its compression through the Krein metric is the open question), or a Froese–Herbst exponential bootstrap for the genuinely self-adjoint scalar A with the explicit conjugate D_g, pushed past the nonlocal rate ceiling. After iteration 23, its natural home is the Pontryagin channel: execute the on-shell nondegeneracy argument inside the type-π_- local spectral structure, where the open channel's constant negative Krein density is the lever.

Plus the standing pair from the conditional Mourre theorem: **(H1)**, compactness of $K = A - A_0$ [INFERENCE, medium — resolvent-probe supported, three lattice falsifiers named and ready]; and **(H2)**, the $C^{1,1}(D_g)$ regularity of K on compact windows `[OPEN]`.

The chain is: transfer lemma and transversality lemma, plus (H1) and (H2) where the Mourre route needs them ⇒ LEM-A1'' ⇒ no modular eigenvalue $\lambda \neq 1$ ⇒ the free-field gate (E_O) ⇒ hedge rung R7 advances to R8. Four methods-no-gos stand guard over the remaining routes: the BKT family-level obstruction, the Gaussian-BIC decay-class obstruction, the Krein-symmetrizer pointwise no-go, and the global non-definitizability. Every soft path is proved closed. What remains is one operator — bounded, positive, self-adjoint, explicitly constructed, numerically diagonalized to three independent signatures of purely continuous spectrum — and one question the program has so far been unable to answer in either direction. The verdict after twenty-two consecutive confirmations is what it has been since the beginning: PARTIAL; the structure encodes, and has not been shown to generate; not yet physics. The honest state of the mathematics is that the answer is almost certainly "no eigenvalues" — and *almost certainly* is exactly the phrase this book was written to refuse.

CHAPTER 12

The Verdict

What twenty-three adversarial iterations refused to say turns out to be the most valuable thing they said.

Every research program dreams of ending with a discovery. This one ends with a refusal — repeated, sharpened, and stress-tested until the refusal itself became the result. The standing verdict, unchanged through twenty-two consecutive refereed confirmations, reads in full: coherence of the unification programs is **partial**; the strongest program **encodes geometry but does not generate it**; the central wager is **not yet physics**; and the object-level question — is there a final theory, and is it this one? — is **not forecastable**. This chapter states each clause honestly, explains why the verdict is nevertheless not a theorem, lays out exactly what would reopen it, and then walks the full distance from where the program stands to what a universal theory would actually require. The distance is the point. Most accounts of quantum gravity either compress it to nothing ("we're close") or inflate it to despair ("never"). The refereed record supports neither.

The four clauses

Partial coherence. The survey that opened this book found not one unified program but two internally coherent ones joined by thin bridges: Program A, geometry-from-algebra — the modular, holographic, and entropic-gravity constructions in which spacetime is supposed to condense out of entanglement structure — and Program B, the reconstruction-and-tabletop cluster asking whether the quantum framework itself is final. They share exactly two things: no-signaling with a single causal order, and the logical prerequisite that gravity be non-classical. The tempting fusion — "it's all information" — was examined and refused, because "information" names non-interchangeable objects across the two clusters: entanglement entropy of a type III algebra in one, operational distinguishability of preparations in the other. A slogan that equivocates between them is not a synthesis; it is a pun. This clause is graded `[INFERENCE, high]` in the wiki's tagging convention — the standard conclusion of careful reasoning, logically defeasible, not a proven fact.

Encodes, does not generate. Chapter by chapter, the load-bearing results of Program A — Ryu–Takayanagi, the entanglement-first-law-to-linearized-Einstein chain, the crossed-product entropy constructions, Jacobson's thermodynamic derivation — were found to presuppose, by their own architects' concessions, a fixed background with a Killing or conformal symmetry. The recurring quantitative symptom is that the area-entropy coefficient in

$$S = \frac{A}{4G}$$

— the statement that a horizon's entropy S is its area A in units of four times Newton's constant G — enters every known derivation as an *input*, never an *output*. A program that must be told the value of $1/4G$ before it can derive Einstein's equations has encoded gravity, the way a JPEG encodes a photograph: faithfully, compactly, and without any capacity to tell you what the camera saw that the file format didn't. Across the iterations this diagnosis hardened from an observation into a readout-independent structural obstruction, closing at the same signature-installation step in every one of the five known readout families. The one apparent counterexample was manufactured by the program itself: a canonical indefinite form

$$\eta_0 = \operatorname{sgn}(\ln \Delta),$$

the sign of the logarithm of the modular operator Δ, *does* exist, derived smuggle-free from pure algebra-and-state data — and is provably geometry-void: identical across every algebra in its class, carrying no localization, no light cones, nothing. The program can generate an indefinite signature; it cannot pin the signature to places. That gap has a name — the carrier problem — and it is where this entire book's mathematics now lives.

Not yet physics. By the project's own anti-crank standard, a proposal is physics only if some observation could in principle distinguish it from its rivals. The central wager — *causal, algebraic, entanglement structure precedes metric geometry, and this is true of our* $\Lambda > 0$ *universe* — has no such observation, and iteration 6 upgraded this from a survey of the experimental landscape to a *corollary* of the encoding obstruction: an encode-only program's observational content equals that of the effective field theory it encodes, so every candidate signal factors through physics equally available to geometry-first descriptions. A class-separating test would require what the wiki calls a reverse Weinberg–Witten observable — a measurable quantity realizable *only* in a theory with no fundamental metric — and none exists, none has even been attempted, and iteration 9 proved the theorem-form of that escape is equivalent to the carrier problem itself. The wager is a sharp, coherent research *strategy*. It is not yet a physical *claim*.

Not forecastable. The popular framing blames the energy desert — the fifteen orders of magnitude between colliders and the Planck scale. The refereed record locates the obstruction elsewhere: no experiment at any energy, with any technology, currently bears on the central wager, so no amount of experimental progress along existing lines converges on an answer. A question with no observational channel has no forecast horizon. This is a stronger and stranger statement than "it's hard," and it is the clause most worth sitting with.

Two clarifications keep the verdict from being misread as pessimism. First, on whether a universal theory is *possible*: yes, very plausibly, in the framework sense — the seventeen catalogued clashes between quantum field theory and general relativity contain not one proven logical inconsistency, and AdS/CFT stands as a strongly-evidenced worked example (modulo the AdS/CFT conjecture itself) that *some* consistent quantum gravity exists. Whether a final *substrate* exists, and whether any theory could fix its own parameters, are separately graded and progressively less likely — but the door is open. Second, the project explicitly declines the fashionable overreach that Gödel's theorems forbid a final theory: undecidability of a theory's consequences is not incompleteness of its laws.

Why this is not a theorem

The verdict rests on explicit hedges, and the book's method requires displaying them rather than footnoting them. The principal hedge is the hypothesis the wiki calls HYP-CKV-VACUITY, now in its seventh revision (R7): the claim that every currently known way of reading geometry out of modular data is vacuous ranks as high-confidence *conditional* on two named residuals. The first is n_1, the localization template — the fact that "localized" only means anything relative to the net's index-set of regions and their causal order, and that order *is* the geometry; iteration 9 proved this input irreducible, and iteration 18's attempt to derive it from a smaller seed found that the candidate seed, half-sided modular inclusion, simply *is* the causal order under another name. The second is the ergodicity hypothesis

$M(O)_\omega = \mathbb{C}\,1,$

labeled (E_O): the statement that the vacuum state ω, viewed on the algebra $M(O)$ of a bounded spacetime diamond O in a massive theory, has trivial centralizer — no nontrivial operator in the local algebra is exactly fixed by the vacuum's modular flow. If (E_O) holds (plus weak additivity), the carrier no-go becomes a theorem on its stated hypotheses and the "encodes" clause stops being a bet. If (E_O) fails, a carrier *might* live on the failure — necessary, the referees insisted, but not sufficient. Everything now funnels through this single unproven statement and its irreducible companion. A verdict resting on a named, attackable open hypothesis is not a theorem. It is, however, the rarest kind of non-theorem: one that knows exactly what it is missing.

The reopening specification

Because the verdict is not a theorem, the record specifies — with contractual precision — what would overturn it. One analytical development flips the headline from "encodes" to "generates": **a construction that outputs a Lorentzian causal or metric structure from genuinely non-symmetric (algebra, state) data, installing by hand no fundamental symmetry, no η-form, no foliation, no twist, no indefinite-pairing or (n, n) operator-class axiom, and no signature-valued parameter template.** The banned list is not pedantry; it is the accumulated forensic record of five families of constructions, each of which was caught installing the Lorentzian signature through exactly one of those doors. In compressed form: derive an algebra-compatible, *localized* indefinite pairing from modular data. Solve the carrier problem with a counterexample, and geometry becomes an output of algebra plus state; the wager converts from strategy to candidate result; the reverse Weinberg–Witten witness becomes possible in principle; and this book's conclusion is void. The full specification, with every established reduction and every adjacent open problem, is posed as a self-contained dossier for external solvers — an operator algebraist can pick it up cold, with no physics context — in the iteration-15 carrier-problem dossier in the project repository.

The other direction reopens the analysis too. Prove the no-go — discharge (E_O) and the naturality theorem — and "encodes, does not generate" becomes a theorem, the no-test corollary becomes unconditional, and the program is *permanently* not-physics absent a genuinely different idea. The record is symmetric about this in a way partisan literature rarely is: either resolution of the open gate would be a major result, and the project's own expected outcome (the no-go) is flagged as exactly that — an expectation, not a finding.

The surviving lemmas

What makes the current state unusual is how far the gate has been compressed. In iteration 16 the residual was an abstract hypothesis about von Neumann algebras. Seven iterations of reduction later, the free-field half of it is a single concrete question about a single self-adjoint operator — and the chain deserves to be displayed, because each link is `[ESTABLISHED]` and each was adversarially refereed.

For the free massive scalar field, (E_O) reduces to a spectral question: does the one-particle modular Hamiltonian of a double cone have purely continuous spectrum? Half of that — the absence of a modular eigenvalue at zero energy — is now a theorem, proved two independent ways, one of which turned out to have been proved already by Figliolini and Guido (1989); the project's referee downloaded the 1989 paper and verified the claim on page 429 before the tag was upgraded. The other half, the so-called $\lambda \neq 1$ exclusion, was transformed in iteration 20 into a statement with *no operator-algebra content at all*: a putative eigenvalue $\mu^2 > 1$ exists if and only if the fractional operator $\omega = (m^2 - \Delta)^{1/2}$ — the relativistic energy operator for mass m, with Δ the Laplacian — admits a certain bilateral-antilocality configuration with coupling

$$c = \frac{\mu^2}{\mu^2 - 1} > 1,$$

the 1989 result being precisely the degenerate boundary case of this family. Iteration 21 collapsed the whole question again: it is equivalent to whether the single operator

$$A = S_I^{1/2} R_I S_I^{1/2}$$

— built from the restriction operators S_I and R_I that the project's own numerical scripts diagonalize — has any eigenvalue in the interval $(1, \infty)$. The lemma target and the numerics are formally the same object, and three independent numerical signatures (a density-of-states ladder, an embedded-eigenvalue pinning scan, and an edge-divergence probe) all say the answer is no. Along the way the reduction produced a small jewel: the corner indicial equation $\cos^2(\pi s) = c$, whose roots recover the Bisognano–Wichmann temperature relation $\tau_0 = \varepsilon/2\pi$ — the deep link between modular flow and thermal physics — from nothing but a boundary-value computation at the endpoints of an interval, lattice-verified twice.

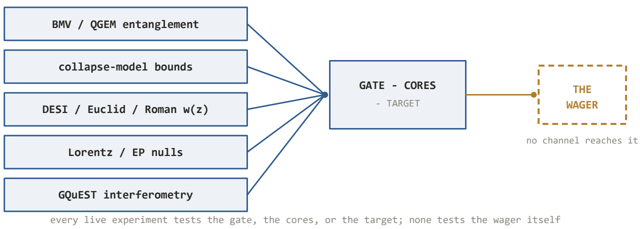
What survives, as of iteration 23, is a short named ladder: LEM-A1'''-T, a concrete Riesz-basis Gram bound that no off-the-shelf theorem covers (the pencil's repeated characteristic roots defeat every standard chain — a genuinely new small problem in non-self-adjoint spectral theory), and behind it LEM-A1'''', an on-shell transversality statement in a local Pontryagin-space channel. Flanking them stand four proved *methods no-gos* — theorems establishing that whole strategy classes can never decide the question: no soft embedded-eigenvalue theorem (such eigenvalues genuinely occur in the ambient operator family), no decay-or-analyticity-class theorem (a Gaussian rank-one counterexample lives inside every exponential class), no invariant-level argument, no global Krein definitizability. The gate is small, named, numerically probed from three directions, and provably approachable only through the structural specifics of this one operator. Proving the ladder closes free-field (E_O), fires the hedge from R7 to R8, and hardens the verdict on the free subclass. It would *not* flip anything — the flip requires a carrier, not a proof of its absence — and even full (E_O) closure would leave n_1 standing as the irreducible geometric input.

And beyond the free field lies the honest wall: for general interacting theories in the relevant class, all three known structure-theory routes to (E_O) — clustering-to-weak-mixing, wedge-to-double-cone reduction, and genericity arguments — were shown in iteration 17 to fail for identified, specific reasons. There is at present *no known mathematical technology* for the interacting case. That sentence is not rhetoric; it is the referee-confirmed content of a dedicated assault.

The 2027 clock

One dated external event sits on the calendar: the DESI five-year dark-energy analysis, expected in 2027. DESI's second data release showed a 2.8–4.2σ preference for a dark-energy equation of state $w(z)$ that varies with redshift — parameterized by (w_0, w_a), the present-day value and its evolution — over the cosmological constant's fixed $w = -1$; under low-redshift supernova-calibration systematics the preference deflates below 2σ, which is why the row is tagged `[CONTESTED]` and why 2027, with the full survey and Euclid's cosmology release converging, is the decision point.

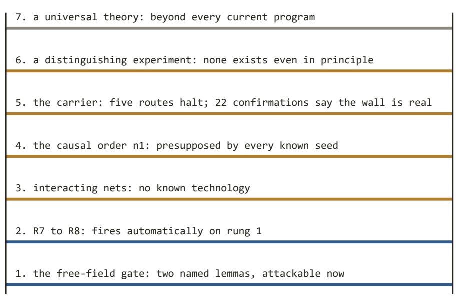
Be precise about what this clock does and does not test, because it is the single most tempting misreading of the program's empirical situation. If dynamical dark energy is confirmed, the universe is not settling into the eternal de Sitter state that the holographic and emergence constructions take as their target — the *motivation* for a strict $\Lambda > 0$ entanglement first law weakens, and the program's target moves. That is real and consequential. But it tests the wager's **target**, not the wager. A universe with rolling quintessence is exactly as compatible with "entanglement precedes geometry" — and exactly as compatible with its negation — as an eternal de Sitter one. The same discipline governs the entire experiment watchlist, whose standing caveat is worth quoting in spirit: no channel on the page decides the central wager. The gravitationally-induced-entanglement proposals (BMV/QGEM) test the program's *gate* — is gravity non-classical at all? — and are decades from decisive masses. Objective-collapse bounds test a *core* — is quantum linearity exact? — and have now killed the parameter-free models while leaving engineered corners alive. Lorentz-invariance and equivalence-principle nulls anchor other cores. GQuEST, the one funded experiment wearing the program's vocabulary, probes a conjecture *inspired by* modular fluctuations whose inference chain re-imports $\eta = 1/4G$ at the crucial step and whose predicted signal is reproduced identically by conformal-field-theory, dilaton, and hydrodynamic models — the encoding screen, published as an experimental design. There is even an asymmetric route by which the gate could close from the classical side: complete falsification of the explicit classical-quantum diffusion kernels, already under data-driven pressure, would eliminate the principal classical alternative years before any entanglement witness runs. A positive anywhere on the watchlist would be enormous physics. None of it would confirm the wager.



The experimental funnel: every live channel tests the program's gate, cores, or target. None reaches the wager itself.

The distance ladder

How far, then, from here to a universal theory of physics — if the geometry-from-algebra road is the road? The record supports an exact answer, rung by rung, and the honest function of this section is to let the reader count.



The distance ladder from here to a universal theory. The bottom two rungs are attackable now; the upper rungs have no known technology.

Rung one: the free-field gate. Close LEM-A1'''-T and LEM-A1'''' — or find the eigenvalue. This is the current frontier: a bounded, well-posed problem in spectral theory, checkable by a specialist with no physics background, with three numerical signatures pointing one way. It is the *easiest* rung on the ladder, and it has now absorbed eight consecutive iterations of professional-grade assault without closing.

Rung two: interacting nets. Extend (E_O) from the free field to the interacting class. Every known route provably fails; no technology exists. For calibration: rigorous control of interacting quantum field theory in four dimensions is itself a Millennium-Prize-adjacent open frontier. This rung is not a harder version of rung one; it is a different mountain range.

Rung three: the causal order n_1. Even with (E_O) closed everywhere, the no-go theorem would establish that geometry cannot be *generated* while the index-set order — which regions are causally disjoint from which — is presupposed. Deriving n_1 from a smaller seed was attempted directly in iteration 18 and failed structurally: every candidate seed examined *is* the causal order in disguise. No smaller seed is known, and iteration 9 proved the localization half of the naturality question irreducible. To flip the verdict rather than confirm it, someone must produce a structure in which the causal order and the signature are outputs. Nothing in the current mathematical literature — modular theory, Krein spaces, factorization algebras, index theory, higher categories, all tried — offers a starting point. Each reproduced the screen.

Rung four: the carrier. The algebra-compatible, localized, indefinite pairing itself — the full reopening specification. Rungs three and four are two faces of the same open problem, and they are the program's Everest: five independent routes converge on the obstruction, and the dossier poses it because reasoning inside the program is exhausted.

Rung five: the missing experiment. A generative construction is a *prerequisite* for a distinguishing test, not a guarantee of one. The reverse Weinberg–Witten witness — an observable realizable only without fundamental metric degrees of freedom — is provably at least as hard as rung four, and no one has ever attempted to build one. Until it exists, no measurement made by any instrument bears on the wager, and physics cannot render a verdict.

Rung six: a theory. Only past all five rungs does "a universal theory of physics" even become a *candidate* description of some construction — which would then face the ordinary gauntlet: reproduce the Standard Model's chiral anomaly-free structure, general relativity to gravitational-wave precision, ΛCDM, and the thermodynamic spine, and survive whatever the experiment on rung five measures.

The program stands below rung one, with a hand on it. That is what twenty-three iterations of maximal effort by a process optimized for progress actually bought — and stating it plainly is worth more than any of the alternatives on offer, because each rung is now *named*, which is more than could be said when the survey began.

The strongest kind of knowledge this question admits

It is fair to ask what twenty-two consecutive confirmations are worth, given that they confirm an absence. The answer depends entirely on how they were produced, which is why this book has dwelt on procedure. These were not twenty-two polite reviews. The referees' standing instruction was to refute. Iteration 14 inverted the burden entirely — three tracks ordered to *assume the verdict false* and build the strongest 2024–2026 case against each pillar; all three failed, and the strongest published counterclaim of the year collapsed because the Lorentzian signature it "derived" is an unconditional algebraic fact of the qubit carrier it chose, installed at the choice of Hilbert space. Iteration 15 imported three branches of mathematics never before aimed at the problem; each independently reproduced the obstruction. Iteration 18, at the operator's explicit instruction, stopped attacking and tried to *build* the generative step four different ways; both referees returned the same verdict on all four, and the project's first numerical computation, commissioned as a genuine test with a live chance of finding the carrier's entry ticket, came back against it. The verdict survived being defended, being assaulted, being flanked by new mathematics, and being abandoned in favor of construction. Reasoning, construction, and computation converged.

A verdict of this shape — negative, hedged, precise about its own conditions, and repeatedly survivable under adversarial pressure designed to kill it — is not a consolation prize. For a question that currently admits no experiment, it is the strongest form of knowledge available: not *the wager is false*, which no one has proved, and not *the wager is true*, which nothing supports, but *here is exactly where every road ends, here is the named lemma the next road must pass through, and here is the specification any claimed breakthrough must meet*. The claim "encodes, not generates" is `[INFERENCE, high]`, one open hypothesis from theoremhood. The claim that no experiment yet conceived can decide the wager inherits the same grade. The claim that this was worth knowing carries no hedge at all.

CHAPTER 13

How This Book Was Made

A research program run by machines, kept honest by machinery — and what that does and does not prove.

This book has, until now, kept its narrator offstage. The convention was deliberate: the physics had to stand or fall on the refereed record, not on the novelty of who produced it. But a book whose central claim is that *method is the result* owes the reader a final chapter on the method itself — including the parts that failed, because the failures, and the fact that they were caught, are the strongest evidence the rest can be trusted at all.

Here is the plain statement. The investigation chronicled in this book was conducted by an AI system — large language models, orchestrated in structured workflows, directed by a single human operator who set objectives, sanctioned iterations, and made the final integration decisions, but who did not supply the mathematics. The wiki that is this book's source, the twenty-three iterations, the lemma ladder, the retractions, the numerics: machine-generated, machine-refereed, human-governed. Whether that provenance makes the results more interesting or less is a question the reader may weigh; this chapter's job is to describe the machinery precisely enough that the weighing is informed.

The shape of the machine

Every analytical iteration had the same anatomy. A small number of *finder* tracks — typically three to five — were launched in parallel, each an independent model instance given one attack angle, the relevant wiki state, and no sight of the other finders. Their outputs went to *adversarial referees*: separate model instances whose standing instruction was not "review this" but "refute this," whose default verdict was rejection, and whose corrections were **binding** — by protocol, a referee's downgrade governed over a finder's claim, always, with no appeal to the finder's confidence. An *integrator* then assembled what survived into the permanent record, logging every promotion and demotion of every claim in an append-only changelog, under an epistemic regime — described in Chapter 2 and enforced on every page — in which each nontrivial claim carries exactly one of five tags, from `[ESTABLISHED]` down through `[INFERENCE]` and `[SPECULATIVE]` to `[OPEN]` and `[CONTESTED]`, and in which the cardinal rule was written before any physics was: *never let a speculative idea wear the clothes of an established fact.*

The human role deserves its own precise sentence, because both exaggerations — "a person really did this" and "no person touched this" — are false. The operator chose what to attack and when: sanctioning iterations against the roadmap's clock rule, which after saturation forbade launching new analytical work without a genuinely new external input; deciding, against the grain of the record, that iteration 18 would abandon critique and attempt outright construction of the generative step (the instruction is logged verbatim, and the attempt's four-fold failure became one of the verdict's strongest confirmations); and serving as the integrator of last resort — it was the human-directed assembly pass, not any model, that caught iteration 22's miscounted confirmation tally. What the operator did not do is supply arguments, proofs, or citations. Direction was human; mathematics was machine; authority over truth-claims belonged to neither, but to the protocol.

Three further disciplines ran through everything. Every load-bearing citation was verified live against the actual source — arXiv, journal pages, in one memorable case a scanned 1989 French annals paper — before it could bear load; the sourcing rules state flatly that a missing reference is acceptable and a fake one is a disqualifying error. Wherever a claim could be computed, it was: lattice numerics, machine-checked determinants, identities verified to a part in 10^{15}. And the whole record was cumulative and public — pages revised in place, with the *why* of every revision preserved in the changelog, so that the project's mistakes are as inspectable as its results.

What worked

Parallel lenses. Independent finders attacking one target from different angles turned out to be worth far more than one long serial effort, for a reason specific to the failure modes of language models: a single line of reasoning can drift into a self-consistent error and stay there, but five lenses drift in five different directions, and the referee sees the disagreement. The decisive iterations were all fan-outs — five lenses on the ergodicity hypothesis in iteration 17, four construction angles in iteration 18, three falsification tracks in iteration 14 — and in each case the *pattern* across tracks (every route failing at the same step, every framework smuggling the same input) carried more evidential weight than any single track's conclusion.

Binding referees. The single most consequential design choice. A referee that advises is a rubber stamp with extra steps; a referee whose corrections override the finder changes what the finders can get away with, and — more subtly — what the integrator can. The record's most important negative results exist because a referee said no: iteration 10's attempted closure of the commutant gap, submitted as a major advance, was struck when the referee found the argument had conflated the modular centralizer with the representation-space multiplicity gap — the two prematurely minted hedge advances were revoked, and the honest R7 had to be re-earned six iterations later on a correct argument. Iteration 11's submitted "logical equivalence" to Connes' bicentralizer problem was major-corrected down to "structural connection," and in the process the referee discovered that the iteration's own framing premise — that the bicentralizer problem had been resolved in the literature — was simply false. A weaker protocol would have written both errors into the permanent record, and every later chapter of this book would have been quietly wrong.

Machine verification. Wherever an argument touched something computable, computing it was the cheapest insurance available. The corner indicial equation $\cos^2(\pi s) = c$ of iteration 21 — the little boundary computation that unexpectedly recovered the Bisognano–Wichmann relation $\tau_0 = \varepsilon/2\pi$ — was lattice-verified at two precisions before it was believed; the iteration-23 characteristic quadratic was confirmed by a machine identity at 10^{-15} *and* an independent four-by-four determinant evaluated at the roots. The habit paid off most sharply in reverse: in iteration 23, a submitted monotonicity claim, graded ESTABLISHED by its finder, was struck in final assembly because it contradicted a machine-verified computation from a parallel track. Two model instances disagreed; the arithmetic voted; the claim died. That is the correct order of authority, and maintaining it required no judgment at all — only the prior decision that computations outrank prose.

Live citation checks. The project treated bibliography as a load-bearing wall, and the returns were concrete. The best moment in the record is the iteration-19 referee downloading Figliolini and Guido's 1989 paper from the numdam archive and finding, on page 429, that the lemma the project had just proved independently was already there — converting a two-year-old [unverified] tag into an `[ESTABLISHED]` attribution and an honest sentence: *we verified and re-proved a 1989 theorem* rather than *we discovered a new one*. The same discipline ran outward: a claimed experimental figure attributed to a 2026 paper was struck when verification showed the paper contains no such number; a 2019 journal reference was corrected to its actual 2014 volume; a claimed "equivalence" in a cited preprint was downgraded to the implication the preprint actually states.

Numerics as arbiter. For seventeen iterations the project was pure reasoning. Iteration 18 changed that, and the change was epistemically clarifying out of proportion to the code involved: a few hundred lines computing the modular spectrum of a lattice scalar field — a genuine test, with a live chance of exhibiting the zero-mode that would have been the carrier's entry ticket. It found none, and the subsequent iterations built a three-signature numerical case (a density-of-states ladder, an embedded-eigenvalue pinning scan, an edge-divergence probe) that now constrains the analytical work from outside. When iteration 21 proved the entire gate equivalent to the point spectrum of the operator $A = S_I^{1/2} R_I S_I^{1/2}$, it turned out this was *formally the same object the script had been diagonalizing* — the lemma and the numerics had converged on one matrix from opposite directions, which is the kind of consistency no amount of internal reasoning can manufacture.

What failed, and was caught

The catches matter more than the successes, so they get their own accounting, uncushioned.

A pipeline stage fabricated a quotation. During the first watch-mode sweep, an intermediate PDF-summarization step attributed to a 2026 Krein-space paper a verbatim sentence its authors never wrote. The fabrication was caught before entry — by checking the claimed quote against the paper's actual text — and the corrected entry carries a permanent integrity note. This is the failure mode that most disqualifies language models from unsupervised scholarship, it happened *inside this project*, and the only reason it is a footnote rather than a corruption is that the protocol assumed it would happen and checked.

A finder misattributed a paper — crediting a 2022 numerical study to Figliolini and Guido, authors of a different, 1989 work — and the referee caught it before it was written into any page. A synthesis stage miscounted the project's own headline statistic, reporting the twenty-first consecutive confirmation as the twentieth; the integrator's audit caught the arithmetic. A numerical probe was found to be *non-discriminating by design* — a bulk-fraction diagnostic that could not, even in principle, distinguish the hypothesis from its negation on a fixed interval — and rather than quietly replaced, it was documented in the script as a cautionary artifact alongside its corrected successor, because a probe that flatters the expected answer is exactly the contamination a record like this must display to be believed.

And the record contains genuine retractions, at every scale: iteration 10's struck closure (the largest — a claimed end to the whole program's central gap, revoked wholesale); iteration 11's demoted equivalence; iteration 23's assembly-struck monotonicity claim. The changelog format made each cheap to perform honestly — a dated entry, a reason, a re-tag — where in most research cultures a retraction is expensive enough that the temptation is to patch and move on. Even the infrastructure failed visibly and recoverably: one iteration's synthesis stage crashed against an output-format retry cap and was reconstructed from the surviving agent transcripts; another workflow was killed at a session boundary and resumed from cache. None of this is elegant. All of it is in the record, which is the point.

Discipline over intelligence

The models involved were, by any reasonable description, extremely capable: they executed reductions in Tomita–Takesaki modular theory, spectral analysis of nonlocal operators, and fractional unique-continuation arguments that would occupy a competent graduate student for weeks apiece. That capability was *not* the operative variable, and the record shows it directly: the very same class of model that proved the corner indicial theorem also fabricated a quotation, conflated a centralizer with a multiplicity gap while announcing a breakthrough, and graded its own erroneous monotonicity claim ESTABLISHED. Fluency and failure came from the same source, often in the same session.

What separated this project from an armchair theory-of-everything document was therefore never the intelligence of any single pass — it was the architecture that assumed every pass might be wrong: referees who could not be overridden, tags that could not be inflated without a logged justification, citations that could not bear load unverified, computations that outranked arguments, and a changelog in which errors were cheaper to admit than to hide. Twenty-two consecutive confirmations of an unwelcome verdict is the machinery's signature achievement, and it is worth stating why:

a system optimized to *please* would have confirmed the wager; a system optimized to *produce results* would have declared victory at iteration 10, and again at 21. A system optimized to be *correct* spent twenty-three iterations narrowing the precise sense in which nothing had been achieved. The discipline was not a constraint on the intelligence. The discipline was the product.

What this does and does not demonstrate

An honest paragraph, then, of the kind the project's own epistemics would demand. What this record demonstrates: that an AI system under adversarial protocol can hold a genuinely difficult research front coherently across months and dozens of sessions; that it can perform real mathematical reduction — the compression of an abstract operator-algebraic hypothesis to a single named spectral question is real work by any standard; that it can catch its own fabrications, retract its own claims, and maintain calibrated uncertainty under pressure to conclude; and that machine verification and live source-checking can be integrated tightly enough to make the output *auditable*, which is the property that matters. What it does not demonstrate: that AI has proved a hard open theorem (the gate is still open — the project's deepest results are reductions, rigidity theorems, and no-gos, refereed internally but not by the professional community); that the internal referee standard equals journal peer review (it does not, and no claim here should be read as though it does — much of the spine is `[INFERENCE]`-grade by the project's own tags); that the approach scales to fields without the dense, checkable literature that operator algebras and mathematical physics provide; or that any of this replaces the external mathematician the record itself says it now needs. The fairest one-line summary: this is an existence proof that AI-assisted mathematics can be *honest at scale*, not yet a proof that it can be *decisive*.

The problems, posed

A book that ends in an open problem should end by handing it over properly, and the project's final analytical act was exactly that: distilling the one open gate into a self-contained dossier — the carrier problem, posed cold for an operator algebraist, AQFT theorist, or noncommutative geometer, with every established reduction, every adjacent named problem, and the exact geometric input that every framework tried so far has been caught smuggling. It lives in the repository as the iteration-15 carrier-problem dossier, and its sharpest current forms need no physics at all: LEM-A1'''-T is a Riesz-basis Gram bound for an explicit exponential system that no off-the-shelf theorem covers, numerically probeable this afternoon by anyone with the taste for it; LEM-A1' is a coupled Zaremba corner problem for the operator $(m^2 - \Delta)^{1/2}$, checkable by a fractional-unique-continuation specialist with the standard toolset. Prove the ladder and the encodes-not-generates verdict becomes a theorem on the free field. Find the eigenvalue — the numerics say you won't, but the numerics are not a proof — and you hold the first carrier candidate, and Chapter 12's reopening clause fires.

As of the last recorded sweep, external engagement with the posed problem stood at zero. The repository is public: the wiki, the dossier, the changelog with every retraction intact, the numerical scripts with their non-discriminating probe documented beside the corrected one, and the full refereed record of twenty-three iterations that tried to break the verdict this book reports and could not. The wager — that the causal and algebraic skeleton of the world precedes its geometry — remains what it was on the first page: not forbidden, not delivered, and now, at least, exactly specified. Someone will close the gate or walk through it. The record is arranged so that either way, whoever does will know precisely what they did.

Glossary

This page collects precise, cross-linked definitions for the mathematical objects, physical concepts, and named results used throughout this wiki. Entries are alphabetical. Each definition is meant to be *correct as stated* within its domain of validity; where a term carries genuine subtlety or dispute, an epistemic marker (`[ESTABLISHED]`, `[INFERENCE]`, `[SPECULATIVE]`, `[OPEN]`, `[CONTESTED]`) flags it (see `EPISTEMICS.md` (in the repository) for the calibration scheme). For the map of how domains relate, see `THEORY_MAP.md` (in the repository); for the canonical-source pointers, see the Bibliography (p. 175).

A

Action — The functional $S[\gamma] = \int L\,dt$ (or $\int \mathcal{L}\,d^4x$ in field theory) assigning a number to each history γ. Physical trajectories make S *stationary*, not necessarily least `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

Action–angle variables — Canonical coordinates (I, ϕ) for an integrable system in which $\dot{I} = 0$ and $\dot{\phi} = \omega(I)$; motion is linear winding on invariant tori. The cleanest picture of integrable order, against which chaos is the failure `[ESTABLISHED]`.

Adiabatic (cosmological) perturbations — Primordial fluctuations in which all species share a common local time-shift, so number-density ratios are spatially constant. Observed to high accuracy and predicted by single-field inflation `[ESTABLISHED as description]`. See `domains/cosmology.md` (in the repository).

ADM formalism — Arnowitt–Deser–Misner $3+1$ decomposition of the metric into spatial metric h_{ij}, lapse N, and shift N^i, recasting general relativity as constrained Hamiltonian evolution of geometry `[ESTABLISHED]`. Foundation of numerical relativity and canonical quantization attempts. See `domains/general-relativity.md` (in the repository).

AdS/CFT correspondence — Conjectured exact duality between a quantum-gravity theory in asymptotically anti-de Sitter spacetime and a conformal field theory on its boundary `[INFERENCE: overwhelming evidence, no general nonperturbative proof]`. See `domains/information-theory.md` (in the repository), `UNIFICATION_LANDSCAPE.md` (in the repository).

Algebraic QFT (Haag–Kastler) — Formulation of QFT as a *net* of local operator algebras $\mathcal{O} \mapsto \mathfrak{A}(\mathcal{O})$ satisfying isotony, microcausality, and covariance, avoiding any privileged Hilbert-space representation `[ESTABLISHED structural framework]`. See `domains/quantum-field-theory.md` (in the repository).

Anomaly (quantum/chiral) — A classical symmetry broken by quantization; e.g. the Adler–Bell–Jackiw anomaly $\partial_\mu j^{5\mu} = \frac{e^2}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$. Physical for global currents ($\pi^0 \to 2\gamma$); must cancel for gauged currents `[ESTABLISHED]`. See `domains/particle-physics.md` (in the repository).

Asymptotic freedom — The decrease of a gauge coupling at high energy due to a negative beta function; in QCD $\beta_0 = 11 - \frac{2}{3}n_f > 0$ makes quarks weakly coupled at short distance `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Asymptotic safety — Conjecture that gravity (or another theory) is UV-completed by a nontrivial fixed point of the renormalization-group flow `[SPECULATIVE/OPEN]`. See `OPEN_PROBLEMS.md` (in the repository).

Atiyah–Singer index theorem — Equates the analytic index of an elliptic operator (e.g. net chiral zero modes of the Dirac operator) with a topological integral, ind(\slashed$D$) =

$\int_M \hat{A}(M)\,\mathrm{ch}(F)$ `[ESTABLISHED]`. Underlies anomalies and instanton fermion counting. See `domains/mathematics.md` (in the repository).

B

Baryon acoustic oscillations (BAO) — Frozen-in sound-horizon scale in the matter distribution, a standard ruler probing cosmic geometry and expansion history `[ESTABLISHED]`. See `domains/cosmology.md` (in the repository).

Bekenstein bound — Upper bound on the entropy in a weakly gravitating region of radius R and energy E: $S \leq 2\pi k_B RE/\hbar c$ `[INFERENCE; sharp only in weak gravity]`. See `domains/information-theory.md` (in the repository).

Bekenstein–Hawking entropy — Black-hole entropy proportional to horizon *area*: $S_{\mathrm{BH}} = k_B c^3 A/(4G\hbar) = k_B A/4\ell_P^2$ [ESTABLISHED theory; microstate counting `[OPEN]` for generic black holes]. See `domains/thermodynamics.md` (in the repository).

Bell's theorem — No local-hidden-variable theory satisfying measurement independence can reproduce all quantum correlations; the CHSH inequality $|S| \leq 2$ for LHV is violated up to the Tsirelson bound $2\sqrt{2}$ by quantum systems, confirmed loophole-free `[ESTABLISHED]`. The escape via denying measurement independence (superdeterminism) is `[OPEN/CONTESTED]`. See `domains/quantum-mechanics.md` (in the repository).

Beta function — $\beta(g) = \mu\, dg/d\mu$, governing the renormalization-group running of a coupling; its sign and zeros determine asymptotic freedom, Landau poles, and fixed points `[ESTABLISHED]`. See *renormalization group*.

Bianchi identity (contracted) — The geometric identity $\nabla^\mu G_{\mu\nu} \equiv 0$, which forces local energy–momentum conservation $\nabla^\mu T_{\mu\nu} = 0$ and reduces the Einstein equations to 6 independent dynamical equations `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

Black-hole information paradox — Apparent conflict between Hawking's thermal (pure→mixed) evaporation and quantum unitarity. Recent island/replica-wormhole computations reproduce a unitary Page curve `[INFERENCE within their assumptions]`; a complete mechanism for realistic black holes is `[OPEN/CONTESTED]`. See `GAPS_AND_CONTRADICTIONS.md` (in the repository).

Bohmian mechanics (de Broglie–Bohm) — Interpretation restoring definite particle positions guided by the wavefunction; empirically equivalent to standard QM, paying with explicit nonlocality and contextuality `[ESTABLISHED equivalence; CONTESTED as preferred]`. See `domains/quantum-mechanics.md` (in the repository).

Boltzmann entropy — $S_B = k_B \ln W$, the logarithm of the number of microstates compatible with a macrostate; the surface-entropy definition. Contrast the Gibbs (volume) entropy; their inequivalence for small/bounded systems is `[CONTESTED]`. See `domains/statistical-mechanics.md` (in the repository).

Boltzmann equation — Kinetic evolution equation for the one-particle distribution f with a collision integral; valid in the dilute-gas (Grad) limit. Its H-theorem yields irreversibility only after the molecular-chaos assumption `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository).

Born rule — Outcome probability $\Pr(A \in \Delta) = \mathrm{Tr}(\rho\, E_A(\Delta))$. Gleason's theorem makes it the unique noncontextual probability measure on projections for $\dim \geq 3$ `[ESTABLISHED as form]`; whether it is *derivable* from unitary dynamics is `[CONTESTED]`. See `domains/quantum-mechanics.md` (in the repository), `EPISTEMICS.md` (in the repository).

Bousso (covariant entropy) bound — Holographic bound $S_{\text{light-sheet}} \leq A/4G\hbar$ on the entropy crossing a light-sheet of a surface of area A; the covariant, generally applicable form of holography `[INFERENCE]`. See `domains/information-theory.md` (in the repository).

BRST symmetry — A global fermionic symmetry of the gauge-fixed action encoding gauge redundancy, used to prove unitarity and renormalizability of gauge theories `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

C

C*-algebra — A complex Banach $*$-algebra satisfying the C*-identity $\|a^*a\| = \|a\|^2$. The Gelfand–Naimark theorem represents any C*-algebra as bounded operators on a Hilbert space `[ESTABLISHED]`. The natural setting for algebraic QM/QFT and superselection. See `domains/mathematics.md` (in the repository).

Canonical commutation relation (CCR) — $[\hat{x}, \hat{p}] = i\hbar 1$. Has no bounded or finite-dimensional realization; its exponentiated (Weyl) form is essentially unique for finitely many degrees of freedom by Stone–von Neumann but admits inequivalent representations for infinitely many `[ESTABLISHED]`. See *Stone–von Neumann theorem*.

Canonical ensemble — Probability distribution $\rho = Z^{-1}e^{-\beta H}$ for a system in thermal contact with a reservoir at temperature $T = 1/k_B\beta$; $F = -k_B T \ln Z$ `[ESTABLISHED for short-range interactions]`. See `domains/statistical-mechanics.md` (in the repository).

Carnot bound — Maximum efficiency of a heat engine between reservoirs, $\eta = 1 - T_c/T_h$, a direct consequence of the second law `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

Chern number / characteristic class — Topological invariant of a bundle; e.g. the second Chern number $\frac{1}{8\pi^2}\int \mathrm{tr}(F \wedge F) \in \mathbb{Z}$ classifies instanton sectors, and the first Chern number gives the quantized Hall conductance (TKNN) `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

CHSH inequality — See *Bell's theorem*. The four-correlator form $S = |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2$ for LHV, with quantum maximum $2\sqrt{2}$.

CKM matrix — Unitary mixing matrix relating quark mass and weak-interaction eigenstates; its single irreducible phase is the sole source of Standard-Model quark CP violation (Jarlskog $J \approx 3 \times 10^{-5}$) `[ESTABLISHED]`. See `domains/particle-physics.md` (in the repository).

Coarse-graining — Partition of microstate space into macroscopically distinguishable cells; thermodynamic entropy and irreversibility are defined relative to such a partition. The *need* for it is structurally fundamental; the specific choice has a conventional element `[CONTESTED interpretation]`. See `domains/statistical-mechanics.md` (in the repository).

Coleman–Mandula theorem — Under reasonable assumptions (nontrivial S-matrix, mass gap, finitely many particle types), the only Lie-algebra symmetries of the S-matrix that combine spacetime and internal symmetries are direct products of the Poincaré group with internal symmetries — forbidding nontrivial mixing `[ESTABLISHED]`. Supersymmetry evades it by using a graded (super-)algebra. See `UNIFYING_PRINCIPLES.md` (in the repository).

Conformal field theory (CFT) — A QFT invariant under the conformal group; sits at renormalization-group fixed points and anchors the UV/IR endpoints of RG flows. Most rigorously controlled interacting QFTs (especially in 2D) `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Connection — On a fiber bundle, a rule for parallel transport / covariant differentiation; for a principal G-bundle it is a $\mathfrak{g}$-valued one-form whose curvature is the gauge field strength $F = dA + A \wedge A$ `[ESTABLISHED]`. The geometric content of the gauge principle. See `domains/mathematics.md` (in the repository).

Constraint (Hamiltonian) — A phase-space relation that must vanish (weakly) on physical states, e.g. the GR momentum and Hamiltonian constraints. First-class constraints generate gauge transformations; their quantization yields the Wheeler–DeWitt equation $H\Psi = 0$ `[ESTABLISHED]`. See *problem of time*.

Contextuality (Kochen–Specker) — No noncontextual assignment of definite values to all observables (for $\dim \geq 3$) reproduces quantum predictions; measured values can depend on which compatible observables are measured alongside `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Cosmological constant (Λ) — The unique term Lovelock's theorem permits beyond the Einstein tensor; drives late-time accelerated expansion. Its observed value lies $\sim 10^{120}$ below naive QFT vacuum-energy estimates — the cosmological-constant problem `[OPEN]`. See `GAPS_AND_CONTRADICTIONS.md` (in the repository), `domains/cosmology.md` (in the repository).

Cosmological principle — Spatial homogeneity and isotropy on large scales ($\gtrsim$ 100 Mpc); the symmetry assumption underlying FLRW `[ESTABLISHED statistically; INFERENCE as exact]`. See `ASSUMPTIONS_LEDGER.md` (in the repository).

CPT theorem — Any local, Lorentz-invariant QFT with a Hermitian Hamiltonian is invariant under the combined operation of charge conjugation, parity, and time reversal `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Critical exponents / universality — Power-law behavior of thermodynamic quantities near a continuous phase transition (e.g. $\xi \sim |t|^{-\nu}$), with exponents depending only on dimension and symmetry (universality classes), explained by RG fixed points `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository).

Crooks fluctuation theorem — Detailed symmetry $P_F(+W)/P_R(-W) = e^{\beta(W-\Delta F)}$ between forward and reverse driving protocols; quantifies the exponential rarity of transient second-law violations `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

D

Dark energy — The component (constant or dynamical) driving cosmic acceleration; modeled as Λ with equation of state $w = -1$. Acceleration is `[ESTABLISHED]`; whether dark energy is exactly Λ is `[OPEN/CONTESTED]`. See `domains/cosmology.md` (in the repository).

Dark matter — A non-baryonic, cold, gravitating component required by rotation curves, lensing, the CMB peaks, and structure formation. Its existence (beyond baryons + GR) is near-`[ESTABLISHED]`; its particle identity is `[OPEN]`. See `domains/cosmology.md` (in the repository), `domains/particle-physics.md` (in the repository).

Decoherence — Suppression of interference in a subsystem through entanglement with an uncontrolled environment, selecting a quasi-classical pointer basis and yielding effective classicality. The dynamics is `[ESTABLISHED]`; its foundational significance for the measurement problem is `[CONTESTED]`. See `domains/quantum-mechanics.md` (in the repository).

Density matrix — Positive, unit-trace operator ρ encoding pure ($\rho^2 = \rho$) or mixed states; the reduced state $\rho_A = \mathrm{Tr}_B\, \rho_{AB}$ captures all locally accessible statistics `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Diffeomorphism invariance — Invariance under smooth coordinate changes; in GR physical states are equivalence classes under diffeomorphisms (background independence), making spacetime points have no physical identity apart from fields (hole argument) `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository).

Dirac–Bergmann constraint analysis — Hamiltonian treatment of singular (gauge) Lagrangians via first- and second-class constraints and Dirac brackets; the classical skeleton of gauge theory `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

Domain of validity — The regime (energies, scales, couplings, idealizations) in which a theory is empirically reliable. Distinguishing a true inconsistency from a domain-of-validity mismatch is central to this wiki's epistemics. See `EPISTEMICS.md` (in the repository), `GAPS_AND_CONTRADICTIONS.md` (in the repository).

E

Effective field theory (EFT) — A theory valid below a cutoff Λ, organizing all symmetry-allowed operators by mass dimension into relevant/marginal/irrelevant, with $\mathcal{L}_{\text{eff}} = \sum_i c_i \Lambda^{4-d_i} \mathcal{O}_i$. Renormalizability becomes an emergent low-energy property `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository).

Ehrenfest's theorem — $\frac{d}{dt}\langle\hat{x}\rangle = \langle\hat{p}\rangle/m$, $\frac{d}{dt}\langle\hat{p}\rangle = -\langle\nabla V\rangle$; recovers Newtonian motion for narrow packets but fails for chaotic systems beyond the (logarithmic) Ehrenfest time `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Einstein field equations (EFE) — $G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$; ten coupled quasilinear second-order PDEs relating curvature to stress-energy `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

Eigenstate thermalization hypothesis (ETH) — Ansatz that few-body observables in individual energy eigenstates of a generic non-integrable many-body Hamiltonian already take thermal values, explaining thermalization of isolated quantum systems [INFERENCE: strong numerical support, unproven; fails for integrable and many-body-localized systems]. See `domains/statistical-mechanics.md` (in the repository).

Energy conditions — Pointwise positivity constraints on $T_{\mu\nu}$ (weak/null/strong/dominant) used in singularity and censorship theorems; violated by quantum fields and by a positive Λ, so the pointwise forms are now seen as too strong `[CONTESTED]`, replaced by averaged/quantum versions (ANEC, QNEC). See `domains/general-relativity.md` (in the repository).

Entanglement — Nonseparability of a composite quantum state; the source of Bell violations and a quantifiable resource (teleportation, dense coding). Pure-state bipartite entanglement is measured by the entanglement entropy $S(\rho_A)$ `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository), `domains/information-theory.md` (in the repository).

Entanglement entropy — Von Neumann entropy $S(\rho_A) = -\operatorname{Tr} \rho_A \ln \rho_A$ of a subsystem. In continuum QFT it is UV-divergent (area law); only universal/finite pieces are cutoff-independent, and local QFT algebras are Type III (no density matrices) `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

Envariance — Environment-assisted invariance; Zurek's symmetry argument attempting to derive the Born rule within a no-collapse (Everettian) framework `[CONTESTED]`. See `domains/quantum-mechanics.md` (in the repository).

Equivalence principle — Universality of free fall (weak EP, tested to $\sim 10^{-15}$) and local Lorentz/position invariance (Einstein EP); the basis for geometrizing gravity `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

ER=EPR — Conjecture (Maldacena–Susskind) identifying entanglement (EPR) with geometric wormhole connections (Einstein–Rosen) `[SPECULATIVE]`. See `UNIFICATION_LANDSCAPE.md` (in the repository).

Ergodicity — Equality of long-time and phase-space averages on the energy surface. Generic Hamiltonian systems are *not* ergodic (KAM); statistical mechanics is now justified more by typicality/measure concentration than by strict ergodicity [ESTABLISHED that ergodicity is neither necessary nor sufficient as usually invoked]. See `domains/statistical-mechanics.md` (in the repository).

Euler–Lagrange equations — $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0$; coordinate-invariant equations of motion from stationary action

`[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

Everett (many-worlds) interpretation — Retains only unitary evolution; measurement is branching into decoherent worlds. Pays with branching multiplicity and an unsettled account of probability `[ESTABLISHED equivalence to QM predictions; CONTESTED foundations]`. See `domains/quantum-mechanics.md` (in the repository).

F

Faddeev–Popov / Gribov ambiguity — Gauge-fixing procedure introducing ghost fields; globally obstructed in non-abelian theories (Gribov copies; Singer's theorem shows no global gauge section exists), so it is only locally valid nonperturbatively `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Fermi–Dirac / Bose–Einstein statistics — Occupation numbers $\langle n_k \rangle = [e^{\beta(\epsilon_k - \mu)} \pm 1]^{-1}$ ($+$ fermions, $-$ bosons), following from identical-particle (anti)symmetrization `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository), *spin–statistics theorem*.

Feynman path integral — Propagator as a sum over histories $\int \mathcal{D}[q]\, e^{iS/\hbar}$; the classical equations arise by stationary phase as $\hbar \to 0$. Rigorous via Euclidean (Wiener) measure in QM and some low-dimensional QFTs; the literal Lorentzian measure does not exist `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Fiber bundle — A space locally a product $U \times F$ of base and fiber but possibly globally twisted; principal G-bundles host gauge connections, and the bundle topology is physical (Dirac quantization, Aharonov–Bohm) `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

First law of entanglement — Perturbative equality $\delta S_A = \delta \langle K_A \rangle$ (modular Hamiltonian K_A); in holographic setups it yields the *linearized* Einstein equations — a concrete "gravity from entanglement" result

`[INFERENCE within AdS/CFT; SPECULATIVE as a route to full GR]`. See `domains/information-theory.md` (in the repository).

Fluctuation–dissipation theorem — Relates a system's linear response to its equilibrium fluctuation spectrum; Einstein, Nyquist, and Green–Kubo relations are special cases `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository).

Fluctuation theorems — Exact relations (Jarzynski equality, Crooks theorem, Gallavotti–Cohen) constraining nonequilibrium work and entropy-production distributions, sharpening the second law into an equality over trajectory ensembles `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

FLRW metric — Friedmann–Lemaître–Robertson–Walker line element, the unique homogeneous-isotropic spacetime, with scale factor $a(t)$ and curvature k `[ESTABLISHED]`. See `domains/cosmology.md` (in the repository).

Free energy — Thermodynamic potentials obtained as Legendre transforms of internal energy: Helmholtz $F = U - TS$, Gibbs $G = U - TS + pV$, grand $\Omega = U - TS - \mu N$, each minimized at equilibrium under its control variables `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

Friedmann equations — Dynamical equations for $a(t)$ from the EFE under FLRW symmetry: $H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3}$ and the acceleration equation $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3p) + \Lambda/3$ `[ESTABLISHED]`. See `domains/cosmology.md` (in the repository).

G

Gauge invariance / gauge principle — Invariance under local transformations of internal (or spacetime) symmetry; interactions follow from the covariant derivative $D_\mu = \partial_\mu - igA_\mu^a T^a$. Gauge symmetry is a *redundancy* of description, not a physical symmetry (Elitzur's theorem forbids its spontaneous breaking) `[ESTABLISHED]`; framing it as a fundamental "principle" is `[CONTESTED]`. See `domains/quantum-field-theory.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository).

Generalized probabilistic theories (GPT) — Operational framework reconstructing quantum theory from information-theoretic axioms (local tomography, purification, continuous reversibility); illuminating but with multiple inequivalent axiom sets `[OPEN]`. See `domains/quantum-mechanics.md` (in the repository).

Generalized second law (GSL) — Conjecture that ordinary entropy plus black-hole horizon entropy never decreases `[INFERENCE, strong support]`. See `domains/thermodynamics.md` (in the repository).

Geodesic — Extremal-proper-time worldline of a free-falling body: $\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$; "gravity = inertial motion in curved spacetime" `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

Geodesic deviation — Relative acceleration of nearby geodesics governed by curvature: $D^2\xi^\mu/d\tau^2 = -R^\mu{}_{\alpha\nu\beta}u^\alpha\xi^\nu u^\beta$; the operationally measurable tidal signature of gravity `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

Gibbs entropy — $S_G = -k_B \int \rho \ln \rho \, d\Gamma$ (classical) or $-k_B \, \mathrm{Tr}(\rho \ln \rho)$ (von Neumann, quantum); the volume-entropy definition. Constant under exact dynamics; coarse-graining is required for increase `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository).

Gleason's theorem — For $\dim \geq 3$, the only probability measures on the projection lattice of a Hilbert space are of the form $\mathrm{Tr}(\rho\,\cdot)$, fixing the Born rule's form given noncontextuality `[ESTABLISHED]`. See *Born rule*.

Goldstone bosons — Massless (or pseudo-) excitations accompanying spontaneous breaking of a continuous global symmetry (Goldstone's theorem); when the symmetry is gauged they are "eaten" via the Higgs mechanism `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Grand unified theory (GUT) — Embedding of $SU(3) \times SU(2) \times U(1)$ into a simple group (SU(5), SO(10), Pati–Salam), explaining charge quantization and predicting proton decay

`[INFERENCE/SPECULATIVE; minimal SU(5) excluded by proton-decay bounds]`. See `UNIFICATION_LANDSCAPE.md` (in the repository).

Groenewold–van Hove theorem — No quantization map exists from the full classical Poisson algebra to operators respecting $\{\cdot, \cdot\} \mapsto [\cdot, \cdot]/i\hbar$ on all observables; operator ordering is a genuine ambiguity `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

H

Haag's theorem — In QFT the interaction picture does not exist: the free and interacting field representations are unitarily inequivalent, so the naive perturbative starting point is mathematically inconsistent (yet renormalized perturbation theory works) `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Hamilton–Jacobi equation — $H(q, \partial S/\partial q, t) + \partial S/\partial t = 0$; a single first-order PDE whose complete integral solves the dynamics, and the classical limit of the Schrödinger equation via $S = -i\hbar \ln \psi$ `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

Hamiltonian mechanics — Phase-space dynamics $\dot{q}^i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q^i$ on a symplectic manifold `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

Hawking radiation — Thermal emission from a black-hole horizon at temperature $T_H = \hbar c^3/(8\pi G k_B M)$, derived from QFT on a curved background `[INFERENCE: not directly observed, but theoretically robust]`. See `domains/general-relativity.md` (in the repository).

Higgs (Brout–Englert–Higgs) mechanism — Spontaneous breaking of a gauge symmetry by a scalar vacuum expectation value, giving mass to gauge bosons ($M_W = \frac{1}{2}gv$, $v \approx 246$ GeV) without breaking gauge invariance explicitly `[ESTABLISHED]`. See `domains/particle-physics.md` (in the repository).

Hilbert space — Complete complex inner-product space; states of a quantum system are rays (unit vectors up to phase), and observables are self-adjoint operators on it `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository), `domains/mathematics.md` (in the repository).

Holographic principle — Conjecture that the degrees of freedom in a region are bounded by its boundary area in Planck units, so a gravitational theory is encoded on a lower-dimensional boundary `[INFERENCE; realized concretely in AdS/CFT]`. See `domains/information-theory.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository).

Hubble tension — ~ 4–6σ discrepancy between the early-universe ($H_0 \approx 67$–68) and local distance-ladder ($H_0 \approx 73$ km/s/Mpc) determinations of the expansion rate `[CONTESTED: systematics vs new physics]`. See `domains/cosmology.md` (in the repository), `GAPS_AND_CONTRADICTIONS.md` (in the repository).

I

Identical particles — Indistinguishable quanta whose multiparticle states must be symmetric (bosons) or antisymmetric (fermions); in 2D the exchange structure is richer (anyons, braid group) `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Inflation — A hypothesized epoch of slow-roll accelerated expansion solving the horizon, flatness, and monopole problems and seeding near-scale-invariant fluctuations [INFERENCE bordering SPECULATIVE; no primordial B-mode detection, inflaton unidentified]. See `domains/cosmology.md` (in the repository).

Instanton — A finite-action solution of the Euclidean field equations representing tunneling between topologically distinct gauge vacua, labeled by an integer winding (second Chern) number `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Integrability (Liouville–Arnold) — A Hamiltonian system with as many independent Poisson-commuting conserved quantities as degrees of freedom; motion confines to invariant tori `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

J

Jacobson's derivation — Demonstration that the Einstein equations follow as an equation of state from horizon thermodynamics ($\delta Q = T\delta S$ on local Rindler horizons) `[SPECULATIVE/INFERENCE: suggestive of emergent gravity]`. See `UNIFYING_PRINCIPLES.md` (in the repository).

Jarzynski equality — Exact nonequilibrium identity $\langle e^{-\beta W}\rangle = e^{-\beta \Delta F}$, valid arbitrarily far from equilibrium, implying $\langle W\rangle \geq \Delta F$ `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

K

KAM theorem — Kolmogorov–Arnold–Moser: most invariant tori of an integrable system survive small perturbations, provided their frequencies satisfy a Diophantine (non-resonance) condition; resonances seed chaos `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

KMS condition — Kubo–Martin–Schwinger characterization of thermal equilibrium states via analyticity of correlation functions in imaginary time; via Tomita–Takesaki it identifies "temperature" and modular "time" with a common algebraic origin `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Kochen–Specker theorem — See *contextuality*.

Kolmogorov complexity — Length $K(x)$ of the shortest program generating a string x; machine-independent up to an additive constant but provably uncomputable `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

Kraus (operator-sum) representation — General quantum channel $\rho \mapsto \sum_k K_k \rho K_k^\dagger$ with $\sum_k K_k^\dagger K_k = 1$; via Stinespring/Naimark every channel and POVM is a unitary/projective measurement on a dilated space `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

L

Lagrangian — Function $L = T - V$ (or density $\mathcal{L}$) whose action governs the dynamics; the specific $T - V$ form is a low-energy shadow of deeper (relativistic/path-integral) structure `[ESTABLISHED, INFERENCE as to origin]`. See `domains/classical-mechanics.md` (in the repository).

Landauer's principle — Erasing one bit of information dissipates at least $k_B T \ln 2$ of heat, linking logical irreversibility to thermodynamic cost; experimentally confirmed near the bound `[ESTABLISHED; its independent status vs the second law is CONTESTED]`. See `domains/information-theory.md` (in the repository).

Landau pole — Energy at which a coupling (e.g. in QED or ϕ^4) formally diverges, signaling that the theory is at most an EFT with a finite UV cutoff (triviality) `[INFERENCE, strong]`. See `domains/quantum-field-theory.md` (in the repository).

ΛCDM — The concordance cosmological model: GR with a cosmological constant and cold dark matter, specified by six base parameters, fitting the CMB, BBN, BAO, and supernovae [ESTABLISHED as a fit; constituents `[OPEN]`]. See `domains/cosmology.md` (in the repository).

Legendre transform — Map exchanging a variable for its conjugate (e.g. $L \to H, U \to F$); underlies the Hamiltonian formalism and thermodynamic potentials `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository), `domains/thermodynamics.md` (in the repository).

Levi-Civita connection — The unique torsion-free, metric-compatible connection on a (pseudo-)Riemannian manifold, with Christoffel symbols $\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$

[ESTABLISHED; the torsion-free choice is conventional, cf. Einstein–Cartan]. See `domains/general-relativity.md` (in the repository).

Lindblad (GKSL) equation — Most general generator of completely-positive, trace-preserving Markovian dynamics: $\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$; governs open-system evolution and decoherence `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Liouville's theorem — Hamiltonian phase-space flow preserves volume ($d\rho/dt = 0$), so fine-grained entropy is conserved — the technical reason coarse-graining is needed for irreversibility `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository).

Lovelock's theorem — In 4D, the only divergence-free symmetric 2-tensor built from the metric and its first two derivatives (with second-order field equations) is $aG_{\mu\nu} + bg_{\mu\nu}$, making the Einstein equations plus Λ essentially unique `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

LSZ reduction — Lehmann–Symanzik–Zimmermann formula relating S-matrix elements to amputated correlation functions, justified for isolated mass shells with a mass gap; fails for confined or massless asymptotic states `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Lyapunov exponent — Rate λ of exponential divergence of nearby trajectories, $|\delta x(t)| \sim |\delta x(0)| e^{\lambda t}$; $\lambda > 0$ is the signature of deterministic chaos `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

M

Majorana mass / neutrino — A fermion identical to its own antiparticle; Majorana neutrino masses (via the seesaw or Weinberg operator) would violate lepton number, testable by neutrinoless double-beta decay. Dirac vs Majorana nature of neutrinos is `[OPEN]`. See `domains/particle-physics.md` (in the repository).

Mandelstam–Tamm bound — Time–energy inequality bounding the rate of state evolution by the energy spread, reinterpreting $\Delta E\,\Delta t \gtrsim \hbar$ (which is *not* a Robertson relation, since there is no time operator) `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Margolus–Levitin theorem — Quantum speed limit: a state of mean energy E above the ground state needs at least $t_\perp \geq \pi\hbar/2E$ to reach an orthogonal state `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

Maxwell relations — Equalities among partial derivatives of thermodynamic potentials (e.g. $(\partial S/\partial V)_T = (\partial p/\partial T)_V$) following from exactness of the differentials `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

Maxwell's demon — Thought experiment apparently violating the second law by sorting molecules; resolved by the entropy cost of information acquisition and especially erasure (Szilard, Landauer, Bennett) `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

Measurement problem — The structural incompatibility between linear unitary evolution (which never selects a single outcome) and the stochastic projection postulate, with no rule specifying what triggers collapse ("the shifty split") `[OPEN/CONTESTED]`. The deepest internal problem of QM. See `domains/quantum-mechanics.md` (in the repository), `GAPS_AND_CONTRADICTIONS.md` (in the repository).

Microcausality — Vanishing of commutators of local observables at spacelike separation, $[\phi(x), \phi(y)] = 0$ for $(x-y)^2 < 0$; the QFT encoding of relativistic causality and no-signaling `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Microcanonical ensemble — Uniform (Liouville) measure on the energy shell; $S_B = k_B \ln \Omega(E)$. The equal-a-priori-probability postulate bridges micro to macro `[INFERENCE/OPEN as to first-principles derivation]`. See `domains/statistical-mechanics.md` (in the repository).

Modular Hamiltonian — Generator $K_A = -\ln \rho_A$ of the Tomita–Takesaki modular flow of a region's state; central to relative entropy, energy conditions, and the entanglement first law `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

Molecular chaos (Stosszahlansatz) — Assumption that pre-collision velocities are uncorrelated; the precise time-asymmetric input that lets the Boltzmann H-theorem produce irreversibility `[ESTABLISHED as the locus of time-asymmetry]`. See `domains/statistical-mechanics.md` (in the repository).

Moyal star product — Noncommutative product $f \star g = fg + \frac{i\hbar}{2}\{f, g\} + O(\hbar^2)$ deforming the Poisson algebra; the phase-space (deformation-quantization) route from classical to quantum mechanics `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

Mutual information — $I(X;Y) = H(X) + H(Y) - H(X,Y) = D(p(x,y)\|p(x)p(y)) \geq 0$; the shared information between two systems, finite even when individual entropies diverge `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

N

Negative temperature — For systems with a bounded energy spectrum, population inversion gives $dS/dU < 0$, hence $T < 0$, which is "hotter" than any positive T; experimentally realized [ESTABLISHED; admissibility tied to the CONTESTED Boltzmann-vs-Gibbs entropy debate]. See `domains/thermodynamics.md` (in the repository).

No-cloning theorem — Linearity of quantum mechanics forbids a universal device that copies an unknown state; underlies quantum cryptography and sharpens the information paradox `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

No-signaling theorem — Local marginal statistics are independent of a distant party's measurement choice, so entanglement cannot transmit usable information faster than light `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Noether's theorem — Every continuous symmetry of an action yields a conserved current/charge; in Hamiltonian form, $\dot{J} = 0 \iff \{J, H\} = 0$ `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository).

O

Objective-collapse models (GRW/CSL, Diósi–Penrose) — Modifications of QM adding stochastic nonlinear terms that spontaneously localize macroscopic superpositions, predicting tiny, *testable* deviations from unitary QM `[OPEN, empirically constrained]`. See `domains/quantum-mechanics.md` (in the repository), `HYPOTHESES.md` (in the repository).

Osterwalder–Schrader reconstruction — Theorem relating Euclidean (reflection-positive) correlation functions to a Lorentzian Wightman QFT; underlies rigorous and lattice work but requires reflection positivity (which can fail at finite density — the sign problem) `[ESTABLISHED where conditions hold]`. See `domains/quantum-field-theory.md` (in the repository).

P

Page curve — The entanglement entropy of Hawking radiation as a function of time, rising then falling for a unitarily evaporating black hole; recently reproduced via quantum-extremal-surface/island computations `[INFERENCE within assumptions]`. See `domains/information-theory.md` (in the repository).

Partition function — Generating functional $Z = \operatorname{Tr} e^{-\beta \hat{H}}$ (statistical mechanics) or $\int \mathcal{D}\phi \, e^{-S_E}$ (Euclidean QFT); all equilibrium thermodynamics follows by differentiating $\ln Z$, and the Euclidean path integral *is* a partition function (Wick-rotation isomorphism) `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository), `domains/quantum-field-theory.md` (in the repository).

Past hypothesis — The posit of an extraordinarily low-entropy initial cosmological state, required to ground the thermodynamic arrow of time given time-symmetric microdynamics `[INFERENCE that it is necessary; OPEN why it holds]`. See `domains/statistical-mechanics.md` (in the repository), `domains/cosmology.md` (in the repository).

Pauli's theorem (no time operator) — No self-adjoint operator is canonically conjugate to a Hamiltonian bounded below; time is treated as a c-number parameter, not an observable `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

PBR theorem — Pusey–Barrett–Rudolph: under the assumption that independently prepared systems have independent physical states, the quantum state cannot be a mere probability distribution over deeper "ontic" states (a ψ-epistemic model) — the wavefunction is "real" `[INFERENCE, conditional on the preparation-independence assumption]`. See `domains/quantum-mechanics.md` (in the repository).

Penrose–Hawking singularity theorems — Under energy conditions and a trapped surface (or expansion), spacetime is geodesically incomplete, proving singularities are generic in gravitational collapse and cosmology `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

Phase transition — Non-analyticity of a thermodynamic potential, occurring only in the thermodynamic limit (Lee–Yang); classified by order and universality class `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository).

PMNS matrix — Pontecorvo–Maki–Nakagawa–Sakata lepton mixing matrix governing neutrino oscillations; its large mixing angles and undetermined CP phase contrast with the nearly diagonal CKM matrix

`[ESTABLISHED that mixing is large; CP phase OPEN]`. See `domains/particle-physics.md` (in the repository).

Poincaré group — The isometry group of Minkowski spacetime (Lorentz transformations + translations); its unitary irreducible representations classify particles by mass and spin (Wigner) via the Casimirs $P^2 = m^2$, $W^2 = -m^2 s(s+1)$ `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Poisson bracket — Lie-algebraic structure $\{f, g\} = \sum_i (\partial_{q^i} f\, \partial_{p_i} g - \partial_{p_i} f\, \partial_{q^i} g)$ on phase space; evolution is $df/dt = \{f, H\} + \partial_t f$. Survives (deformed) into the quantum commutator `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

POVM — Positive operator-valued measure $\{E_i \geq 0, \sum_i E_i = 1\}$; the most general quantum measurement, generalizing projective (von Neumann) measurements `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Problem of time — In canonical gravity the Hamiltonian is a sum of constraints, so $H\Psi = 0$ (Wheeler–DeWitt) describes a "timeless" quantum universe; reconciling this with observed time evolution is unresolved `[OPEN/CONTESTED]`. See `domains/general-relativity.md` (in the repository), `GAPS_AND_CONTRADICTIONS.md` (in the repository).

Q

QBism — Quantum Bayesianism: reads the wavefunction and Born rule as an agent's subjective degrees of belief, dissolving the measurement problem at the cost of a non-objective ψ `[CONTESTED interpretation]`. See `domains/quantum-mechanics.md` (in the repository).

Quantum error correction (QEC) — Encoding logical information redundantly across physical degrees of freedom to protect against noise; in AdS/CFT it models how bulk operators are reconstructed from boundary subregions (entanglement-wedge reconstruction)

`[ESTABLISHED as code theory; INFERENCE as a model of holography]`. See `domains/information-theory.md` (in the repository).

R

Reduced density matrix — See *density matrix*.

Reeh–Schlieder theorem — The vacuum is cyclic and separating for every local algebra: acting with operators in an arbitrarily small region densely spans the whole Hilbert space, exhibiting ubiquitous vacuum entanglement (no contradiction with no-signaling) `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Relational quantum mechanics — Rovelli's interpretation in which physical facts are relative to the systems interacting, with no observer-independent absolute state `[CONTESTED]`. See `domains/quantum-mechanics.md` (in the repository).

Renormalization — Procedure absorbing cutoff dependence into a finite set of measured parameters; modern Wilsonian view treats it as RG flow of an effective theory rather than infinity-subtraction `[ESTABLISHED]`. See *renormalization group*, `domains/quantum-field-theory.md` (in the repository).

Renormalization group (RG) — The flow of couplings with energy scale, $\beta(g) = \mu\, dg/d\mu$; fixed points define scale-invariant (conformal) theories, and irrelevant operators are suppressed at low energy, explaining universality and emergent renormalizability `[ESTABLISHED]`. See `domains/statistical-mechanics.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository).

Ricci / Riemann curvature — $R^{\rho}{}_{\sigma\mu\nu}$ is the tensorial measure of gravity ($[\nabla_\mu, \nabla_\nu]V^\rho = R^{\rho}{}_{\sigma\mu\nu}V^\sigma$); its contraction (Ricci) sources the Einstein tensor `[ESTABLISHED]`. See `domains/general-relativity.md` (in the repository).

Rigged Hilbert space — Gelfand triple $\Phi \subset \mathcal{H} \subset \Phi'$ making rigorous the Dirac formalism of continuous-spectrum "eigenstates" (e.g. position, momentum) that lie outside $\mathcal{H}$ `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Robertson uncertainty relation — $\Delta A\,\Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|$, a statement about ensemble spreads in a fixed state (not measurement disturbance), specializing to $\Delta x\,\Delta p \geq \hbar/2$ `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Ryu–Takayanagi formula — In holography, the entanglement entropy of a boundary region equals the area of the minimal homologous bulk surface, $S(A) = \text{Area}(\gamma_A)/4G_N$; covariant (HRT) and quantum-corrected (QES) versions extend it `[INFERENCE within AdS/CFT]`. See `domains/information-theory.md` (in the repository).

S

Sakharov conditions — The three requirements for dynamically generating a baryon asymmetry: baryon-number violation, C and CP violation, and departure from thermal equilibrium `[ESTABLISHED as necessary conditions]`. See `domains/particle-physics.md` (in the repository).

Schmidt decomposition — Canonical form $|\psi\rangle_{AB} = \sum_i \sqrt{p_i}\,|i\rangle_A|i\rangle_B$ of a bipartite pure state, with the p_i the eigenvalues of either reduced density matrix `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Schrödinger equation — $i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle$; deterministic, linear, unitary evolution of a closed quantum system, with $U(t) = e^{-iHt/\hbar}$ `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Second law of thermodynamics — Entropy of an isolated system does not decrease, $\Delta S \geq 0$ (Clausius $dS \geq \delta Q/T$); phenomenologically absolute, but statistically a probabilistic statement refined by fluctuation theorems `[ESTABLISHED at macroscale; refined by statistical mechanics]`. See `domains/thermodynamics.md` (in the repository).

Self-adjoint operator — A symmetric operator equal to its adjoint *including domains*; required (not mere symmetry) for a real spectrum, a spectral measure, and unitary dynamics (Stone). Inequivalent self-adjoint extensions encode different physics (boundary conditions) `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Shannon entropy — $H(X) = -\sum_x p(x)\log_2 p(x)$, the unique measure of average uncertainty (up to base), equal to the optimal lossless compression rate `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

Spectral theorem — Every self-adjoint operator decomposes as $A = \int_{\sigma(A)} \lambda\, dE_A(\lambda)$ against a projection-valued measure on its spectrum; underwrites both the value set and the Born rule `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Spin–statistics theorem — In relativistic QFT, integer-spin fields are bosonic (commuting) and half-integer-spin fields are fermionic (anticommuting), forced by microcausality, positive energy, and Lorentz invariance; an *added postulate* in non-relativistic QM `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Spontaneous symmetry breaking (SSB) — A symmetric dynamics with a non-symmetric ground state; the symmetry is realized nonlinearly (Goldstone modes) rather than on the vacuum. Requires infinitely many degrees of freedom (inequivalent vacua) `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository).

Standard Model — The renormalizable, anomaly-free chiral gauge theory with group $SU(3)_c \times SU(2)_L \times U(1)_Y$ spontaneously broken to $SU(3)_c \times U(1)_{\text{em}}$; the most precisely tested theory in science `[ESTABLISHED, with confirmed incompleteness via neutrino mass]`. See `domains/particle-physics.md` (in the repository).

Stinespring dilation — Theorem that every completely-positive map is a compression of a $*$-homomorphism (unitary) on a larger space; the structural basis for Kraus/POVM purification `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

Stone's theorem — A strongly continuous one-parameter unitary group has a unique self-adjoint generator, tying unitary dynamics to self-adjointness of the Hamiltonian `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Stone–von Neumann theorem — For finitely many canonical pairs, all irreducible representations of the (Weyl form of the) CCR are unitarily equivalent; uniqueness *fails* for infinitely many degrees of freedom, the technical root of inequivalent vacua, Haag's theorem, and SSB `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Strong CP problem — The unexplained smallness ($\bar{\theta} \lesssim 10^{-10}$) of the CP-violating QCD theta angle, naturally $O(1)$; the Peccei–Quinn axion is the leading proposed resolution `[OPEN; axion SPECULATIVE]`. See `domains/particle-physics.md` (in the repository).

Strong subadditivity — The deep entropy inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$, equivalent to nonnegativity of conditional mutual information; source of most holographic entropy constraints `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

Superselection sector — A subspace between which coherent superpositions are unobservable (e.g. different charge or, for infinite systems, distinct phases), arising from inequivalent representations `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Supersymmetry (SUSY) — Proposed graded symmetry relating bosons and fermions (evading Coleman–Mandula via the Haag–Łopuszański–Sohnius extension), addressing the hierarchy problem and improving gauge unification `[SPECULATIVE; no superpartners observed]`. See `UNIFICATION_LANDSCAPE.md` (in the repository).

Symplectic form — Closed, nondegenerate two-form $\omega = dq^i \wedge dp_i$ on phase space defining Hamiltonian vector fields via $\iota_{X_H}\omega = dH$; the rigid geometric core of classical mechanics (Darboux's theorem) `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

T

Tensor product — The composition rule $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ for quantum subsystems, encoding local tomography and entanglement; singled out among operationally reasonable rules in GPT reconstructions `[ESTABLISHED operationally; INFERENCE as necessity]`. See `domains/quantum-mechanics.md` (in the repository).

Thermodynamic limit — The idealization $N, V \to \infty$ at fixed density, in which extensivity holds, ensembles are equivalent, and genuine phase transitions exist `[ESTABLISHED idealization; fails for long-range/gravitating systems]`. See `domains/statistical-mechanics.md` (in the repository).

Third law of thermodynamics — Entropy approaches a constant (conventionally zero) as $T \to 0$ for a non-degenerate ground state, and absolute zero is unattainable; a quantum consequence of discrete spectra `[ESTABLISHED]`. See `domains/thermodynamics.md` (in the repository).

Tomita–Takesaki modular theory — For a cyclic separating state, the operator $S = J\Delta^{1/2}$ generates a canonical modular automorphism flow $\sigma_t(a) = \Delta^{it} a \Delta^{-it}$ and identifies the commutant ($JMJ = M'$); thermal (KMS) states are modular states `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Topological quantum field theory (TQFT) — A QFT whose observables depend only on topology; formalized as a symmetric monoidal functor $Z : \mathrm{Bord}_n \to \mathcal{C}^{\otimes}$ (Atiyah–Segal), with the cobordism hypothesis classifying fully extended TQFTs `[ESTABLISHED for topological theories]`. See `domains/mathematics.md` (in the repository).

Triviality — The conjecture/theorem that the continuum limit of a theory (e.g. ϕ^4 in 4D, proven by Aizenman–Duminil-Copin) is the free theory, so it survives only as a cutoff EFT [ESTABLISHED for ϕ^4_4; INFERENCE for QED]. See `domains/quantum-field-theory.md` (in the repository).

Tsirelson bound — The quantum maximum $2\sqrt{2}$ of the CHSH expression, exceeding the classical 2 but below the no-signaling (PR-box) maximum 4 `[ESTABLISHED]`. See `domains/quantum-mechanics.md` (in the repository).

U

Unitarity — Conservation of total probability under time evolution / the S-matrix; non-negotiable within quantum mechanics, enforced order-by-order via Ward/BRST identities and the optical theorem `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Universality — See *critical exponents / universality*.

Unruh effect — A uniformly accelerated observer perceives the Minkowski vacuum as a thermal bath at temperature $T = \hbar a / 2\pi k_B c$; demonstrates the observer-dependence of "particle" in QFT `[INFERENCE: theoretically robust, not directly observed]`. See `domains/quantum-field-theory.md` (in the repository).

V

Vacuum (QFT) — The lowest-energy state; for interacting theories it is not the free Fock vacuum (Haag's theorem), is entangled across regions (Reeh–Schlieder), and carries energy whose gravitational weight is the cosmological-constant problem `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Von Neumann entropy — $S(\rho) = -\operatorname{Tr}(\rho \ln \rho)$; the quantum analogue of Shannon entropy, zero for pure states and maximal for the maximally mixed state, and the entanglement entropy of a bipartite pure state `[ESTABLISHED]`. See `domains/information-theory.md` (in the repository).

Von Neumann algebra — A $*$-subalgebra of bounded operators closed in the weak operator topology; classified into types I, II, III. Local QFT algebras are Type III_1 (no trace, no local density matrix) `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

W

Ward identities — Relations among correlation functions enforcing a (gauge) symmetry at the quantum level; ensure renormalizability and unitarity `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

Weinberg operator — The unique dimension-5 Standard-Model EFT operator, $\frac{c_{ij}}{\Lambda}(\bar{L}^c_{Li}\tilde{\phi}^*)(\tilde{\phi}^\dagger L_{Lj})$, giving Majorana neutrino masses $m_\nu \sim v^2/\Lambda$ — the leading window to physics beyond the SM `[ESTABLISHED as the leading operator]`. See `domains/particle-physics.md` (in the repository).

Weinberg–Witten theorem — Forbids, under broad assumptions, massless particles of spin > 1 carrying a Lorentz-covariant conserved current, and spin > 2 carrying a covariant stress-energy tensor — constraining how gravitons or composite gauge bosons can be *emergent* (the graviton must evade it, e.g. via the absence of a covariant boundary stress tensor) `[ESTABLISHED]`. See `UNIFICATION_LANDSCAPE.md` (in the repository).

Wheeler–DeWitt equation — The quantum Hamiltonian constraint $H\Psi = 0$ of canonical gravity, formally timeless; see *problem of time* `[ESTABLISHED as the constraint; interpretation OPEN]`. See `domains/general-relativity.md` (in the repository).

Wick rotation — Analytic continuation $t \to -i\tau$ relating Lorentzian QFT to Euclidean statistical field theory; rigorous under reflection positivity (Osterwalder–Schrader) but problematic in generic curved/Lorentzian backgrounds

`[ESTABLISHED where conditions hold]`. See `domains/mathematics.md` (in the repository).

Wightman axioms — Axiomatic characterization of a relativistic QFT in terms of operator-valued distributions, Poincaré covariance, the spectrum condition, microcausality, and a cyclic vacuum `[ESTABLISHED framework; 4D interacting realizations OPEN]`. See `domains/quantum-field-theory.md` (in the repository).

Wigner's theorem — Every symmetry preserving transition probabilities is implemented by a unitary or antiunitary operator (antiunitary for time reversal) `[ESTABLISHED]`. See `domains/mathematics.md` (in the repository).

Wilson loop — Gauge-invariant trace of the holonomy of a connection around a closed loop; an order parameter for confinement and the natural gauge-invariant observable `[ESTABLISHED]`. See `domains/quantum-field-theory.md` (in the repository).

WKB approximation — Semiclassical expansion in $\hbar$ with phase $e^{iS/\hbar}$; recovers classical mechanics by stationary phase and captures tunneling ($\sim e^{-S/\hbar}$), but the $\hbar \to 0$ limit is singular and non-uniform `[ESTABLISHED]`. See `domains/classical-mechanics.md` (in the repository).

Y

Yang–Mills theory — Non-abelian gauge theory with field strength $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ and self-interacting gauge bosons; the basis of QCD and the electroweak sector. Rigorous 4D existence with a mass gap is a Clay Millennium Problem `[OPEN]`. See `domains/quantum-field-theory.md` (in the repository), `OPEN_PROBLEMS.md` (in the repository).

References

For canonical textbooks and landmark papers underlying these definitions, see the Bibliography (p. 175).

Bibliography

This page is the consolidated source list for the wiki. It contains **only real, standard references** that the author is confident exist: canonical textbooks (author + title) and genuinely landmark papers (author, year). Where a precise detail (volume, page, exact year) is uncertain, the entry carries an inline `[unverified]` flag and gives author + title + approximate year only — per the `EPISTEMICS.md` (in the repository) rule that we never fabricate citations, numbers, or theorems.

Scope of the epistemic tags here. A reference's *existence* is treated as `[ESTABLISHED]` unless flagged `[unverified]`. The tag attached to an entry instead grades the *epistemic status of the physics the source reports*: a textbook of confirmed theory is `[ESTABLISHED]`; a foundational-interpretation paper is `[CONTESTED]` or `[SPECULATIVE]`; a research-frontier program is `[SPECULATIVE]` or `[OPEN]`. This lets the bibliography double as a quick map of *how firm the underlying claim is*, consistent with the conventions in `EPISTEMICS.md` (in the repository).

> **How to cite from wiki pages.** Domain pages should link here rather than re-listing full citations. Cite as, e.g., "Weinberg, *QTF* I–III (see the Bibliography (p. 175))." Numerical values should always be sourced to the Particle Data Group or Planck (below), never stated from memory — see `CONSTANTS_AND_SCALES.md` (in the repository).

A handful of sources are load-bearing in more than one domain (MTW, Wald, Reed–Simon, Nielsen–Chuang, Haag, Glimm–Jaffe, Peskin–Schroeder, Weinberg's *QTF*, Landau–Lifshitz, the Hawking 1975 and Bekenstein 1973 papers, Jarzynski/Crooks, Wilson–

Kogut). Each is listed once under its **primary** domain and cross-referenced from the others to avoid duplication.

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1. Quantum mechanics and foundations

Primary support for `domains/quantum-mechanics.md` (in the repository).

Textbooks and monographs

P. A. M. Dirac, *The Principles of Quantum Mechanics* (Oxford Univ. Press, 1930; 4th ed. 1958). `[ESTABLISHED]` — Origin of the bra–ket formalism, representation-independent state/operator picture, and transformation theory underlying the Hilbert-space axioms.

J. von Neumann, *Mathematical Foundations of Quantum Mechanics* (Springer, 1932; Princeton Univ. Press transl. 1955). `[ESTABLISHED]` — The rigorous Hilbert-space axiomatization: self-adjoint operators, the spectral theorem, density matrices, the projection postulate, and the first serious analysis of measurement.

A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer, 1995). `[ESTABLISHED]` — Operationally rigorous text emphasizing measurement, Kochen–Specker, Bell, and careful statement of the formalism.

J. S. Bell, *Speakable and Unspeakable in Quantum Mechanics* (Cambridge Univ. Press, 1987; 2nd ed. 2004). `[ESTABLISHED]` (results) / `[CONTESTED]` (interpretive morals) — Collected papers including the Bell inequality, the critique of the measurement "shifty split," and incisive analyses of nonlocality and hidden variables.

M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge Univ. Press, 2000; 10th anniversary ed. 2010). `[ESTABLISHED]` — Standard reference for density matrices, POVMs, Kraus/CPTP maps, entanglement

measures, Tsirelson's bound, and no-cloning. *(Also primary for §7; cross-listed there.)*

Landmark papers

A. M. Gleason, "Measures on the closed subspaces of a Hilbert space," *J. Math. Mech.* **6**, 885 (1957). `[ESTABLISHED]` — Derives the Born-rule form as the unique noncontextual probability measure on projections for $\dim \mathcal{H} \geq 3$.

S. Kochen and E. P. Specker, "The problem of hidden variables in quantum mechanics," *J. Math. Mech.* **17**, 59 (1967). `[ESTABLISHED]` — The Kochen–Specker theorem establishing quantum contextuality for $\dim \mathcal{H} \geq 3$: no noncontextual value assignment $v : \mathcal{P}(\mathcal{H}) \to \{0, 1\}$ exists.

J. F. Clauser, M. A. Horne, A. Shimony, R. A. Holt, "Proposed experiment to test local hidden-variable theories," *Phys. Rev. Lett.* **23**, 880 (1969). `[ESTABLISHED]` — The CHSH inequality $|\langle S \rangle| \leq 2$ classically (vs. Tsirelson's $2\sqrt{2}$ quantum bound); foundation of all subsequent Bell tests.

V. Gorini, A. Kossakowski, E. C. G. Sudarshan (1976) and **G. Lindblad**, "On the generators of quantum dynamical semigroups," *Commun. Math. Phys.* **48**, 119 (1976). `[ESTABLISHED]` — Independent derivations of the GKSL/Lindblad generator, the most general Markovian (CPTP-semigroup) open-system master equation.

P. Busch, "Quantum states and generalized observables: a simple proof of Gleason's theorem," *Phys. Rev. Lett.* **91**, 120403 (2003); and **C. M. Caves, C. A. Fuchs, K. Manne, J. M. Renes**, "Gleason-type derivations of the quantum probability rule for generalized measurements," *Found. Phys.* **34**, 193 (2004; arXiv:quant-ph/0306179). `[ESTABLISHED]` — The Gleason–Busch (effect/POVM) theorem, extending the Born-form uniqueness to $\dim \mathcal{H} = 2$ at the price of additivity over all compatible effects. *(Fourth author is Manne, not Schack.)*

Recent preprints (2026, not yet peer-reviewed)

E. O. Torres Alegre, "Causal Rigidity of Born-Type Probability Rules in Infinite-Dimensional Operational Theories," arXiv:2602.09056 (Pontifical Catholic Univ. of Chile, submitted 7 Feb 2026). [arXiv preprint — **verified real**, not peer-reviewed] — A **rigidity** theorem (Thm 1, "Causal Rigidity"): under (i) no-signaling, (ii) normal steering via σ-additive purification — an infinite-dimensional generalization of the Hughston–Jozsa–Wootters/GHJW theorem — and (iii) σ-affinity of probability assignments under countable preparation mixtures (plus implicit continuity/monotonicity of Φ), the post-processing map in $P = \Phi(\tau)$ must be the identity, making the Born rule the **unique** no-signaling-compatible probability assignment on **infinite-dimensional normal operational state spaces** (lifting the finite-dimensional Gleason–Busch form-rigidity). **Caveat:** §6 *presupposes* a von Neumann-algebra representation and then recovers $\tau = |\langle\phi|\psi\rangle|^2$ — it does NOT derive the Hilbert space or the algebraic scaffold, so it sharpens the Born-*form* rigidity rather than grounding the formalism (relocates, not dissolves; cf. the GPT-reconstruction verdict KA-5 in Working note: *Is the Born Rule Fundamental or Derivable, and Is Quantum Mechanics Exactly Linear?* (repo: `notes/2026-06-08-born-rule-and-quantum-linearity.md`)). Authors flag normal steering as "the strongest assumption." — *Note (iteration-4, web-confirmed):* the other 2026 preprint IDs cited in the Born-rule note are **all real** (promoted from `[unverified]`): arXiv:2603.06211 (J. Zhang, *Summing to Uncertainty* — additivity as the shared load-bearing premise across Born-rule derivations); arXiv:2601.18856 (V. Pronskikh, *Operationally induced preferred basis in unitary quantum mechanics*); arXiv:2601.13012 (A. K. Pati, *No-Signalling Fixes the Hilbert-Space Inner Product* — fixes the **inner product**, *not* linearity, which it assumes). No fabrication was found anywhere in the wiki's citations.

Iteration-4 verified additions

M. B. Fröb, "Modular Hamiltonian for de Sitter diamonds," *JHEP* **12** (2023) 074 (arXiv:2308.14797). `[ESTABLISHED]` — *Single-authored.* The dS-diamond modular Hamiltonian is the Casini–Huerta–Myers conformal-Killing-vector flow, geometric only in the conformal vacuum, reducing to the static-patch boost in the large-diamond limit. (Corrects an iteration-3 "Forste et al." misattribution.) Bears on the OP-41 dS first-law analysis (see §9, `HYPOTHESES.md` (in the repository)).

Muon $g-2$ Theory Initiative (WP2025), "The anomalous magnetic moment of the muon in the Standard Model" (arXiv:2505.21476, 2025). `[ESTABLISHED]` — Lattice-QCD HVP-based SM prediction $a_\mu^{\text{SM}} = 116\,592\,033(62) \times 10^{-11}$, $\Delta a_\mu = 38(63) \times 10^{-11}$ (SM-consistent); supersedes the WP2020 data-driven value for the $g-2$ anomaly status. See `OPEN_PROBLEMS.md` (in the repository) OP-21 and `domains/particle-physics.md` (in the repository).

Iteration-6 verified additions (2026-06-10)

All arXiv IDs below were re-resolved by the iteration-6 referees against arXiv/journal listings on 2026-06-10. Journal references flagged `[unverified]` rest on search-snippet or Springer-sighting provenance only and must not be promoted until a DOI is checked. Per EPISTEMICS.md §4, no citation here is invented; titles are quoted only where verified.

Closed-universe / dS holography (OP-50, HYP-dS-CARDINALITY-R1):

M. Usatyuk, Z. Wang, Y. Zhao, *SciPost Phys.* **17**, 051 (2024). `[ESTABLISHED, within 2D models]` — Including spacetime wormholes in the inner product degenerates the Gram matrix of perturbatively-distinct closed-universe states to rank 1 in JT gravity + matter and a topological model: a unique closed-universe state per theory. The proven core of the OP-50 debate.

D. Harlow, M. Usatyuk, Y. Zhao, arXiv:2501.02359 [JHEP ref **unverified** — Springer sighting only]. `[CONTESTED premise]` — The "observer rescue": an internal observer with S_{Ob} degrees of freedom is described by an effective QM on a Hilbert space of dim ~$e^{S_{Ob}}$ (gravitational-path-integral argument, not a theorem). Counter-positions: Antonini–Rath–Sasieta–Swingle–Vilar López (arXiv:2507.10649); H. Liu (arXiv:2509.14327); Engelhardt–Gesteau (arXiv:2504.14586).

J. Maldacena, arXiv:2412.14014 (2024). `[ESTABLISHED as the paper's claim]` — An observer mostly cancels the dimension-dependent phase i^{D+2} of the dS sphere partition function; the $A/4G$ coefficient still comes from the classical saddle (η input).

Algebraic frontier (OP-44, OP-48c):

K. Jensen, S. Raju, A. J. Speranza, arXiv:2412.21185; *JHEP* **06** (2025) 242. `[ESTABLISHED]` — The AdS boundary time-band algebra strictly has no commutant nonperturbatively; coarse-grained observer version reduces to the modular crossed product at leading order; uniqueness: the modular crossed product is the only crossed product of a III1 algebra with a trace.

J. De Vuyst, S. Eccles, P. A. Höhn, J. Kirklin, "Linearization (in)stabilities and crossed products," arXiv:2411.19931; *JHEP* (2025) 211. `[ESTABLISHED]` — Locates the order-G_N^2 obligation at the quadratic (boost) linearization-instability constraints; "essentially unambiguous" for spatially closed spacetimes. Companion: PW=CLPW + QRF-relative entropies, arXiv:2405.00114, *JHEP* (2025) 146.

M. S. Klinger, J. Kudler-Flam, G. Satishchandran, "Generalized Entropy is von Neumann Entropy II: The complete symmetry group and edge modes," arXiv:2601.07910 (2026, 104 pp). `[ESTABLISHED]` — Perturbative Type II structure survives the complete constraint family with boost-supertranslation edge modes; entropy = S_{gen} + edge-mode + constant term.

V. Chandrasekaran, É. É. Flanagan, "Subregion algebras in classical and quantum gravity," arXiv:2601.07915 (2026, 111 pp). `[ESTABLISHED]` — GSL for non-stationary linearized perturbations of Killing horizons; quantum focusing conjecture proven within the linearized/perturbative regime.

J. Kudler-Flam, E. Witten, "Emergent Mixed States for Baby Universes and Black Holes," arXiv:2510.06376. `[ESTABLISHED]` — Baby-universe state sequences fail to converge at large N; appropriate averaging over N yields mixed-state convergence with nontrivial commutants (the OP-44 "survives-after-averaging" option). Companion: Mellin averaging over N, arXiv:2605.15180.

A. Herderschee, J. Kudler-Flam, "Stringy algebras, stretched horizons, and quantum-connected wormholes," arXiv:2510.01556. `[ESTABLISHED]` — Hagedorn spectrum breaks the split property for regions within a string length; algebraic definition of stretched horizons; Type III ER=EPR factors.

H. Liu, "Lectures on entanglement, von Neumann algebras, and emergence of spacetime," arXiv:2510.07017 (2025, 119 pp). `[ESTABLISHED, review]` — The actual 2025 synthesis of the algebraic program (corrects the garbled "KLS-trio review" lead).

L. van Luijk, A. Stottmeister, H. Wilming, "Multipartite Embezzlement of Entanglement," *Quantum* **9**, 1818 (2025); arXiv:2409.07646. `[ESTABLISHED]` — A multipartite system of commuting III_1 factors on which every state is embezzling: the consistent non-Type-I composition endpoint in the OP-48(c) horn-C residue. Synthesis: van Luijk, arXiv:2510.07563 (type classification ↔ operational entanglement properties).

Pseudo-entropy / OP-49:

W.-z. Guo, S. He, Y.-X. Zhang, arXiv:2209.07308; *JHEP* **05** (2023) 021. `[ESTABLISHED]` — Reality of pseudo-(Rényi-)entropy trace moments is necessary-and-sufficiently characterized by η-pseudo-Hermiticity of the transition matrix (diagonalizable case), with a Tomita–Takesaki reality class. The reality theorem grounding the OP-49 near-no-go.

K. Doi, J. Harper, A. Mollabashi, T. Takayanagi, Y. Taki, *Phys. Rev. Lett.* **130**, 031601 (2023). `[ESTABLISHED]` — dS3/CFT2 pseudo-entropy: Re S = $S_{\mathrm{dS}}/2$, finite, region-independent; all region-dependence and divergence imaginary. (Source of the iter-5 Re/Im inversion correction.)

Li–Xiao–He–Sun, arXiv:2511.17098 (2025). [ESTABLISHED re paper content, equation-level verified; no verbatim cancellation quote — paraphrase only] — Timelike-EE first law in AdS/CFT ⇔ linearized Einstein equations; the imaginary part is a state- and interval-independent constant dropping out of variations — the unitary mirror of the dS case under $c \to ic$.

A. J. Parzygnat, T. Takayanagi, Y. Taki, Z. Wei, arXiv:2307.06531; *JHEP* **12** (2023) 123. `[ESTABLISHED]` — SVD entanglement entropy: the only known intrinsically real, non-negative two-state entropy; the OP-49 residual protocol (no first law exists as of 2026-06).

Phenomenology (watchlist):

E. Verlinde, K. Zurek, *JHEP* **04** (2020) 209; and *Phys. Rev. D* **106**, 106011 (2022). `[INFERENCE within AdS/CFT]` — Modular-fluctuation area law $\langle \Delta K^2 \rangle = \langle K \rangle = A/4G$ and its shockwave reproduction: the anchor of the geontropic chain (η input at the modular→metric step).

S. M. Vermeulen et al. (GQuEST), *Phys. Rev. X* **15**, 011034 (2025). `[ESTABLISHED design]` — Photon-counting Michelson design; 5σ on the pixellon benchmark α=1 in O(10^5) s; pixellon derivation from a UV-complete theory "still underway" (stated admission). Sibling bound: QUEST, *PRL* **135**, 101402 (2025), arXiv:2410.09175.

D. Carney, S. Karydas, A. Sivaramakrishnan, *Phys. Rev. D* **113**, 106002 (2026). `[ESTABLISHED]` — Minimal graviton-EFT null prediction: arm-length variance ~ℓ_p^2, no IR divergences; a VZ-magnitude detection would "signal a severe breakdown of effective quantum field theory in low energy quantum gravity" (verbatim verified). Confound: Freidel–Oberfrank, arXiv:2601.17849 (PSD depends on the graviton quantum state).

J. Oppenheim, "A postquantum theory of classical gravity?," arXiv:1811.03116; *Phys. Rev. X* **13**, 041040 (2023). `[ESTABLISHED framework / CONTESTED physics]` — CPTP classical–quantum gravity; stochastic metric + matter, no measurement postulate.

J. Oppenheim, C. Sparaciari, B. Šoda, Z. Weller-Davies, arXiv:2203.01982; *Nat. Commun.* **14**, 7910 (2023). `[ESTABLISHED, theorem within CQ]` — The decoherence–diffusion trade-off: coherence forces classical-sector diffusion observable as force noise.

T. Janse, D. G. Uitenbroek, J. van Everdingen, S. Plugge, B. Hensen, T. H. Oosterkamp, arXiv:2403.08912; *Phys. Rev. Research* **6**, 033076 (2024). `[ESTABLISHED]` — Force-noise compilation lowering the spacetime-diffusion bound ≥15 orders; ultralocal-continuous dead, ultralocal-discrete below the upper bound (with stated caveats), non-local continuous within ~1 order.

A. Biggs, J. Maldacena, arXiv:2405.02227. `[ESTABLISHED as published counter-reading]` — Ordinary finite-temperature matter reproduces the qualitative Danielson–Satishchandran–Wald horizon-decoherence effect: thermality, not horizon-ness, does the work (the DSW deflation). (Bibliographic fix from the A1 referee: arXiv:2505.17450 is a Lifshitz moving-mirror holographic analogue by Kawamoto–Lee–Yeh, NOT an independent reproduction of the DSW rate.)

V. Vilasini, R. Renner, *Phys. Rev. A* **110**, 022227 (2024); companion arXiv:2408.13387. `[ESTABLISHED]` — Indefinite-causal-order processes embedded in acyclic spacetime admit a definite, acyclic fine-grained causal-order explanation: process matrices do not generate causal order.

Generation-construction survey (OP-46 / HYP-CKV-VACUITY-R3):

S.-S. Lee, *JHEP* **06** (2020) 070; arXiv:1912.12291 (full text checked).
`[SPECULATIVE as physics; ESTABLISHED that the paper exhibits it]` — Matrix model in which signature is a state-dependent dynamical output (dS-like Lorentzian regions bridged by Euclidean ones) on a posited carrier + signature-parametrized hypersurface-deformation template. The nearest sketch to CONCLUSION §8's spec; fails it (selection-type).

F. Finster et al. (causal fermion systems) — program review arXiv:1709.04781 (full text checked); CUP textbook listing arXiv:2411.06450 (2024/25); status paper arXiv:2603.05018 (Farnsworth, Finster, Paganini, Singh — future-dated ID verified real).
`[ESTABLISHED definitions; INFERENCE on the η-identification]` — The $(\leq n, \leq n)$-eigenvalue axiom + indefinite spin scalar product as the relocated Krein η (fourth readout family); minimizer existence only finite-dimensional; all worked examples Dirac-sea encodes.

L. Rodina, *Phys. Rev. Lett.* **134**, 031601 (2025).
`[ESTABLISHED — with the binding hypotheses caveat]` — Hidden zeros ⟺ "subset" enhanced BCFW UV scaling; zeros uniquely fix $\mathrm{Tr}(\varphi^3)$ tree amplitudes **assuming an ordered local propagator structure and trivial numerators** (locality is a hypothesis, not derived — must not be entered as "locality as derived"). Companion conjecture line: Zhou, arXiv:2604.07195 (also assumes locality+unitarity).

O. Müller, "On the Hauptvermutung of Causal Set Theory," arXiv:2503.01719 (v2 Dec 2025; preprint, not peer-reviewed, proofs not independently checked). [web-confirmed real] — Two precise inequivalent formulations: one false, one true; countable-set version holds.

Experiment refresh:

LIGO–Virgo–KAGRA, GW250114 area-law companion: arXiv:2509.08054; *Phys. Rev. Lett.* **135**, 111403 (2025).
`[ESTABLISHED]` — Hawking's area law $A_f > A_1 + A_2$ confirmed

at high credibility (the correct citation — NOT the ringdown-spectroscopy paper 2509.08099, which carries the $(2, 2, 0)$ +overtone Kerr-consistency at tens-of-percent level; SNR 80 discovery / 76.9 catalog).

LHCb, $B^0 \to K^{*0}\mu^+\mu^-$ comprehensive amplitude analysis: arXiv:2512.18053; *PRL* 2026, DOI 10.1103/24g9-yn9d. `[ESTABLISHED measurement / CONTESTED interpretation]` — ~4σ SM tension [σ-figure from press coverage of the PRL] on full Run 1+2; the flagship surviving flavor anomaly (OP-21).

KiDS-Legacy, arXiv:2503.19441 (cosmic shear: $S_8 = 0.815^{+0.016}_{-0.021}$, 0.73σ from Planck) and arXiv:2503.19442 (joint analysis, Planck-consistent). `[ESTABLISHED]` — The S8 easing on the KiDS axis; DES Y6 remains in tension per arXiv:2602.12238 `[CONTESTED]`.

S. Pedalino, A. Ramírez-Galindo, S. Ferstl, K. Hornberger, M. Arndt, S. Gerlich (Vienna), arXiv:2507.21211; *Nature* **649**, 866 (2026). `[ESTABLISHED]` — Interference of sodium nanoclusters >7,000 atoms / >1.7×10^5 Da; macroscopicity μ=15.5 — the most stringent generic-macrorealism exclusion to date.

LHAASO, GRB 221009A LIV bounds: arXiv:2402.06009, *PRL* (v3 Feb 2026). `[ESTABLISHED]` — $E_{\mathrm{QG},1} > 10\,E_{\mathrm{Pl}}$, $E_{\mathrm{QG},2} > 6 \times 10^{-8}\,E_{\mathrm{Pl}}$.

`[unverified]` journal-reference ledger (binding; do not promote without a DOI check):

From R1 (Springer-sighting-only "JHEP" refs, six): Harlow–Usatyuk–Zhao 2501.02359; Higginbotham 2512.17993 (arXiv page: "Submitted to JHEP"); MSS 2506.18054; Nomura–Vafa-camp NV 2310.16994; ARTW 2511.04511; CCGPR 2303.16316.

From R3 (search-snippet JHEP issue/page refs, five): the colored-Yukawa loop-integrated paper arXiv:2406.04411 is now **promoted** — De–Pokraka–Skowronek–Spradlin–Volovich, *JHEP* **09** (2024) 160, DOI 10.1007/JHEP09(2024)160 (web-verified against OSTI/arXiv listings 2026-06-19); enter the remaining four flagged R3 journal refs as arXiv-ID-only with

`[unverified]` journal fields. Attribution fix: the gravituhedron paper arXiv:2012.15780 is single-author **Trnka**.

Other riders: Nimmrichter et al. arXiv:2502.13821 "accepted PRL [journal ref **still unverified** — arXiv carries no journal-ref/DOI as of 2026-06-19; leave flagged]"; diamagnetic microchip trap arXiv:2411.02325 ["Accepted for publication in *Phys. Rev. A*" per arXiv, **no volume/page yet** — leave flagged]. **Promoted (web-verified 2026-06-19):** Gonzalo–Montero–Obied–Vafa arXiv:2209.09249 = *JHEP* **11** (2023) 109, DOI 10.1007/JHEP11(2023)109 (ADS 2023JHEP...11..109G); the Barrett–Crane entry **PRD 111, 026014 (2025)** is Dekhil–Jercher–Pithis, *"Phase transitions in tensorial group field theory: Landau-Ginzburg analysis of the causally complete Lorentzian Barrett-Crane model,"* DOI 10.1103/PhysRevD.111.026014. **Over-attribution removed:** the "$\sim 10^{-13}$ d displacement" figure is **not** in arXiv:2604.16276 (Xue et al., verified real) — its abstract/body argue the Aziz–Howl effect is "essentially classical" / "negligibly small"; the unsourced number has been struck wiki-wide. Bibliographic fixes carried from referees: Sharmila et al. is *Nat. Commun.* **17**, 701 (2026); Bub et al. arXiv:2408.11094 is published as *PRL* **134**, 121501 (2025).

Iteration-7 verified additions (2026-06-10)

All references below were re-verified live by the iteration-7 referees (arXiv/journal pages fetched; verbatim quotes checked against primary PDFs where load-bearing). No `[unverified]` riders were introduced by this iteration.

Modular theory ↔ Krein structures (A1, the verdict-flipper track):

H. Gottschalk, "Die Momente gefalteter Gauß-Maße und ihre Anwendung in der Quantenfeldtheorie" / related Krein-space QFT work, *J. Math. Phys.* **43**, 4753 (2002). `[ESTABLISHED]` — Prior art running **Krein → modular**: modular structure built atop a posited indefinite-metric space; the reverse direction

(modular → Krein) is absent from the literature (the A1 direction-map finding).

L. Jakóbczyk, on Tomita–Takesaki theory in indefinite-metric (Krein-space) QFT, 1985 (OSTI 5135323). `[ESTABLISHED]` — Same direction: Krein structure assumed, modular objects derived.

A. Devastato, S. Farnsworth, F. Lizzi, P. Martinetti, "Lorentz signature and twisted spectral triples," *JHEP* **03** (2018) 089 (arXiv:1710.04965). `[ESTABLISHED]` — The twisted-spectral-triple route: the Krein/η datum installed via the twist (author list corrected by the iteration-7 referee from a truncated form).

Crossed products / constraints (A2):

J. Kirklin, all-orders generalized second law, arXiv:2412.01903; *JHEP* **07** (2025) 192. `[ESTABLISHED]` — The nearest existing discharge machinery for higher-order constraint imposition; does not execute the dS closed-slice cubic step.

S. Ali Ahmad, R. Jefferson (AAJ), perturbative crossed products beyond subleading order, arXiv:2501.01487. `[ESTABLISHED]` — First controlled step past leading order (carried from iteration 6; re-anchored here for the A2 descent argument).

Closed-universe / OP-50 watch (A4):

D. Harlow, M. Usatyuk, Y. Zhao, *JHEP* **02** (2026) 108 (arXiv:2501.02359). `[CONTESTED premise]` — Now journal-published; upgrades the iteration-6 `[unverified]` JHEP flag for this entry.

K. Higginbotham, observer complementarity, *JHEP* **03** (2026) 183 (arXiv:2512.17993). `[CONTESTED]` — Now journal-published; upgrades the iteration-6 `[unverified]` flag.

Y. Nomura, T. Ugajin, partial-observability follow-up, arXiv:2602.13387. `[ESTABLISHED as claims]`

Miyata–Horikoshi–Tajima–Ugajin, arXiv:2606.04575. `[ESTABLISHED as the paper's construction]` — The third OP-

50 position: a pre-gauge-constraint hyperfinite II_1 dominant sector with a post-constraint polynomial-dimension physical Hilbert space (scope per the iteration-7 referee).

Distinguishing-test-hunt verified additions (2026-06-10)

The 2024–2026 experimental/phenomenology sweep + the reverse-Weinberg–Witten / net-invariant analysis. Every arXiv ID below resolves to the stated authors and title; decisive primary-source quotes verified verbatim where load-bearing.

The verified lead (mechanism-blind taxonomy — positive ENCODING-SCREEN evidence):

B. Sharmila, S. M. Vermeulen, A. Datta, "Signatures of correlation of spacetime fluctuations in laser interferometers," *Nat. Commun.* **17**, 701 (2026) (online 23 Dec 2025); arXiv:2505.22892. `[ESTABLISHED]` — Three-class taxonomy of spacetime-fluctuation two-point functions: class (a) factorised (Oppenheim CQ, CSL, Diósi–Penrose), (b) inverse/polynomial (Károlyházy, Verlinde–Zurek geontropic, EFTs), (c) exponential (entanglement-holographic/mesoscopic). Load-bearing verbatim quote confirmed present: *"Our approach is agnostic to the microscopic origins of the SFs."* The model→class map is many-to-one (fundamental and emergent models co-classed), so the taxonomy classifies *gravity models*, not fundamental-vs-emergent ontology — a published worked instance of HYP-ENCODING-SCREEN Corollary 2. (Bibliographic note already on record in the [unverified]-ledger fixes; entered here as a standalone source.)

Author-side concessions of observational degeneracy (independent screen confirmations):

Sidan A, T. Banks, holographic-spacetime inflation / CMB, arXiv:2502.15108 (2025); *Phys. Rev. D* **112** (2025) [volume per submission, not independently re-checked this session]. `[ESTABLISHED re content]` — Inflationary scalar+tensor predictions *"fit extant data equally well"* (verbatim) as field-

theoretic inflation; the differences are *"radically different"* conceptually (no transplanckian problem), not observational. **Quotation-hygiene correction (binding):** the string "No claim is made that the CMB could observationally distinguish emergent/holographic spacetime from conventional metric inflation" is a PARAPHRASE, not a verbatim quote — record it as such (EPISTEMICS §4: no fabricated quotes).

J. Erlich, stochastic composite gravity (GRF 2025 essay), arXiv:2504.03084 (2025). `[ESTABLISHED]` — Predictions (BH entropy; low-ℓ CMB modification; a stochastic-vs-QFT out-of-equilibrium disagreement) are explicitly portable: the low-ℓ effect is *"not unique."* Nuance on record (not verdict-changing): the out-of-equilibrium stochastic-vs-QFT test is framed as a positive experimental test of the framework, but it bears on exact linearity of QM (the gate / objective-collapse axis), NOT on geometry-priority — so it is not a reverse-WW.

Other 2024–2026 sweep channels (all screened — dead/gate/member-level for the wager):

R. Schützhold, "Stimulated Emission or Absorption of Gravitons by Light," *Phys. Rev. Lett.* (2025), HZDR. `[ESTABLISHED, verified]` — ~10^6 km effective-path interferometer probing stimulated graviton absorption/emission; tests the radiative-sector **gate** (is the gravitational field quantized / do gravitons exist), not derivational priority. Self-stated: *"would not directly prove the existence of gravitons ... strong supporting evidence."*

L. Aalsma, S.-E. Bak, "Modular Fluctuations in Cosmology," arXiv:2503.04886; *Phys. Rev. D* **112**, 026017 (2025) [journal ref carried from iteration 6, arXiv identity + quote re-verified]. `[ESTABLISHED]` — Causal-diamond modular fluctuations → near-scale-invariant CMB spectrum, requires quantizing an s-wave mode by hand; model-agnostic. Verbatim: *"whether such a mode exists in the vacuum of any theory of quantum gravity remains an open question."*

G. Calcagni, L. Modesto, primordial GWs from Weyl-invariant QG, arXiv:2206.07066; *JHEP* **12** (2024) 024. `[ESTABLISHED]` — $r_{0.05} \approx 0.01$ (BICEP/LiteBIRD) + blue-tilted SGWB (DECIGO) from a single UV-complete construction; a member-level parameter prediction, not class-level — the quasi-scale-invariant spectrum is also produced by fundamental-metric quantized gravity.

K. Hashimoto, K. Matsuo, K. Murata, G. Ogiwara, D. Takeda, ML emergent-spacetime from linear response, arXiv:2411.16052. `[ESTABLISHED]` — A neural net *assuming AdS/CFT* reconstructs a bulk metric as *"interpretable weights"*; emergence is presupposed by training on the holographic dictionary, not tested (the encode-only architecture made literal).

D. Jafferis et al., traversable-wormhole quantum simulation (Google Sycamore), *Nature* **612**, 51 (2022), DOI 10.1038/s41586-022-05424-3; criticism **G. Kobrin, T. Schuster, N. Y. Yao**, arXiv:2302.07897. `[ESTABLISHED]` — The gravitational dual is an interpretive dictionary over an engineered system; a simulation cannot distinguish emergence-in-nature from simulation by construction.

Reverse-Weinberg–Witten / net-invariant analysis (H2 track):

C. Casini, J. M. Magán, "A generalization of the DHR theorem for higher form symmetries," arXiv:2511.21810 (submitted 26 Nov 2025). `[ESTABLISHED, verified live]` — Generalized order/disorder parameters in $D > 2$ (Wilson/'t Hooft loops) are labeled by groups attached to Haag-duality-violation data of the net — i.e. the global form / line-operator spectrum is a **net (DHR/HDV) invariant**, hence EFT-shared between algebra-first and geometry-first descriptions (load-bearing only under the net-sharing assumption A1).

C. D. Carone, T. V. B. Claringbold, D. Vaman, "Composite graviton self-interactions in a model of emergent gravity," arXiv:1710.09367. [ESTABLISHED — author list and physics confirmed this session; corrects a prior author-less citation] — A worked Weinberg–Witten evasion (composite massless spin-2 via emergent gravity); the evasion is symmetric across both theory classes, so it supplies no class-separating observable.

J. J. Heckman, M. Hübner, C. Murdia, "On the Holographic Dual of a Topological Symmetry Operator," *Phys. Rev. D* **110**, 046007 (2024); arXiv:2401.09538. `[ESTABLISHED]` — Supports "no global p-form symmetry in AdS" and the SymTFT-inside-holography picture; a no-global-symmetry result under A1/holography does NOT upgrade reverse-WW's literature-absence to a structural impossibility.

A. Parnachev, N. Poovuttikul, "Topological Entanglement Entropy, Ground State Degeneracy and Holography," arXiv:1504.08244 (2015). `[ESTABLISHED]` — TEE ↔ GSD ↔ holography; TEE $\gamma = \log \mathcal{D}$ is the total quantum dimension of the DHR superselection category, a net+state invariant — diagnoses topological order, not derivational order.

B. Czech, J. de Boer, D. Espíndola, B. Najian, J. van der Heijden, J. Zukowski, "Changing states in holography: From modular Berry curvature to the bulk symplectic form," arXiv:2305.16384; and **W.-z. Huang, M.-z. Ma**, "Emergence of Kinematic Space from Quantum Modular Geometric Tensor," arXiv:2110.08703. `[ESTABLISHED]` — Modular Berry / kinematic-space data are net+state invariants (Tomita–Takesaki–built); the one emergent Lorentzian readout (Huang–Ma dS2) inherits its signature from the symmetric vacuum (iter-5 closure).

*(Vilasini–Renner, PRA **110**, 022227 (2024), already on record at BIBLIOGRAPHY line 85 — indefinite causal order reduces to definite acyclic order — is reused by the H2 track; no new entry needed.)*

`[unverified]` **riders introduced by this hunt (do not promote without a DOI/page check):**

Banks–Sidan A *Phys. Rev. D* **112** volume number (arXiv identity + the "fit extant data equally well" quote ARE verified; the journal volume is not independently re-checked this session).

Schützhold PRL (2025) volume/page (APS Physics + HZDR coverage confirm the result; the exact PRL coordinate is not pinned this session).

Delhom–Giacomelli analog-gravity lecture notes arXiv:2512.14209 and the Steinhauer-line analog-Hawking PRL **126**, 111301 (2021) — cited in the analog-gravity dead-by-construction note; loose but immaterial (the verdict does not depend on either resolving precisely).

Iteration-8 verified additions (2026-06-10)

All re-verified live by the iteration-8 referees (arXiv/journal pages fetched; load-bearing quotes checked against primary PDFs). No `[unverified]` *riders introduced.*

Net/family carrier no-go (A1):

Y. Lashkari (M. Lashkari), C. Leung, M. Moosa, A. Ouseph (as listed), "...local representation of the boundary algebras / AdS_2 emergence," *JHEP* **10** (2025) 153; arXiv (verified). `[ESTABLISHED]` — Verbatim: *"AdS_2 emerges from insisting on a local representation of the boundary algebras that satisfy Haag's duality"* — localization is an imposed input, the load-bearing "net-index-set IS the geometry" datum.

V. Morinelli, K.-H. Neeb, G. Ólafsson, Euler-element / Bisognano–Wichmann for standard-subspace nets. `[ESTABLISHED]` — The Euler element (wedge localization) is an *input*, not derived from modular data.

(Borchers–Buchholz–Longo positivity⇔isotony, and the chiral standard-pair machinery, already on record.)

OP-44 R1/R2 (A2):

P. Hintz, "...solvability / linearization," arXiv:2306.07715. `[ESTABLISHED]` — Thm 1.1 solvability for smooth, divergence-free, spatially-compact sources; Remark 1.2 extends the *linear* theorem to $\Lambda > 0$ "with purely notational modifications" (verbatim, verified).

A. Pound, second-order self-force / puncture regularization, arXiv:1506.06245. `[ESTABLISHED]` — $\delta^2 G[h_1, h_1] \sim 1/r^4$ non-integrable; no solution to the fully nonlinear point-particle-sourced equation; the puncture/effective-source scheme.

(KLS-II, arXiv:2601.07910, already on record — edge modes "address only second-order perturbations.")

OP-49 junction (A3):

C. Bachas, I. Brunner, D. Roggenkamp, conformal-interface transmissivity (transmissive-defect criterion $T^1 = T^2$). `[ESTABLISHED]`

J. Fuchs, M. R. Gaberdiel, I. Runkel, C. Schweigert, arXiv:0705.3129. `[ESTABLISHED]` — Topological-defect / orientation-reversal machinery (corrects a prior "Bachas et al." attribution of this ID).

A. Karch, Y. Kusuki, H. Ooguri, H.-Y. Sun, M. Wang, interface effective central charge / transmission-dependent log coefficient (SSA, c-theorem, positive c assumed — hence inapplicable at imaginary c). `[ESTABLISHED, unitary regime]`

OP-48c X1 (A4):

M. Takesaki, *On the direct product of W^*-factors*, Tôhoku Math. J. **10**, 116–119 (1958). `[ESTABLISHED]` — The normal-product-state ⇔ W^*-isomorphism theorem (Rédei–Summers Prop. 2): the obstruction that forces singular product states across a Type III1 Haag-dual pair to be **non-normal** ($\mathcal{M} \vee \mathcal{M}' = B(\mathcal{H})$ is Type I while $\mathcal{M} \mathbin{\overline{\otimes}} \mathcal{M}'$ is Type III). This is the non-normality obstruction — *not* Florig–Summers; the iteration-8 submission's joint "Takesaki/Florig–Summers" attribution is corrected here.

M. Florig, S. J. Summers, *On the statistical independence of algebras of observables*, J. Math. Phys. **38**, 1318–1328 (1997). `[ESTABLISHED]` — The C*-side **faithful**-product-state ⇔ C*-independence-in-the-product-sense characterization (Rédei–Summers Prop. 4 = ref [14]); a separate, stronger result, *not* the non-normality obstruction and not load-bearing for X1-existence.

H. Roos, *Independence of local algebras in quantum field theory*, Commun. Math. Phys. **16**, 238–246 (1970). `[ESTABLISHED]` — On commuting C*-algebras the C*-independence extension state may be chosen a **product state** (the existence input for X1, via the Rédei–Summers post-Def-1 sentence). **Plain** C*-independence suffices — no product-sense isomorphism needed (Rédei–Summers Prop. 1: plain ⇏ product-sense in general).

M. Rédei, S. J. Summers, *When are quantum systems operationally independent?*, arXiv:0810.5294 (2009). `[ESTABLISHED]` — Assembles the Roos / Takesaki / Florig–Summers independence results (Props 1/2/4); the primary carrier for the X1 existence theorem. **H. Halvorson** (math-ph/0602036) supplies the Schlieder ⟺ C*-independence implication box (Def 3.4 is the *definition* of Schlieder, not the proof of Step 1).

The **central-carrier fact** — every nonzero projection in a factor has central carrier I — is **Kadison–Ringrose**, *Fundamentals of the Theory of Operator Algebras* (1983/1986); it (not Halvorson Def 3.4) is the proof of Step 1, factor ⟹ Schlieder.

Iteration-9 verified additions (2026-06-10)

Referee-audited live this session; no `[unverified]` riders introduced.

Carrier naturality + reverse-WW (A1, A3):

Z. Weiner, "An algebraic version of Haag's theorem," arXiv:1006.4726. `[ESTABLISHED]` — Under the split property + geometric modular action, the unitary-equivalence class of the vacuum-sector net of double-cone algebras on a fixed Cauchy surface **completely determines the AQFT model**: the load-bearing "net + vacuum is a complete invariant" theorem behind the reverse-WW ⟺ carrier-problem identification.

D. Buchholz, O. Dreyer, M. Florig, S. J. Summers, "Geometric modular action and spacetime symmetry groups," arXiv:math-ph/9805026. `[ESTABLISHED]` — GMA: the geometry is a functor of the modular data (the constructibility-naturality input; "Buchholz–Summers" is incomplete shorthand for this four-author paper).

S. J. Summers, R. White, arXiv:hep-th/0304179. `[ESTABLISHED]` — GMA reconstruction of the spacetime symmetry group from modular data.

D. Bisch, arXiv:math/0304340. `[ESTABLISHED]` — Subfactor / planar-algebra standard-invariant reference (attribution corrected: **Bisch, not Jones**).

K.-H. Neeb, G. Ólafsson, arXiv:1704.01336. `[ESTABLISHED]` — Antiunitary representations / Euler element as the localization datum (the "fiber bundles over ordered symmetric spaces" reading: geometry in the base, no signature in the fibers).

arXiv:2312.12182 — partial-converse Euler-element generation (Euler element *derived* from net + Bisognano–Wichmann + regularity, which already encode the causal/Lie structure). `[ESTABLISHED]`

Modular-Berry holonomy (A2):

B. Czech et al., arXiv:2305.16384 (modular Berry curvature ↔ bulk symplectic form); **W.-z. Huang, M.-z. Ma**, arXiv:2110.08703 (kinematic space from the modular geometric tensor — the Lorentzian dS_2 metric, vacuum-inherited); supporting: arXiv:1812.02176, arXiv:2505.04682, arXiv:2111.05345. `[ESTABLISHED]` — Establish the modular-Berry curvature as the signature-free symplectic Crofton form and the kinematic-space metric's signature as symmetry-inherited.

Iteration-10 verified additions (2026-06-10)

Connes, A.; Takesaki, M. — "The flow of weights on factors of type III." *Tôhoku Mathematical Journal* **29** (1977), 473–575. — Cited for: the modular centralizer of an integrable weight on a Type III_1 factor is a Type II_∞ factor (trivial flow of weights ⟺

III1), so the commutant-to-generated gap is real and wide for degenerate modular spectrum (iteration-10 A1-F1).

Popa, S. — "On a class of type II_1 factors with Betti numbers invariants." arXiv:math/0011084. — Cited for: a free-product/Popa-type realization in which any prescribed finite von Neumann algebra (N, τ) occurs as the centralizer of a state on a Type III1 factor (the centralizer can be a Type II_1 factor or any prescribed large algebra). *Replaces the digest-flagged mis-citation arXiv:1804.05706 for this point* (iteration-10 A1-F1).

Shlyakhtenko, D. — free-quasi-free-states structure theory of free Araki–Woods factors. — Cited for: the centralizer-triviality fact for Lebesgue-spectrum (geometric/BW) modular flow — $\mathcal{M} \cap \{\Delta^{it}\}' = \mathbb{C}1$ when Δ has no eigenvectors. This, not Connes–Størmer / Houdayer / Boutonnet–Houdayer, is the source of the biconditional (iteration-10 A2-F2 referee correction). *(Attribution per the iteration-10 referee; pin the precise Shlyakhtenko reference — "Free quasi-free states," Pacific J. Math. 177 (1997) 329–368 — on the next bibliography pass if confirming the exact paper.)*

Houdayer, C. — bicentralizer triviality for type III1 factors. arXiv:0809.3827. — Cited as a *corroborating* (not biconditional) result: proves the *bicentralizer* trivial, distinct from the centralizer-triviality used in A2-F2. Attribution-precision entry to forestall the same over-attribution the iteration-8/9 referees flagged.

Boutonnet, R.; Houdayer, C. — amenable modular-invariant subalgebras lie in the almost-periodic summand. arXiv:1602.01741. — Cited as *corroborating* the fork in A2-F2 (amenable modular-invariant subalgebras / almost-periodic summand), again explicitly NOT the source of the centralizer biconditional.

Iteration-11 verified additions (2026-06-10)

Operator algebras — Connes' bicentralizer problem (the named open problem the hedge re-grounds on):

Haagerup, U. — "Connes' bicentralizer problem and uniqueness of the injective factor of type III_1." *Acta Math.* **158** (1987) 95–148. — The biconditional linking bicentralizer-triviality to the existence of a state with an irreducible (large) modular centralizer. Cited for the genuine content of the iteration-11 A1 re-grounding (NOT for a reduction of the carrier no-go).

Marrakchi, A. — "Kadison's similarity problem and the bicentralizer conjecture" (arXiv:2308.15163). — Solves Kadison's problem *only for* type III_1 factors that *satisfy* the bicentralizer conjecture; treats the conjecture itself as OPEN. Load-bearing source for "Connes' bicentralizer problem is open in general as of 2026" — the correction to the mission's false HMT-resolved premise.

Marrakchi, A. & Vaes, S. — *Crelle* (J. reine angew. Math.) **809** (2024) 247. — Theorem A: the modular centralizer $M_\psi = \mathbb{C}1$ is achievable on *every* type III_1 factor, **unconditionally** (does not invoke the bicentralizer conjecture). The fact that breaks the submitted logical-equivalence reduction (killing a carrier needs trivial centralizer, achievable for free; the bicentralizer condition is the opposite).

Ando, H., Haagerup, U., Houdayer, C. & Marrakchi, A. — "Structure of bicentralizer algebras and inclusions of type III factors." *Math. Ann.* **376** (2020) 1145–1194 (arXiv:1804.05706). — The relative bicentralizer $BC_\varphi(\mathcal{N} \subset \mathcal{M})$ of an inclusion with expectation. Cited only as the *suggestive structural connection* for subregion-localized carriers (tagged SPECULATIVE — not an established reduction).

Holography of information and its failure mode (A2):

Geng, H., Jafferis, D. L., Shrivastava, A. & Tata, S. — "The Fate of Information Localizability and Holography in Quantum Gravity" (arXiv:2512.18912, 21 Dec 2025). — Keystone new datum. Verbatim-verified: *"holography cannot be proven at the perturbative level in Newton's constant G_N via the non-localizability of information,"* via *"the emergence of a perturbatively localized observer."* The failure mechanism is re-localization via dressing to a nontrivial background / emergent observer-clock — co-locating holographic completeness with the carrier gap.

Laddha, A., Prabhu, S., Raju, S. & Shrivastava, P. — holography of information at null infinity (arXiv:2002.02448). — Massless information localized in an infinitesimal neighbourhood of the past boundary of future null infinity. (Self-correction from the digest: fourth author is Shrivastava, not "Sasmal.")

Raju, S. — holography of information in AdS / Wheeler–DeWitt (arXiv:2107.14802). — Two states agreeing at the boundary for an infinitesimal time agree everywhere in the bulk (perturbative in G_N).

Chakraborty et al. / de Sitter holography of information (arXiv:2303.16316). — Cosmological correlators in an arbitrarily small region completely specify the state. (Cited for the de Sitter instance of the conditional state-completeness result.)

Operational / Tsirelson (A2):

Ji, Z., Natarajan, A., Vidick, T., Wright, J. & Yuen, H. — "MIP*=RE" (arXiv:2001.04383). — $C_{qa} \subsetneq C_{qc}$ (negative answer to Tsirelson's problem; Connes embedding refuted). Cited for: operational correlation data does not pin down the operator algebra — but the underdetermination is in the split/tensor (Type-I) structure (the OP-48 collar), NOT in net-completeness.

(T-duality/mirror symmetry in A2-F3 rests on standard textbook/established results — exact CFT isomorphism between geometrically distinct targets — and on the nLab "no complete construction of these SCFTs" status note for the full mirror-SCFT-isomorphism caveat; no new primary arXiv citation is load-bearing there, so none is added beyond the inline note.)

Iteration-12 verified additions (2026-06-19)

Operator algebras (carrier-lever / bicentralizer).

Ando, Haagerup, Houdayer, Marrakchi, *Structure of bicentralizer algebras and inclusions of type III factors*, Math. Ann. 376 (2020) 1145-1194 (arXiv:1804.05706). [VERIFIED] — defines/studies the **relative** bicentralizer of inclusions; its relative-bicentralizer theorem is restricted to **discrete** inclusions with N amenable (the discrete-only scope is the load-bearing fact for B1; physics local inclusions are non-discrete). Already in the standing iter-11 corpus; re-cited here for the discrete-only scope.

Marrakchi, Vaes, *Ergodic states on type III_1 factors and ergodic actions*, Crelle (J. reine angew. Math.) 809 (2024) 247 (arXiv:2305.14217). [VERIFIED verbatim] — Theorem A: ergodic states (trivial centralizer M_psi = C1) form a dense G_delta on **any** III_1 factor, **UNCONDITIONAL**, no bicentralizer hypothesis. (Corrects the iter-11 A1 misattribution that called Thm A "conditional on the bicentralizer conjecture.")

Haagerup, *Connes' bicentralizer problem and uniqueness of the injective factor of type III_1*, Acta Math. 158 (1987) 95-148. [VERIFIED] — the large-centralizer biconditional (trivial bicentralizer <=> exists a faithful normal state with (M_psi)' cap M = C1) — the carrier-admitting direction. Already in the standing corpus.

Ando, Goldbring, *(model-theory / Connes bicentralizer)*, arXiv:2605.12776 (May 2026). [VERIFIED] — re-confirms Connes' bicentralizer problem is OPEN in general as of 2026. (Authors are Ando-Goldbring, not "I. Goldbring et al.")

AQFT / encoding-screen (external-math sweep).

Morinelli, Neeb, Olafsson, *(Euler-element / geometric Bisognano-Wichmann / Spin-Statistics program)*, arXiv:2508.10960; and Part II arXiv:2603.26390. [VERIFIED by abstract read] — BW / Spin-Statistics derived FROM the Euler element (symmetry is INPUT); a cleaner statement of the encoding screen, NOT a carrier, NOT a reverse-WW inversion, NOT a sixth route.

Morinelli (single-author), arXiv:2412.20410. [VERIFIED — single-author] — Euler-program; part of the same encoding-screen family.

arXiv:2511.09360 (Euler-program). [VERIFIED real per sweep] — per-paper authorship to be confirmed before bibliography insertion (not all three-author); mark [authorship unverified] until checked.

Liu, *(lecture notes, modular/AQFT)*, arXiv:2510.07017. [VERIFIED real] — presupposes geometry as input; fails the encode-not-generate screen.

Naaijkens, Penneys, Wallick, arXiv:2605.10693. [VERIFIED real] — LTO-internal Euclidean-side reflection-positivity axiom + Tomita-Takesaki for Haag duality on cone regions; ORTHOGONAL — constructs no Lorentzian metric, no indefinite pairing, no carrier witness; makes NO claim that J carries the Euclidean datum.

Weiner, *(algebraic Haag theorem),* arXiv:1006.4726. [VERIFIED verbatim, standing corpus] — split + geometric modular action => the vacuum-sector net is a complete invariant (conditional on the geometric inputs).

Sheikh-Jabbari, Taghiloo, *"GR from RG"*, arXiv:2602.11806. [VERIFIED real] — RG-induced Einstein-Hilbert term with a UV-Dirichlet-fixed metric; explicitly engages Weinberg-Witten; an EVADE-not-invert WW instance (discharges the iter-11 "WW gloss not confirmable" minor defect).

Experiment-watchlist (target/core, not wager).

VIP / XENONnT objective-collapse bounds, primary arXiv:2506.05507 (PRL, March 2026). [VERIFIED real] — CSL ~2 orders, DP ~5x; tighten the A-3 core. Adjacent/auxiliary: arXiv:2512.15393, arXiv:2604.21705 [both VERIFIED real].

Huang, Cai, Wang, *(low-z SN calibration-bias analysis for DESI w(z)),* arXiv:2502.04212. [VERIFIED — abstract states dynamical-dark-energy preference "less than 2 sigma"] — Efstathiou in acknowledgments only (not an author).

Iteration-14 verified additions (2026-06-19)

From the maximal adversarial assault (the strongest 2024–2026 falsification attempts on the three pillars). The four load-bearing entries below were independently web-verified this session; the remaining encoding-screen instances surfaced by the assault are catalogued in the iteration-14 D-track notes (all referee-verified live).

Fullwood, J., Vedral, V. & Guzmán-González, E. — "On Lorentzian symmetries of quantum information." arXiv:2604.07471 (Apr 2026). [VERIFIED] — Derives the Minkowski form and $SO^{+}(1,3)$ from "preservation of linear entropy" on qubit observables (linear entropy $S_L = 2\,\det$; det = the SL(2,$\mathbb{C}$)-invariant on $\mathrm{Herm}_2(\mathbb{C})$). The strongest 2026 "signature-as-output" claim — but the (1,3) signature is intrinsic to the *complex* carrier (the determinant on 2×2 complex-Hermitian matrices has signature exactly (1,3), unconditionally), with no locality/net/causal order. Recorded as a 2026 instance of HYP-ENCODING-SCREEN (the indefinite-pairing-in-carrier smuggle), **not** a carrier and **not** a sixth route.

Schneider, M. D. — "Spacetime emergence: an (in)effective story." arXiv:2410.00932 (Philosophy of Physics, 2024). [VERIFIED] — Argues GR's spacetime "irreducibly include[s] global physical content, which is not effective" and that this global content "acts like a superselection rule that must be independently specified." An *external* (philosophy-of-physics) corroboration that locates the encode-screen's sole structural seam — the global/topological net-sharing premise — exactly where this wiki names its own soft spot; constructs no distinguishing observable.

Chou, K.-H. & Chang, P.-Y. — "Emergent de Sitter Space and Non-Unitary Tensor Networks from Non-Hermitian Quantum Criticality." arXiv:2606.17983 (Jun 2026). [VERIFIED] — A non-unitary cMERA built on a *pre-selected* non-Hermitian critical fermion chain; the de Sitter output emerges relative to the installed criticality (the coefficient is reproduced, not

derived). Confirms the iteration-12 closure that the indefinite-metric / Krein route is route-5 in Krein vocabulary, not a sixth route.

Verlinde, E. & Zurek, K. (GQuEST / geontropic): "Modular fluctuations from shockwave geometry" / GQuEST design, arXiv:2404.07524 (PRX 15, 011034 (2025)); EFT baseline "Response of interferometers to the vacuum of quantum gravity," arXiv:2409.03894. [VERIFIED] — The geontropic interferometer signal; its proponents concede "conformal field theory, dilaton theory, and hydrodynamics" yield the same $\alpha = O(1)$ amplitude, so it co-classes emergent and fundamental descriptions (screen-passing) and its amplitude re-imports $\eta = 1/4G$ at the modular→metric step. The freshest worked instance of the §7 no-test corollary.

Iteration-15 verified additions (2026-06-19) — the un-tried mathematical frameworks

From the final analytical iteration (factorization algebras / functorial QFT; index theory / K-theory; higher-categorical AQFT) — all reproduce the encode-not-generate screen. The Krein-spectral-triple entry was independently web-verified this session; the framework references are referee-verified, with attributions corrected per the referees.

Bizi, N., Brouder, C. & Besnard, F. — "Space and time dimensions of algebras with applications to Lorentzian noncommutative geometry and quantum electrodynamics." arXiv:1611.07062. [VERIFIED] — Defines indefinite (pseudo-Riemannian) spectral triples "over Krein spaces instead of Hilbert spaces." The index-theory confirmation of the iteration-12 Krein closure: a Lorentzian spectral triple installs a Krein space whose indefinite inner product **is** the (n, n) carrier η — installed, not generated.

Benini, M., Carmona, V., Grant-Stuart, A. & Schenkel, A. — the categorical equivalence AQFT $\simeq$ time-orderable prefactorization algebra on globally hyperbolic Lorentzian manifolds (arXiv:2412.07318; building on arXiv:1903.03396).

[referee-verified] — A new precise object but **not** a handle: both sides are functors over the same causal-disjointness / orthogonality relation (= the localization template n_1). A fourth independent witness to the same obstruction.

Gwilliam, O. & Rejzner, K. — "The observables of a perturbative algebraic quantum field theory form a factorization algebra." arXiv:2212.08175. [referee-verified] — pAQFT observables form a factorization algebra on a *fixed* Lorentzian manifold (the geometry is the base).

Bunk, S., MacManus, J. & Schenkel, A. — "Lorentzian bordisms in algebraic quantum field theory." arXiv:2308.01026 (Lett. Math. Phys. 115 (2025)). [referee-verified] — the Lorentzian bordism *site* for functorial AQFT (geometry in the site).

Müller, L. & Stehouwer, L. — reflection / dagger structures inducing an indefinite signature in (non-unitary) categorical field theory. arXiv:2301.06664. [referee-verified] — the categorical-vocabulary form of the iteration-12 installed-Krein problem.

Watch-sweep #1 verified additions (2026-07-02) — post-saturation watch-mode

From the first watch-mode sweep (window 2026-06-15 → 2026-07-02; Working note: Watch-mode sweep #1 — external-input channels, window 2026-06-15 → 2026-07-02 (repo: `notes/2026-07-02-watch-sweep-01.md`)). Zero referee-surviving strong findings; these entries are watch-channel records, not verdict movers. All web-verified live this session (un-refereed v1 preprints unless noted).

Marrakchi, A. — "Ergodicity of the bicentralizer flow and Kadison's problem." arXiv:2606.23636 (v1 2026-06-22, math.OA). [VERIFIED] — Unconditional: the relative bicentralizer flow of a type III_1 irreducible subfactor **with expectation** is always ergodic; consequence: such subfactors contain a maximal abelian subalgebra, completing Kadison's 1967 problem. NOT bicentralizer triviality (Connes' problem stays OPEN); no carrier content (output is abelian/definite); does not transfer to the expectation-free physics-net inclusions (the iter-12 blocker). Watch value: the tracked adjacent cluster is actively moving.

Bateman, S. & Turok, N. — "Escape from Ostrogradsky via Hidden Ghost Parity." arXiv:2607.00096 (v1 2026-06-30, hep-th). [VERIFIED] — Four-derivative scalar QFT on a Krein space with a Krein Born rule and tree-level positivity via a "hidden ghost parity" κ from an O(1,1) embedding; the paper itself: "no preferred choice of ghost parity." A new instance of the iteration-12 Krein closure, channel (B) installed-not-derived (n1 = fixed Minkowski commutator; κ global and non-unique); useful probabilistic-consistency machinery for indefinite carriers, no naturality content.

Wakakuwa, E., Petruzziello, L., Lantaño, T. B., Huelga, S. F. & Plenio, M. B. — "Relativistic Gravity-Induced Entanglement via Frame Dragging." arXiv:2606.31678 (v1 2026-06-30). [VERIFIED] — First genuinely post-Newtonian GIE channel (Lense–Thirring), derived two independent ways (quantized frame-dragging Hamiltonian; on-shell linearized-QG action). Gate-channel item (BMV/QGEM row); η and the causal order installed by design, openly.

External clocks (news, verified live): ESA Euclid DR1 restructured (published 2026-06-15) — DR1-Foundation Nov 2026 (no cosmology products), complete DR1 with GC+WL **mid-2027** (cosmos.esa.int/web/euclid/dr1-timeline); NASA Roman at KSC 2026-06-21, launch NET 2026-08-30, Falcon Heavy (science.nasa.gov/blogs/roman/2026/06/21/).

Dated in-window markers, inside the screen (not load-bearing): arXiv:2606.17951 (one-parameter phantom-crossing w(z) form); arXiv:2606.19427 (AI-in-the-loop w(z) parametrization search); arXiv:2606.21826 (DESI-era dark-energy review); arXiv:2606.30928 (Cardy-type open/closed CFT from conformal nets — the net is input); arXiv:2606.28868 (finite-dim Krein metric-congruence — η installed per representation).

Iteration-16 verified additions (2026-07-03) — the multi-wedge iteration

From the clock-(1)-triggered iteration (tracks G1–G4 + referees R1/R2/R3; Working note: Iteration 16 synthesis: the multi-wedge closure — Lemma G, the (E_O) residual, and the first hedge advance since iteration 9 (repo: `notes/2026-07-03-iter16-synthesis.md`)). All web-verified live this session.

Doplicher, S. & Longo, R. — "Standard and split inclusions of von Neumann algebras." *Invent. Math.* **75** (1984) 493–536, DOI 10.1007/BF01388641. [VERIFIED — Springer/ADS/EuDML] — Standard split inclusions, the canonical interpolating type I factor, and the canonical implementation machinery. **Volume repair:** the iter-12 ledger's "Invent. Math. 73" was a typo, corrected in place this iteration.

Morinelli, V. & Neeb, K.-H. — "From local nets to Euler elements." *Adv. Math.* **458** (2024) 109960 (arXiv:2312.12182). [VERIFIED — published] — The two-directional, **BW-conditional** bridge between the two smuggle items: given BW, regularity forces Euler-generated modular groups; conversely an Euler element plus BW yields regularity and localizability. Theorem-level confirmation that n1 and η are interchangeable *relative to an installed BW/wedge structure* — deriving neither from (algebra, state). Hardens dossier §6.

Borchers, H.-J. — "The CPT-theorem in two-dimensional theories of local observables." *Commun. Math. Phys.* **143** (1992) 315–332, DOI 10.1007/BF02099011. [VERIFIED] — The Borchers commutation relations (spectrum condition + half-sided translations ⟹ modular scaling), the engine of LEM-NET-NATURALITY-G's (G-loc) branch.

Marrakchi, A. — arXiv:2606.23636 **full-source audit** (this iteration; abstract entry above under watch-sweep #1). [VERIFIED — full TeX source, twice independently downloaded] — The "with expectation" hypothesis is pervasive AND definitional (the relative bicentralizer flow is constructed against φ = φ∘E_N via Takesaki's theorem; the author flags the hypothesis "essential"); toolkit = averaging + ultrapower-implemented binormal states + approximate eigenstates of log Δ. Internal-bibliography caution: the preprint's "[Ha86] Acta Math. 69 (1986)" is a misprint (correct: Acta Math. **158** (1987) 95–148); its [Ma18] year is also suspect — import nothing from its bibliography unverified.

Houdayer, C. & Marrakchi, A. — "Selfless W*-probability spaces and Connes' bicentralizer problem." arXiv:2511.11409 (v2 2026-05-01; to appear J. Math. Soc. Japan per comments field). [VERIFIED — abstract verbatim] — States an **implication** (selflessness ⟹ bicentralizer-related structure), not an equivalence; the watch-sweep-#1 "equivalence" wording is repaired this iteration. Single-factor, no inclusion content; LOW relevance to expectation-free inclusions.

*AGHS / Arulseelan (continuous model theory of W-algebras, 2025 axiomatizations)** — arXiv:2508.17241 and companions. [VERIFIED — abstract level] — The axiomatizability infrastructure making the commutant-to-generated gap posable as a *definability* question (the DCNG watch item). Includes the II∞-undecidability sentence verified verbatim.

Cipriani, F. & Sauvageot, J.-L. — derivations from completely Dirichlet forms (the first-order differential calculus). [VERIFIED at listing level per the G4 referee] — The constitutive-positivity (g6) closure reason for the NC-Dirichlet-form readout: the form is completely *positive*-definite by construction; no indefinite version exists in the theory's foundations.

Longo, R. & Xu, F. — "Von Neumann Entropy in QFT." arXiv:1911.09390. [VERIFIED — attribution repaired per referee R2: Longo–Xu, not single-author] — The F-formula used in the G2 split-inclusion angle.

Interpretation, measurement, and dynamical-collapse models (mostly `[CONTESTED]` / `[SPECULATIVE]`)

H. Everett III, "Relative state formulation of quantum mechanics," *Rev. Mod. Phys.* **29**, 454 (1957). `[CONTESTED]` — The no-collapse (relative-state / many-worlds) interpretation retaining only unitary evolution.

D. Bohm, "A suggested interpretation of the quantum theory in terms of hidden variables, I & II," *Phys. Rev.* **85**, 166 & 180 (1952). `[CONTESTED]` — The de Broglie–Bohm pilot-wave theory: an explicit deterministic, contextual, nonlocal hidden-variable completion empirically equivalent to standard QM for the usual observables.

G. C. Ghirardi, A. Rimini, T. Weber, "Unified dynamics for microscopic and macroscopic systems," *Phys. Rev. D* **34**, 470 (1986). `[SPECULATIVE]` (empirically testable, not yet decided) — The GRW objective-collapse model: an explicit modification of unitary dynamics with experimentally bounded parameters.

W. H. Zurek, "Decoherence, einselection, and the quantum origins of the classical," *Rev. Mod. Phys.* **75**, 715 (2003). `[ESTABLISHED]` (decoherence mechanism) / `[CONTESTED]` (envariance derivation of the Born rule) — Authoritative review of decoherence, pointer-basis einselection, and the emergence of classicality.

Functional-analysis underpinnings (Reed–Simon) are listed under §8; rigorous operator-algebra results (Gelfand–Naimark, Bargmann, Gleason) are cross-referenced from §8.

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2. Quantum field theory and particle physics

Primary support for `domains/quantum-field-theory.md` (in the repository) and `domains/particle-physics.md` (in the repository).

Textbooks

S. Weinberg, *The Quantum Theory of Fields*, Vols. I–III (Cambridge Univ. Press, 1995–2000). `[ESTABLISHED]` — The deepest first-principles treatment: derives QFT from Poincaré invariance, unitarity, and cluster decomposition; authoritative on Wigner's classification, renormalization, gauge theories, electroweak symmetry breaking, and anomalies.

M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Addison-Wesley, 1995). `[ESTABLISHED]` — Standard graduate text for canonical + path-integral methods, renormalization, RG, gauge theories, and the chiral anomaly; the working calculational reference (and the QFT backbone for the Standard Model).

M. D. Schwartz, *Quantum Field Theory and the Standard Model* (Cambridge Univ. Press, 2014). `[ESTABLISHED]` — Modern text tying QFT formalism to SM phenomenology, EFT, and precision calculations.

A. Zee, *Quantum Field Theory in a Nutshell*, 2nd ed. (Princeton Univ. Press, 2010). `[ESTABLISHED]` — Path-integral-first and physically motivated; strong on the Wilsonian/EFT picture and conceptual structure.

C. Itzykson and J.-B. Zuber, *Quantum Field Theory* (McGraw-Hill, 1980). `[ESTABLISHED]` — Comprehensive classic on perturbation theory, renormalization, functional methods, and QED precision diagrammatics.

C. Quigg, *Gauge Theories of the Strong, Weak, and Electromagnetic Interactions*, 2nd ed. (Princeton Univ. Press, 2013). `[ESTABLISHED]` — Focused, physically motivated exposition of the SM gauge structure, electroweak theory, and the Higgs mechanism.

R. F. Streater and A. S. Wightman, *PCT, Spin and Statistics, and All That* (Benjamin, 1964; Princeton reissue 2000). `[ESTABLISHED]` — Canonical reference for the Wightman axioms, the reconstruction theorem, spin–statistics, and CPT — the rigorous operator/correlation foundation.

R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed. (Springer, 1996). [ESTABLISHED] — Definitive text on algebraic QFT (Haag–Kastler nets), DHR superselection, modular theory, Reeh–Schlieder, Bisognano–Wichmann, and Haag's theorem. *(Also primary for §8.)*

J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed. (Springer, 1987). [ESTABLISHED] — Authoritative account of constructive QFT: rigorous construction of ϕ^4 in 2 and 3 dimensions, the Osterwalder–Schrader axioms, and the methods/limits of the program. *(Also primary for §8.)*

Data reference

Particle Data Group (R. L. Workman *et al.*), *Review of Particle Physics*, *Prog. Theor. Exp. Phys.* (updated regularly; latest edition supersedes prior ones). [ESTABLISHED] — The standard source for measured SM parameters, masses, mixing matrices, and review articles. **Use for any numerical value** (see `CONSTANTS_AND_SCALES.md` (in the repository)). `[unverified]` exact lead-author/year for the current edition; cite the edition in hand.

Landmark papers — structure of QFT

E. P. Wigner, "On unitary representations of the inhomogeneous Lorentz group," *Annals of Mathematics* **40**, 149 (1939). [ESTABLISHED] — Classifies particles as unitary irreducible representations of the Poincaré group (labeled by mass m and spin/helicity).

K. G. Wilson and J. Kogut, "The renormalization group and the ε expansion," *Physics Reports* **12**, 75 (1974). [ESTABLISHED] — The Wilsonian RG/EFT reframing of renormalization; foundational for relevant/marginal/irrelevant operators and universality. *(Also primary for §5; cross-listed there.)*

H. Reeh and S. Schlieder, *Nuovo Cimento* **22**, 1051 (1961); **J. J. Bisognano and E. H. Wichmann**, *J. Math. Phys.* **16**, 985 (1975). `[ESTABLISHED]` — Vacuum cyclicity/entanglement (Reeh–Schlieder) and modular-flow-equals-boost (Bisognano–Wichmann); structural theorems linking locality, modular theory, and the Unruh effect.

R. D. Sorkin, "Impossible measurements on quantum fields," in *Directions in General Relativity*, Vol. 2 (1993); **C. J. Fewster and R. Verch**, "Quantum fields and local measurements," *Commun. Math. Phys.* **378**, 851 (2020). `[OPEN]` (problem) / `[INFERENCE]` (proposed framework) — The relativistic measurement problem (impossible measurements / superluminal signaling) and a modern consistent local-measurement framework.

Landmark papers — the Standard Model

S. L. Glashow, *Nucl. Phys.* **22**, 579 (1961); **S. Weinberg**, *Phys. Rev. Lett.* **19**, 1264 (1967); **A. Salam** (1968). `[ESTABLISHED]` — Founding papers unifying the electroweak interactions.

G. 't Hooft and M. Veltman, *Nucl. Phys. B* **44**, 189 (1972). `[ESTABLISHED]` — Renormalizability of spontaneously broken non-Abelian gauge theories, making the electroweak theory predictive (Nobel 1999).

P. W. Higgs, *Phys. Rev. Lett.* **13**, 508 (1964); **F. Englert and R. Brout**, *Phys. Rev. Lett.* **13**, 321 (1964); **G. Guralnik, C. R. Hagen, T. W. B. Kibble**, *Phys. Rev. Lett.* **13**, 585 (1964). `[ESTABLISHED]` — The Brout–Englert–Higgs mechanism; the boson was discovered by ATLAS and CMS in 2012 (Nobel 2013).

D. J. Gross and F. Wilczek, *Phys. Rev. Lett.* **30**, 1343 (1973); **H. D. Politzer**, *Phys. Rev. Lett.* **30**, 1346 (1973). `[ESTABLISHED]` — Asymptotic freedom in non-Abelian gauge theory, the foundation of perturbative QCD (Nobel 2004).

S. L. Adler, *Phys. Rev.* **177**, 2426 (1969); **J. S. Bell and R. Jackiw**, *Nuovo Cimento A* **60**, 47 (1969). `[ESTABLISHED]` — The chiral (ABJ) anomaly: a classical symmetry broken by quantization, with physical consequences and gauge-anomaly-cancellation constraints.

M. Kobayashi and T. Maskawa, *Prog. Theor. Phys.* **49**, 652 (1973). `[ESTABLISHED]` — Three quark generations permit CP violation via a single CKM phase (Nobel 2008).

R. D. Peccei and H. R. Quinn, *Phys. Rev. Lett.* **38**, 1440 (1977); **S. Weinberg** (1978); **F. Wilczek** (1978). `[ESTABLISHED]` (mechanism) / `[OPEN]` (axion existence) — The Peccei–Quinn solution to the strong-CP problem and the prediction of the axion.

A. D. Sakharov, *JETP Lett.* **5**, 24 (1967). `[ESTABLISHED]` (necessary conditions) / `[OPEN]` (realized mechanism) — The three conditions (B violation, C/CP violation, departure from equilibrium) for dynamically generating the baryon asymmetry. See `OPEN_PROBLEMS.md` (in the repository).

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3. General relativity and gravitation

Primary support for `domains/general-relativity.md` (in the repository).

Textbooks and treatises

C. W. Misner, K. S. Thorne, J. A. Wheeler, *Gravitation* (Freeman, 1973; reissued Princeton Univ. Press, 2017). `[ESTABLISHED]` — The canonical comprehensive reference ("MTW"); definitive on geometry, the equivalence principle, ADM, and gravitational waves. Source for sign-convention tables (see `domains/general-relativity.md` (in the repository)).

R. M. Wald, *General Relativity* (Univ. of Chicago Press, 1984). `[ESTABLISHED]` — The standard rigorous graduate text; authoritative on causal structure, singularity theorems, energy, asymptotic flatness (ADM/Bondi), and QFT in curved spacetime.

S. M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (Addison-Wesley, 2004). `[ESTABLISHED]` —

Modern, clear graduate text; good on the Einstein–Hilbert variational derivation, diffeomorphism invariance, and observational connections.

S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time* (Cambridge Univ. Press, 1973). [ESTABLISHED] — Definitive treatment of global causal structure, energy conditions, and the Hawking–Penrose singularity theorems. *(Also primary for §6; cross-listed there.)*

Landmark papers and theorems

R. Penrose, "Gravitational collapse and space-time singularities," *Phys. Rev. Lett.* **14**, 57 (1965). [ESTABLISHED] — The original trapped-surface singularity theorem; foundational for black-hole and cosmological singularity results (2020 Nobel).

R. Arnowitt, S. Deser, C. W. Misner, "The dynamics of general relativity" (1962; reprinted as arXiv:gr-qc/0405109). [ESTABLISHED] — The original ADM $3+1$ Hamiltonian formulation, constraints, and definition of ADM energy.

Y. Choquet-Bruhat and R. Geroch, "Global aspects of the Cauchy problem in general relativity," *Commun. Math. Phys.* **14**, 329 (1969). [ESTABLISHED] — Existence and uniqueness of the maximal globally hyperbolic development: the rigorous statement of GR's determinism.

R. Schoen and S.-T. Yau, "On the proof of the positive mass conjecture in general relativity," *Commun. Math. Phys.* **65**, 45 (1979); **E. Witten**, *Commun. Math. Phys.* **80**, 381 (1981). [ESTABLISHED] — Proofs of the positive-energy theorem ($M_{\mathrm{ADM}} \geq 0$), foundational for the consistency of GR's notion of total energy.

D. Christodoulou and S. Klainerman, *The Global Nonlinear Stability of the Minkowski Space* (Princeton Univ. Press, 1993). [ESTABLISHED] — Landmark proof that Minkowski space is nonlinearly stable; a benchmark global-existence result in GR PDE theory.

B. P. Abbott *et al.* (LIGO Scientific & Virgo Collaborations), "Observation of gravitational waves from a binary black hole merger," *Phys. Rev. Lett.* **116**, 061102 (2016). `[ESTABLISHED]` — First direct detection (GW150914); confirmation of GR's dynamical, strong-field, radiative regime.

S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199 (1975). `[ESTABLISHED]` (radiation) / `[OPEN]` (information fate) — Derivation of Hawking radiation; origin of the information paradox at the GR/QFT/thermodynamics interface. *(Load-bearing across §5, §6, §7; primary listing here.)*

C. M. Will, "The confrontation between general relativity and experiment," *Living Reviews in Relativity* **17**, 4 (2014). `[ESTABLISHED]` — Authoritative, regularly updated review of the PPN formalism and the tests defining GR's domain of validity. `[unverified]` exact article number of the most recent revision; cite the current *Living Reviews* version.

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4. Classical mechanics

Primary support for `domains/classical-mechanics.md` (in the repository).

Textbooks

H. Goldstein, C. Poole, J. Safko, *Classical Mechanics*, 3rd ed. (Addison-Wesley, 2002). `[ESTABLISHED]` — Standard graduate text; canonical treatment of Lagrangian/Hamiltonian formalism, canonical transformations, Hamilton–Jacobi theory, rigid bodies, and chaos.

L. D. Landau and E. M. Lifshitz, *Mechanics* (Course of Theoretical Physics, Vol. 1), 3rd ed. (Pergamon, 1976). `[ESTABLISHED]` — Develops all of mechanics from the principle of least action with exceptional economy; authoritative on conservation laws, small oscillations, and the rigid body.

V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed. (Springer, GTM 60, 1989). `[ESTABLISHED]` — The definitive geometric/symplectic treatment: differential forms, symplectic manifolds, Liouville–Arnold integrability, action–angle variables, KAM; the bridge to modern mathematics (`domains/mathematics.md` (in the repository)).

J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed. (Springer, 1999). `[ESTABLISHED]` — Authoritative on the geometry of symmetry: moment maps, Marsden–Weinstein symplectic reduction, Lie-group actions, and the modern form of Noether's theorem.

V. I. Arnold, V. V. Kozlov, A. I. Neishtadt, *Mathematical Aspects of Classical and Celestial Mechanics*, 3rd ed. (Springer, 2006). `[ESTABLISHED]` — Comprehensive modern survey of integrable systems, perturbation theory, KAM, Arnold diffusion, and the rigorous state of celestial mechanics.

R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (McGraw-Hill, 1965). `[ESTABLISHED]` — The path-integral formulation $\propto e^{iS/\hbar}$ and the derivation of the classical least-action principle as its stationary-phase limit.

Landmark papers and original sources

E. Noether, "Invariante Variationsprobleme," *Nachr. Ges. Wiss. Göttingen* (1918). `[ESTABLISHED]` — The original paper establishing the symmetry–conservation correspondence (Noether's first and second theorems).

H. Poincaré, *Les Méthodes Nouvelles de la Mécanique Céleste*, 3 vols. (Gauthier-Villars, 1892–1899). `[ESTABLISHED]` — Landmark origin of qualitative dynamics, the three-body problem, homoclinic tangles, and deterministic chaos / nonintegrability.

A. N. Kolmogorov (1954), **V. I. Arnold** (1963), **J. Moser** (1962) — the original KAM papers; synthesized in **J. Moser**, *Stable and Random Motions in Dynamical Systems* (Princeton Univ. Press, 1973). `[ESTABLISHED]` — Persistence of invariant tori under perturbation; foundational for the integrable–chaotic transition.

P. A. M. Dirac, *Lectures on Quantum Mechanics* (Belfer Graduate School, 1964); and "Generalized Hamiltonian dynamics" (1950). `[ESTABLISHED]` — Constraint analysis of singular Lagrangians (Dirac–Bergmann), Dirac brackets, and the bracket→commutator correspondence.

F. Bayen, M. Flato, C. Fronsdal, A. Lichnerowicz, D. Sternheimer, "Deformation theory and quantization, I & II," *Ann. Phys. (NY)* **111** (1978). `[ESTABLISHED]` (math) — Founding papers of deformation quantization and the star product, formalizing quantization as a deformation of the classical Poisson algebra.

Z. Xia, "The existence of noncollision singularities in Newtonian systems," *Annals of Mathematics* **135** (1992), 411–468. `[ESTABLISHED]` — Rigorous proof that finite-time noncollision singularities exist in the Newtonian N-body problem ($N \geq 5$), bounding global determinism. See `GAPS_AND_CONTRADICTIONS.md` (in the repository).

J. D. Norton, "The dome: an unexpectedly simple failure of determinism," *Philosophy of Science* **75** (2008). `[CONTESTED]` — Sharp articulation of the non-Lipschitz indeterminism debate internal to Newtonian mechanics.

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5. Thermodynamics and statistical mechanics

Primary support for `domains/thermodynamics.md` (in the repository) and `domains/statistical-mechanics.md` (in the repository).

Textbooks

H. B. Callen, *Thermodynamics and an Introduction to Thermostatistics*, 2nd ed. (Wiley, 1985). `[ESTABLISHED]` — The canonical axiomatic (postulatory) treatment: fundamental relation, extremum principles, Legendre transforms, Maxwell relations, stability.

E. Fermi, *Thermodynamics* (Dover, 1956); **M. W. Zemansky and R. H. Dittman**, *Heat and Thermodynamics* (McGraw-Hill; multiple eds.). `[ESTABLISHED]` — Classic clear expositions of the four laws, Carnot's theorem, and the operational definition of absolute temperature. `[unverified]` exact edition/year of the Zemansky–Dittman volume in use.

L. D. Landau and E. M. Lifshitz, *Statistical Physics, Part 1* (Course of Theoretical Physics, Vol. 5), 3rd ed. (Pergamon). `[ESTABLISHED]` — Authoritative bridge from statistical mechanics to thermodynamics: equilibrium ensembles, fluctuations, phase transitions, and the third law from quantum statistics.

K. Huang, *Statistical Mechanics*, 2nd ed. (Wiley, 1987). `[ESTABLISHED]` — Standard pedagogical text: Boltzmann equation and H-theorem, the ensembles, quantum gases, and the Ising model via the transfer matrix.

R. K. Pathria and P. D. Beale, *Statistical Mechanics*, 3rd ed. (Elsevier/Academic Press). `[ESTABLISHED]` — Comprehensive coverage of quantum statistics, the grand canonical ensemble, BEC, fluctuation–dissipation, and an introduction to critical phenomena and the RG.

L. E. Reichl, *A Modern Course in Statistical Physics* (Wiley; multiple eds.). `[ESTABLISHED]` — Strong on nonequilibrium statistical mechanics, kinetic theory, linear response, and the fluctuation–dissipation theorem.

N. Goldenfeld, *Lectures on Phase Transitions and the Renormalization Group* (Addison-Wesley, 1992). `[ESTABLISHED]` — Conceptually careful treatment of scaling, universality, and Wilsonian RG, including the ε-expansion.

A. I. Khinchin, *Mathematical Foundations of Statistical Mechanics* (Dover, 1949). `[ESTABLISHED]` — Rigorous demonstration that the equivalence of time and phase averages for macroscopic sum-functions rests on high dimensionality, not strict ergodicity — a cornerstone of the typicality viewpoint.

Historical origins

S. Carnot, *Réflexions sur la puissance motrice du feu* (1824); **R. Clausius** (1865, introduction of entropy); **L. Boltzmann**, *Lectures on Gas Theory* (1896–98). `[ESTABLISHED]` — Origins of the Carnot bound, the entropy concept, and the kinetic foundation of the second law.

Landmark papers — equilibrium, critical phenomena, foundations

L. Onsager, "Crystal statistics. I. A two-dimensional model with an order–disorder transition," *Phys. Rev.* **65**, 117 (1944). `[ESTABLISHED]` — Exact solution of the 2D Ising model; the landmark demonstration of a genuine phase transition with non-mean-field critical exponents.

K. G. Wilson and J. Kogut, "The renormalization group and the ε expansion," *Phys. Rep.* **12**, 75 (1974). `[ESTABLISHED]` — RG explanation of critical phenomena and universality (Wilson, Nobel 1982). *(Primary entry; also load-bearing in §2.)*

E. T. Jaynes, "Information theory and statistical mechanics," *Phys. Rev.* **106**, 620 (1957). `[CONTESTED]` — The maximum-entropy / inferential reinterpretation of statistical mechanics; central to the disputed information-theoretic foundations. See `domains/information-theory.md` (in the repository).

J. L. Lebowitz, "Boltzmann's entropy and time's arrow," *Physics Today* **46**(9), 32 (1993). `[INFERENCE]` — Influential modern statement of the typicality/Boltzmann-entropy resolution of the irreversibility paradoxes (Loschmidt, Zermelo) and the role of the low-entropy past.

O. E. Lanford III, "Time evolution of large classical systems," in *Dynamical Systems, Theory and Applications*, Lecture Notes in Physics **38** (Springer, 1975). `[ESTABLISHED]` — Rigorous derivation of the Boltzmann equation from Hamiltonian dynamics (Boltzmann–Grad limit) for short times, clarifying the status of molecular chaos.

Landmark papers — nonequilibrium and fluctuation theorems

L. Onsager, *Phys. Rev.* **37**, 405 and **38**, 2265 (1931); **S. R. de Groot and P. Mazur**, *Non-Equilibrium Thermodynamics* (Dover, 1984). `[ESTABLISHED]` — Onsager reciprocal relations and linear irreversible thermodynamics; the near-equilibrium extension and connection to transport.

R. Kubo, "Statistical-mechanical theory of irreversible processes I," *J. Phys. Soc. Jpn.* **12**, 570 (1957). `[ESTABLISHED]` — Linear-response theory and the fluctuation–dissipation theorem (Green–Kubo relations for transport coefficients).

C. Jarzynski, "Nonequilibrium equality for free energy differences," *Phys. Rev. Lett.* **78**, 2690 (1997). `[ESTABLISHED]` — The Jarzynski equality $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$, an exact nonequilibrium identity refining the second law.

G. E. Crooks, "Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences," *Phys. Rev. E* **60**, 2721 (1999). `[ESTABLISHED]` — The Crooks fluctuation theorem relating forward/reverse work distributions; the detailed result from which Jarzynski follows.

U. Seifert, "Stochastic thermodynamics, fluctuation theorems and molecular machines," *Rep. Prog. Phys.* **75**, 126001 (2012). `[ESTABLISHED]` — Comprehensive review of stochastic thermodynamics and trajectory-level entropy production.

Landmark papers — quantum thermalization

J. M. Deutsch, *Phys. Rev. A* **43**, 2046 (1991); **M. Srednicki**, "Chaos and quantum thermalization," *Phys. Rev. E* **50**, 888 (1994). `[INFERENCE]` (hypothesis, strongly supported) — The original formulations of the Eigenstate Thermalization Hypothesis (ETH) explaining thermalization in isolated quantum systems.

L. D'Alessio, Y. Kafri, A. Polkovnikov, M. Rigol, "From quantum chaos and eigenstate thermalization to statistical mechanics and thermodynamics," *Adv. Phys.* **65**, 239 (2016). `[INFERENCE]` — Comprehensive modern review tying ETH, quantum chaos, random-matrix theory, integrability/GGE, and the foundations of quantum statistical mechanics.

Black-hole thermodynamics (interface with §3, §6, §7)

J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* 7, 2333 (1973). `[ESTABLISHED]` (within semiclassical gravity) — Black-hole entropy $S = A/4$ (in $G = \hbar = c = k_B = 1$ units). *(Primary entry; companion to Hawking 1975, §3.)*

J. M. Bardeen, B. Carter, S. W. Hawking, "The four laws of black hole mechanics," *Commun. Math. Phys.* **31**, 161 (1973). `[ESTABLISHED]` — The isomorphism between black-hole mechanics and the four laws of thermodynamics.

A. Strominger and C. Vafa, "Microscopic origin of the Bekenstein–Hawking entropy," *Phys. Lett. B* **379**, 99 (1996). `[ESTABLISHED]` (for the specific BPS cases) / `[OPEN]` (general case) — String-theoretic microstate counting reproducing $S = A/4$ for certain extremal black holes. See `OPEN_PROBLEMS.md` (in the repository).

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6. Cosmology

Primary support for `domains/cosmology.md` (in the repository).

Textbooks

S. Weinberg, *Cosmology* (Oxford Univ. Press, 2008). `[ESTABLISHED]` — Authoritative graduate text; rigorous FLRW dynamics, the CMB and perturbation theory, and the anthropic bound on Λ.

S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed. (Academic Press, 2020). `[ESTABLISHED]` — Standard reference for the Einstein–Boltzmann hierarchy, CMB and matter power spectra, and the formalism behind CAMB/CLASS.

V. Mukhanov, *Physical Foundations of Cosmology* (Cambridge Univ. Press, 2005). `[ESTABLISHED]` (formalism) — Rigorous cosmological perturbation theory and the inflationary generation of primordial fluctuations.

D. Baumann, *Cosmology* (Cambridge Univ. Press, 2022). `[ESTABLISHED]` — Modern pedagogical synthesis of background dynamics, inflation, and perturbations with current observational framing.

E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, 1990). `[ESTABLISHED]` — Classic reference on thermal history, freeze-out, BBN, and relics; standard source for the hot-Big-Bang formalism.

G. F. R. Ellis, R. Maartens, M. A. H. MacCallum, *Relativistic Cosmology* (Cambridge Univ. Press, 2012). `[ESTABLISHED]` — Careful treatment of the cosmological principle, averaging/backreaction (the fitting problem), and the foundations of the FLRW idealization.

Data reference

Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020). `[ESTABLISHED]` — The definitive CMB determination of the six Λ CDM parameters; baseline for the H_0-tension and curvature discussions. **Use for numerical values** (see `CONSTANTS_AND_SCALES.md` (in the repository)); the wiki states such values approximately by design.

Landmark papers and theorems

A. H. Guth, "Inflationary universe: a possible solution to the horizon and flatness problems," *Phys. Rev. D* **23**, 347 (1981). `[INFERENCE]` (paradigm, strongly favored) — Foundational inflation paper articulating the horizon, flatness, and monopole problems.

A. Borde, A. H. Guth, A. Vilenkin, "Inflationary spacetimes are not past-complete," *Phys. Rev. Lett.* **90**, 151301 (2003). `[ESTABLISHED]` (theorem) / `[SPECULATIVE]` (multiverse reading) — The BGV theorem: even eternally inflating spacetimes have a past boundary.

S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989). `[OPEN]` — The canonical statement of the cosmological-constant problem and survey of attempted solutions; still the reference framing. See `OPEN_PROBLEMS.md` (in the repository).

J. Martin, "Everything you always wanted to know about the cosmological constant problem (but were afraid to ask)," arXiv:1205.3365 *C. R. Physique* **13**, 566–665 (2012), DOI 10.1016/j.crhy.2012.04.008. `[ESTABLISHED]` (pedagogy) — The standard regularization-aware treatment: dimensional regularization gives $m^4 \ln \mu$ structure, not M_{cut}^4.

C. P. Burgess, "The Cosmological Constant Problem: Why it's hard to get Dark Energy from Micro-physics," Les Houches lectures, arXiv:1309.4133. `[ESTABLISHED]` (pedagogy) — Naturalness / radiative-instability framing across EFT thresholds.

A. Padilla, "Lectures on the Cosmological Constant Problem," arXiv:1502.05296. `[ESTABLISHED]` (pedagogy) — Works Weinberg's no-go theorem through in detail; radiative instability; surveys sequestering and other exits.

N. Kaloper, A. Padilla, "Sequestering the Standard Model Vacuum Energy," *Phys. Rev. Lett.* **112**, 091304 (2014) (arXiv:1309.6562); **N. Kaloper, A. Padilla, D. Stefanyszyn, G. Zahariade**, "A Manifestly Local Theory of Vacuum Energy Sequestering," *Phys. Rev. Lett.* **116**, 051302 (2016) (arXiv:1505.01492). `[SPECULATIVE]` — Matter-loop vacuum-energy cancellation; the global version predicts transient dark energy and future collapse; graviton loops not covered by these versions.

R. Carballo-Rubio, L. J. Garay, G. García-Moreno, "Unimodular gravity vs general relativity: a status report," *Class. Quantum Grav.* **39**, 243001 (2022) (arXiv:2207.08499). `[ESTABLISHED]` (review) — UG ≡ GR except Λ-as-integration-constant; no further classical/semiclassical/quantum differences found; the "technically natural" reading is `[CONTESTED]`.

A. Salvio, "Unimodular Quadratic Gravity and the Cosmological Constant," *Phys. Lett. B* **856** (2024) 138920 (arXiv:2406.12958). `[ESTABLISHED]` (as literature verdict) — States explicitly that unimodular gravity does not solve the "new" CC problem (the observed value).

DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," *Phys. Rev. D* **112**, 083515 (2025) (arXiv:2503.14738); **K. Lodha et al. (DESI)**, "Extended Dark Energy analysis using DESI DR2 BAO measurements," arXiv:2503.14743. `[ESTABLISHED]` (measurements) / `[CONTESTED]` (dynamical-DE interpretation) — The 3.1σ (BAO+CMB) / 2.8–4.2σ (+SNe) $w_0 w_a$ preference; parameterization-robustness and phantom-crossing preference.

G. Efstathiou, "Evolving Dark Energy or Supernovae Systematics?," arXiv:2408.07175; **D. D. Y. Ong, D. Yallup, W. Handley**, "A Bayesian Perspective on Evidence for Evolving Dark Energy," arXiv:2511.10631. `[CONTESTED]` — The counter-analyses: the preference is SN-sample-driven; in Bayes-factor terms DESI DR2+CMB alone modestly favor Λ CDM, and the signal largely vanishes with recalibrated DES-Dovekie SNe (ln B −0.57±0.26 → −0.30±0.19).

J. de Cruz Pérez, A. Gómez-Valent, J. Solà Peracaula, "Dynamical Dark Energy models in light of the latest observations," arXiv:2512.20616 (accepted 2026). `[CONTESTED]` — Running-vacuum variants vs DESI DR2 + Planck PR4 + SNe; evidence comparable to $w_0 w_a$CDM, strongest with DES-Y5.

L. A. Anchordoqui, I. Antoniadis, D. Lüst, "S-dual Quintessence, the Swampland, and the DESI DR2 Results," arXiv:2503.19428. `[SPECULATIVE]` — Swampland-consistent (distance/dS/TCC) quintessence matching the DESI-fitted potential.

R. Penrose, "Singularities and time-asymmetry," in *General Relativity: An Einstein Centenary Survey*, eds. S. W. Hawking and W. Israel (Cambridge Univ. Press, 1979); and *The Road to Reality* (Jonathan Cape, 2004). `[SPECULATIVE]` — Origin of the Weyl curvature hypothesis and the gravitational-entropy framing of the low-entropy past and the arrow of time. See `HYPOTHESES.md` (in the repository).

S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time* (Cambridge Univ. Press, 1973). `[ESTABLISHED]` — Singularity theorems and the rigorous geometric basis for the breakdown of FLRW at the initial singularity. *(Primary listing in §3.)*

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7. Information theory and quantum information

Primary support for `domains/information-theory.md` (in the repository).

Textbooks

T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, 2006). `[ESTABLISHED]` — The canonical classical reference: AEP, data-processing inequality, capacity, rate–distortion, and the Kolmogorov-complexity chapter.

M. Li and P. Vitányi, *An Introduction to Kolmogorov Complexity and Its Applications* (Springer; multiple eds.). `[ESTABLISHED]` — Definitive treatment of algorithmic information theory: the invariance theorem, incomputability, relation to Shannon entropy, and Chaitin's results.

M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge Univ. Press, 2000/2010). `[ESTABLISHED]` — The standard quantum-information text: von

Neumann entropy, no-cloning, teleportation, entanglement measures, channel capacities, error correction. *(Primary listing in §1.)*

M. M. Wilde, *Quantum Information Theory*, 2nd ed. (Cambridge Univ. Press, 2017). `[ESTABLISHED]` — Modern rigorous treatment of channels, entropies, and the information-theoretic structure underlying the operational generalization of the QM axioms.

Foundational papers — classical and quantum information

C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal* **27**, 379 & 623 (1948). `[ESTABLISHED]` — Founding paper: entropy, mutual information, source and channel coding theorems, channel capacity.

E. H. Lieb and M. B. Ruskai, "Proof of the strong subadditivity of quantum-mechanical entropy," *J. Math. Phys.* **14**, 1938 (1973). `[ESTABLISHED]` — Proof of strong subadditivity, the single most important entropy inequality, underlying both quantum information and holographic entropy constraints.

J. A. Wheeler, "Information, physics, quantum: the search for links," in *Complexity, Entropy and the Physics of Information* (Addison-Wesley, 1990). `[SPECULATIVE]` — The "it from bit" manifesto framing physics as fundamentally informational; the conceptual lodestar of the domain. See `UNIFYING_PRINCIPLES.md` (in the repository).

Thermodynamics of computation (interface with §5)

R. Landauer, "Irreversibility and heat generation in the computing process," *IBM J. Res. Dev.* **5**, 183 (1961). `[ESTABLISHED]` — Landauer's principle: erasing one bit dissipates at least $k_B T \ln 2$ of heat.

C. H. Bennett, "The thermodynamics of computation — a review," *Int. J. Theor. Phys.* **21**, 905 (1982). `[ESTABLISHED]` — Reversible computation and the resolution of the Maxwell-demon paradox via the cost of erasure.

N. Margolus and L. B. Levitin, "The maximum speed of dynamical evolution," *Physica D* **120**, 188 (1998); **S. Lloyd**, "Ultimate physical limits to computation," *Nature* **406**, 1047 (2000). `[ESTABLISHED]` (bounds) — The quantum speed limit / maximum operation rate and Lloyd's synthesis of physical bounds into computational ceilings.

Holography and "spacetime from entanglement"

J. D. Bekenstein, "Universal upper bound on the entropy-to-energy ratio for bounded systems," *Phys. Rev. D* **23**, 287 (1981); **G. 't Hooft** (1993) and **L. Susskind**, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995); **R. Bousso**, "A covariant entropy conjecture," *JHEP* 07 (1999) 004, and "The holographic principle," *Rev. Mod. Phys.* **74**, 825 (2002). `[INFERENCE]` (Bekenstein/Bousso bounds) / `[SPECULATIVE]` (holographic principle as fundamental) — The information-bound backbone of holography. `[unverified]` exact JHEP article number of Bousso 1999.

S. Ryu and T. Takayanagi, "Holographic derivation of entanglement entropy from AdS/CFT," *Phys. Rev. Lett.* **96**, 181602 (2006); **V. E. Hubeny, M. Rangamani, T. Takayanagi**, *JHEP* 07 (2007) 062. `[INFERENCE]` (within AdS/CFT) — The RT formula and its covariant (HRT) generalization: entanglement entropy as a minimal/extremal bulk area.

A. Almheiri, X. Dong, D. Harlow, "Bulk locality and quantum error correction in AdS/CFT," *JHEP* 04 (2015) 163; **F. Pastawski, B. Yoshida, D. Harlow, J. Preskill**, "Holographic quantum error-correcting codes," *JHEP* 06 (2015) 149. `[SPECULATIVE]` — AdS/CFT as a quantum error-correcting code and the HaPPY tensor-network model.

T. Jacobson, "Thermodynamics of spacetime: the Einstein equation of state," *Phys. Rev. Lett.* **75**, 1260 (1995); **M. Van Raamsdonk**, "Building up spacetime with quantum entanglement," *Gen. Rel. Grav.* **42**, 2323 (2010); **J. Maldacena and L. Susskind**, "Cool horizons for entangled black holes," *Fortsch. Phys.* **61**, 781 (2013). `[SPECULATIVE]` — Jacobson's thermodynamic derivation of Einstein's equations, the "spacetime from entanglement" thesis, and the ER=EPR conjecture. See `UNIFICATION_LANDSCAPE.md` (in the repository).

D. N. Page, "Information in black hole radiation," *Phys. Rev. Lett.* **71**, 3743 (1993). `[ESTABLISHED]` — The Page curve. **Island/QES core (coordinates verified 2026-07-02, arXiv abs pages + journal records fetched): G. Penington**, "Entanglement wedge reconstruction and the information paradox," arXiv:1905.08255, *JHEP* **09** (2020) 002; **A. Almheiri, N. Engelhardt, D. Marolf, H. Maxfield**, "The entropy of bulk quantum fields and the entanglement wedge of an evaporating black hole," arXiv:1905.08762, *JHEP* **12** (2019) 063; **G. Penington, S. H. Shenker, D. Stanford, Z. Yang**, "Replica wormholes and the black hole interior," arXiv:1911.11977, *JHEP* **03** (2022) 205; **A. Almheiri, T. Hartman, J. Maldacena, E. Shaghoulian, A. Tajdini**, "Replica wormholes and the entropy of Hawking radiation," arXiv:1911.12333, *JHEP* **05** (2020) 013. `[OPEN]` (paradox mechanism) / `[INFERENCE]` (island Page curve within controlled settings).

Islands — standing objections (all verified 2026-07-02): H. Geng, A. Karch, "Massive islands," arXiv:2006.02438, *JHEP* **09** (2020) 121; **H. Geng, A. Karch, C. Perez-Pardavila, S. Raju, L. Randall, M. Riojas, S. Shashi**, "Information transfer with a gravitating bath," arXiv:2012.04671, *SciPost Phys.* **10**, 103 (2021), and "Inconsistency of islands in theories with long-range gravity," arXiv:2107.03390, *JHEP* **01** (2022) 182; **H. Geng**, "Graviton mass and entanglement islands in low spacetime dimensions," arXiv:2312.13336, *SciPost Phys.* **19**, 146 (2025); "Replica wormholes and entanglement islands in the Karch–Randall braneworld," arXiv:2405.14872, *JHEP* **01** (2025) 063; "Making the case for massive islands," arXiv:2509.22775 (no journal ref); **S. Demulder, A. Gnecchi, I. Lavdas, D. Lüst**, "Islands and light gravitons in type IIB string theory," arXiv:2204.03669, *JHEP* **02** (2023)

016. `[CONTESTED]` — the massive-graviton / gravitational-Gauss-law critique and its string-embedded refinements.

Ensemble-vs-single-theory (verified 2026-07-02): D. Marolf, H. Maxfield, "Transcending the ensemble: baby universes, spacetime wormholes, and the order and disorder of black hole information," arXiv:2002.08950, *JHEP* **08** (2020) 044; **J.-M. Schlenker, E. Witten**, "No ensemble averaging below the black hole threshold," arXiv:2202.01372, *JHEP* **07** (2022) 143. `[CONTESTED]` — whether replica-wormhole entropies are single-theory statements; see also arXiv:2510.06376 / 2605.15180 (OP-44 fourth option) and the OP-50 cluster.

Crossed-product founding trio (verified 2026-07-02): E. Witten, "Gravity and the crossed product," arXiv:2112.12828, *JHEP* **10** (2022) 008; **V. Chandrasekaran, R. Longo, G. Penington, E. Witten**, "An algebra of observables for de Sitter space," arXiv:2206.10780, *JHEP* **02** (2023) 082; **V. Chandrasekaran, G. Penington, E. Witten**, "Large N algebras and generalized entropy," arXiv:2209.10454, *JHEP* **04** (2023) 009; **E. Witten**, "A background independent algebra in quantum gravity," arXiv:2308.03663 (no journal ref listed). `[ESTABLISHED as mathematics / INFERENCE as physics]` — Type $\mathrm{III}_1 \to \mathrm{II}$ via the modular crossed product; vN entropy $= S_{\text{gen}}$ up to an additive constant; the observer/clock is installed. (2024–2026 follow-ups on record in FINDINGS and the iteration-6 note: 2309.15897; 2306.01837; 2406.02116; 2406.01669; 2407.01695; 2501.01487; 2411.19931; 2601.07910; 2601.07915.)

C. Akers, N. Engelhardt, D. Harlow, G. Penington, S. Vardhan, "The black hole interior from non-isometric codes and complexity," arXiv:2207.06536 (no journal ref listed on arXiv). `[SPECULATIVE]` (leading proposal) — leading post-island account of interior reconstruction.

S. Raju, "Lessons from the information paradox," arXiv:2012.05770 [unverified journal-ref: *Phys. Rept.* (2022)]. — The holography-of-information counterpoint: exponentially small correlations + boundary completeness in massless gravity; companion cluster already on record at the A2 entries.

(Hawking 1975 — the origin of the information paradox — is listed in §3.)

8. Mathematical physics

Primary support for `domains/mathematics.md` (in the repository).

Functional analysis and operator algebras

M. Reed and B. Simon, *Methods of Modern Mathematical Physics*, Vols. I–IV (Academic Press, 1972–1979). `[ESTABLISHED]` — The canonical rigorous reference for functional analysis in physics: Hilbert spaces, self-adjointness and self-adjoint extensions (deficiency indices), the spectral theorem, Stone's theorem, and scattering. Vol. I (Functional Analysis), II (Self-Adjointness), III (Scattering), IV (Analysis of Operators). *(Load-bearing for §1.)*

M. Takesaki, *Theory of Operator Algebras*, Vols. I–III (Springer, 1979/2003). `[ESTABLISHED]` — The definitive treatise on C^*-and von Neumann algebras, the Murray–von Neumann/Connes type classification, and Tomita–Takesaki modular theory with the KMS condition.

R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed. (Springer, 1996). `[ESTABLISHED]` — Algebraic QFT, the type-III nature of local algebras, modular theory in physics, DHR superselection. *(Primary listing in §2.)*

Geometry, topology, and representation theory

M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. (IOP, 2003). `[ESTABLISHED]` — Widely used physicist's reference for manifolds, differential forms, fiber bundles, connections and curvature, characteristic classes, the Atiyah–Singer index theorem, and applications to gauge theory, anomalies, and instantons.

A. O. Barut and R. Rączka, *Theory of Group Representations and Applications* (PWN, 1977); **M. E. Taylor**, *Noncommutative Harmonic Analysis* (AMS, 1986). `[ESTABLISHED]` — Standard references for Lie groups, Lie algebras, and unitary representation theory in physics, including induced representations and Wigner's classification of Poincaré irreps. `[unverified]` exact year of the Taylor volume.

Rigorous QFT, constructive results, and modern frameworks

J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed. (Springer, 1987). `[ESTABLISHED]` — Constructive QFT and the rigorous Euclidean path integral, including the Osterwalder–Schrader axioms and the construction of $P(\phi)_2$ and ϕ^4_3. *(Primary listing in §2.)*

A. Jaffe and E. Witten, "Quantum Yang–Mills theory" (Clay Mathematics Institute Millennium Problem description, 2000). `[OPEN]` — The official statement of the Yang–Mills existence and mass-gap problem. See `OPEN_PROBLEMS.md` (in the repository) and `GAPS_AND_CONTRADICTIONS.md` (in the repository).

M. Aizenman (*Phys. Rev. Lett.* 1981; *Commun. Math. Phys.* 1982) and **M. Aizenman and H. Duminil-Copin**, "Marginal triviality of the scaling limits of critical 4D Ising and ϕ^4_4 models," *Annals of Mathematics* **194** (2021), 163–235. `[ESTABLISHED]` — Rigorous proof of triviality of the continuum limit of ϕ^4 in four dimensions: some 4D theories exist only as cutoff effective theories. Central to the domain-of-validity discussion in `domains/quantum-field-theory.md` (in the repository).

J. Lurie, "On the classification of topological field theories," *Current Developments in Mathematics 2008* (2009), 129–280. `[INFERENCE]` (proof program/sketch) — The cobordism hypothesis; central result on fully extended TQFTs and higher-categorical structures.

K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory*, Vols. 1–2 (Cambridge Univ. Press, 2017/2021). `[INFERENCE]` (perturbative scope) — Rigorous formulation of perturbative QFT via factorization algebras and the Batalin–Vilkovisky formalism; a leading bridge between physicists' QFT and rigorous mathematics.

Structural theorems and the meta-question

V. Bargmann, "On unitary ray representations of continuous groups," *Ann. Math.* **59** (1954); **A. M. Gleason** (1957, see §1); **I. M. Gelfand and M. A. Naimark** (1943) and **I. E. Segal** foundational C^*-algebra papers. `[ESTABLISHED]` — Projective/ray representations (Bargmann–Wigner), the near-uniqueness of the Born rule (Gleason), and the representability of C^*-algebras (Gelfand–Naimark–Segal).

E. P. Wigner, "The unreasonable effectiveness of mathematics in the natural sciences," *Comm. Pure Appl. Math.* **13** (1960), 1–14. `[OPEN]` (philosophical) — The original essay framing why mathematics is so effective in physics. See `UNIFYING_PRINCIPLES.md` (in the repository).

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9. Quantum gravity and unification

There is currently **no experimentally confirmed theory of quantum gravity** `[ESTABLISHED]`; accordingly every entry below is `[SPECULATIVE]` or `[OPEN]` as physics, even where the *mathematics* is rigorous. Many key results in this area are also load-bearing in §3 (Hawking 1975, Bardeen–Carter–Hawking), §5 (Bekenstein, Strominger–Vafa), and §7 (holography, RT/HRT, ER=EPR, islands) and are not repeated here. This section collects the additional canonical *unification-program* references. See `UNIFICATION_LANDSCAPE.md` (in the repository), `HYPOTHESES.md` (in the repository), and `OPEN_PROBLEMS.md` (in the repository).

J. M. Maldacena, "The large-N limit of superconformal field theories and supergravity," *Adv. Theor. Math. Phys.* **2**, 231 (1998). `[SPECULATIVE]` (conjecture, very strongly supported within its regime) — The AdS/CFT correspondence; the most concrete realization of holography and the engine behind much of §7. [unverified] exact page; widely cited as *Adv. Theor. Math. Phys.* 2 (1998) 231 / *Int. J. Theor. Phys.* 38 (1999) 1113.

J. Polchinski, *String Theory*, Vols. I–II (Cambridge Univ. Press, 1998). [ESTABLISHED] (as exposition of the framework) — The standard graduate reference for perturbative string theory, D-branes, and dualities. The *physical correctness* of string theory as the theory of nature is [OPEN].

C. Rovelli, *Quantum Gravity* (Cambridge Univ. Press, 2004); **T. Thiemann**, *Modern Canonical Quantum General Relativity* (Cambridge Univ. Press, 2007). [SPECULATIVE] — Canonical references for loop quantum gravity (spin networks, the area/volume operators, spin foams). [unverified] exact year of the Thiemann volume.

Caution (per `EPISTEMICS.md` (in the repository)). The maturity of the *mathematics* in these programs must never be conflated with empirical confirmation of the *physics*. None of the candidate quantum-gravity frameworks has a confirmed prediction distinguishing it from the others or from semiclassical gravity [ESTABLISHED]. Treat all unification claims sourced from this section as [SPECULATIVE] unless a domain page demonstrates otherwise.

Maintenance conventions

Dedup rule. A source load-bearing in multiple domains is listed once under its primary section and cross-referenced (not re-quoted) elsewhere. When adding a reference, search this page first.

No fabrication. Add only sources you are confident exist. If unsure of a detail, supply author + title + approximate year and append [unverified]. Never invent a DOI, volume, page, or theorem name. This mirrors the cardinal rule of `EPISTEMICS.md` (in the repository).

Numbers. All numerical constants and measured parameters trace to the Particle Data Group (§2) or Planck (§6); see

`CONSTANTS_AND_SCALES.md` (in the repository). Do not cite numbers from textbooks of memory.

Provenance. New entries should record which domain dossier or page motivated them; log additions in the repository changelog (repo: `CHANGELOG.md`; condensed here as Publication History (p. 235)).

References

This *is* the references page. Other wiki pages should cite back here rather than maintaining parallel source lists. Cross-domain meta-pages — `UNIFICATION_LANDSCAPE.md` (in the repository), `UNIFYING_PRINCIPLES.md` (in the repository), `FINDINGS.md` (in the repository), `ROADMAP.md` (in the repository) — should link individual entries above by author + short title.

BACK MATTER

About This Book

Universal Physics began as a research wiki, not a manuscript. It was built by a multi-agent process: a human author, Harshit Khemani, directing teams of AI research agents (Claude, built by Anthropic) across six iterations between June 8 and June 10, 2026 — dozens of agents per iteration, each assigned one focused question, each new claim passed through an adversarial referee whose explicit job was to break it.

The process had three standing rules. First, the epistemic-tag discipline (Chapter 2): every nontrivial claim carries exactly one of `[ESTABLISHED]`, `[INFERENCE]`, `[SPECULATIVE]`, `[OPEN]`, or `[CONTESTED]`, and no speculation may wear the clothes of fact. Second, web verification: citations were checked against the actual papers, and the bibliography contains real references only. Third, the append-only log: every change, correction, and demotion is recorded in the repository's changelog — including the places where the investigation reversed itself.

The iterations had distinct characters: 1 built the map; 2 stress-tested the central wager; 3 shifted from surveying to attempting (and red-teamed the project's own posture); 4 measured diminishing returns; 5 executed the final verdict-flipping lever and declared analytical saturation; 6 injected two years of new literature in a deliberate attempt to overturn the verdict, and failed.

The book's chapters reproduce the wiki's pages verbatim; only navigation chrome was trimmed. Roughly thirty long-form working notes behind the Part V syntheses are not reprinted here but remain public, with the full change history, at **github.com/HKTITAN/universal-physics** and **universal-**

 The book and site design are inspired by Dan Hollick's *Making Software* (makingsoftware.com).

Publication History

The wiki's changelog is append-only; this history condenses it to one dated entry per iteration. Per-page micro-edits are omitted — the full log remains in the repository (`CHANGELOG.md`).

2026-06-08 — Iteration 1: initial build. First full pass, bootstrapped by a multi-agent workflow: ten parallel domain dossiers; cross-cutting analyses (contradictions, open-problems registry, assumptions ledger, unification-landscape survey, candidate principles); hypothesis generation through three independent lenses; adversarial verification of every contradiction and hypothesis. Conventions established: the five epistemic tags, the assumption-classification scheme, the anti-crank verification gate, and stable identifiers (OP-n, GC-n, Hn = H0–H8, A-n). Post-build structural QC was clean.

2026-06-08 — Iteration 2: five refereed research tracks + synthesis. First forward-research iteration (38 agents; 0/25 new claims flagged unsound). Tracks: de Sitter / emergent spacetime, Born rule & linearity, tabletop quantum gravity (BMV), entropic / emergent gravity, algebraic background-independence. Synthesis verdict: PARTIAL coherence — two internally-coherent programs joined by thin bridges; the "it's all information" fusion explicitly refused. Registries extended (OP-40…47, A-20…23); one iteration-1 mischaracterization of Verlinde-EG lensing corrected.

2026-06-08 — Iteration 3: attempts + red-team. Shifted from surveying to attempting (31 agents): located the exact obstruction to a de Sitter entanglement first law; delivered the encode-vs-generate criterion; attempted the A+B program merge (narrow success, deep failure — OP-48 added); red-teamed the project's own posture (0 fatal, 4 serious weakenings accepted). Coherence verdict unchanged at PARTIAL; progress negative/structural; the central wager still not yet physics.

2026-06-08 — Iteration 4: attacks + literature refresh + convergence audit. Executed and closed the last open dS escape (a scoped cardinality near-no-go); settled OP-48 as a sharp disjunction; found the encode→generate escape near-vacuous under the standard readout (HYP-CKV-VACUITY). Web-verified literature refresh: muon g−2 now SM-consistent (WP2025). Convergence audit: trend diminishing, analytical saturation ≈1–3 iterations away, object-level confirmation not forecastable. Verdict unchanged for the third consecutive iteration.

2026-06-08 — Iteration 5: closing analytical push → analytical saturation; capstone. The designated verdict-flipping move (the OP-46 Lorentzian non-CKV readout) was executed and did not flip — producing the survey's first emergent Lorentzian metric, with its signature inherited from a symmetric vacuum. OP-48(c) settled as a clean conditional no-go; OP-44 sharpened to a regime boundary; both surviving dS routes failed for identified reasons. Analytical saturation declared per the iteration-4 criterion; the capstone Conclusion (Chapter 27) and the Experiment Watchlist (Chapter 28) were written. A same-day documentation pass closed three propagation gaps (no physics conclusions changed).

2026-06-10 — Iteration 6: new-input iteration; verdict unchanged a fifth consecutive time. First post-saturation iteration: deliberate injection of new 2023–2026 literature (22 agents; 10 adversarially-refereed tracks — 6 research, 4 attack; ~90 findings, all kept, zero fatal). Headline unchanged; hedges restructured: the no-test claim upgraded to a corollary of encode-not-generate (HYP-ENCODING-SCREEN, new third hedge); causal fermion systems added as a fourth readout family closing at the same signature-installation step; OP-48(c) horn C partially discharged; OP-49 closed negative at near-no-go grade; OP-50 added `[CONTESTED]`; watchlist refreshed (w(z) decision point compresses to 2027).